

Healthcare Process Flows — HL7 FHIR / HIMSS Healthcare Capability Model

Generated on 2026-07-08 | Domain: Health Care | Framework: HL7 FHIR R5 / HIMSS Healthcare Capability Model

HIMSS Healthcare Capability Model Reference

The HIMSS Healthcare Capability Model **v1.0** defines the standard business capabilities for healthcare delivery organizations. The table below lists all Level 2 capabilities grouped by Level 1 domain. Each process flow in this document is mapped to its relevant HIMSS capability codes.

 HIMSS Healthcare Capability Model overview

Source: [HIMSS Healthcare Capability Model](#) — used under CC BY-NC-SA 4.0

Patient Administration

Capability	Code	Description
Patient Registration & Demographics	HC001	Register, maintain, and manage patient demographic data and unique identifiers
Patient Identity Management	HC002	Manage patient identity matching, MPI, and record linkage across systems
Consent & Privacy Management	HC003	Manage patient consent directives, HIPAA authorizations, and privacy preferences
Patient Portal & Self-Service	HC004	Provide patient access to health records, appointments, billing, and secure messaging
Advance Directives & Goals of Care	HC005	Capture and manage advance directives, living wills, and goals of care documentation

Clinical Documentation & EHR

Capability	Code	Description
Clinical Note Management	HC010	Create, manage, and store clinical documentation including H&P, progress notes, and summaries
Order Entry & Management (CPOE)	HC011	Computerized provider order entry for medications, labs, radiology, and procedures
Problem List & Medical History	HC012	Maintain patient problem list, past medical history, allergies, and social determinants

Laboratory & Diagnostics

Capability	Code	Description
Lab Order & Specimen Management	HC020	Order laboratory tests, collect specimens, and track throughout the testing lifecycle
Lab Results & Reporting	HC021	Receive, validate, and report laboratory results with critical value alerting
Radiology & Imaging Management	HC022	Order imaging studies, schedule modalities, and distribute reports and images

Pharmacy & Medication Management

Capability	Code	Description
Medication Ordering & Reconciliation	HC030	Order medications, perform drug-drug interaction checks, and reconcile across transitions
Pharmacy Dispensing & Verification	HC031	Dispense medications, perform verification checks, and manage formularies
Medication Administration (MAR)	HC032	Document medication administration, barcode scanning, and dosage tracking

Billing & Revenue Cycle

Capability	Code	Description
Charge Capture & Coding	HC040	Capture charges, apply ICD-10/CPT/HCPCS coding, and validate claim readiness
Claims Submission & Adjudication	HC041	Submit claims to payers, manage electronic clearinghouse exchanges, and track status
Payment Posting & Denial Management	HC042	Post payments, manage denials, process appeals, and reconcile accounts

Scheduling & Access

Capability	Code	Description
Appointment Scheduling & Check-in	HC050	Schedule patient appointments, manage waitlists, and process check-in/check-out
Referral Management & Authorizations	HC051	Manage referrals, obtain prior authorizations, and coordinate specialist access

Care Coordination & Population Health

Capability	Code	Description
Care Plan Development & Management	HC060	Develop, share, and track interdisciplinary care plans across the care continuum
Transition of Care & Discharge	HC061	Manage care transitions, discharge planning, and post-acute follow-up coordination
Population Health Risk Stratification	HC062	Stratify patient populations by risk, manage outreach, and track quality gaps

Health Information Exchange

Capability	Code	Description
HIE Document Sharing & Query	HC070	Query and share clinical documents across organizational boundaries via HIE
Public Health Reporting & Immunization	HC071	Report immunizations, communicable diseases, and vital records to public health agencies

Supply Chain & Inventory

Capability	Code	Description
Medical Supply Procurement	HC080	Manage procurement, vendor contracts, and purchase orders for medical supplies
Inventory & Asset Management	HC081	Track medical inventory levels, manage par levels, and locate medical equipment

Regulatory Compliance & Quality

Capability	Code	Description
Quality Measure Reporting (HEDIS/MIPS)	HC090	Calculate and report quality measures for HEDIS, MIPS, and value-based programs
Regulatory Audit & Compliance	HC091	Manage regulatory audits, accreditation surveys, and compliance documentation
Incident & Patient Safety Reporting	HC092	Report and track patient safety incidents, adverse events, and root cause analyses

Table of Contents

- [Patient Administration - HC001, HC002, HC003, HC004, HC005](#)
 - [1.1 Patient Registration & Demographics](#)
 - [1.2 Patient Identity & MPI Management](#)
 - [1.3 Patient Portal & Self-Service](#)
 - [1.4 Consent & Privacy Management](#)
- [Clinical Documentation & EHR - HC010, HC011, HC012](#)
 - [2.1 Clinical Notes & H&P Documentation](#)
 - [2.2 Order Entry & Management \(CPOE\)](#)
 - [2.3 Problem List & Medical History](#)
- [Laboratory & Diagnostics - HC020, HC021, HC022](#)
 - [3.1 Lab Order & Specimen Collection](#)
 - [3.2 Lab Results Reporting](#)
 - [3.3 Radiology & Imaging Management](#)
- [Pharmacy & Medication Management - HC030, HC031, HC032](#)
 - [4.1 Medication Ordering & Reconciliation](#)
 - [4.2 Pharmacy Dispensing & Verification](#)
 - [4.3 Medication Administration \(MAR\)](#)
- [Billing & Revenue Cycle Management - HC040, HC041, HC042](#)
 - [5.1 Charge Capture & Coding](#)
 - [5.2 Claims Submission & Adjudication](#)
 - [5.3 Payment Posting & Denial Management](#)
- [Appointment Scheduling & Access - HC050, HC051](#)
 - [6.1 Patient Scheduling & Check-in](#)
 - [6.2 Referral Management & Authorizations](#)
- [Care Coordination & Population Health - HC060, HC061, HC062](#)
 - [7.1 Care Plan Development & Management](#)
 - [7.2 Transition of Care & Discharge](#)
 - [7.3 Population Health Risk Stratification](#)
- [Health Information Exchange - HC070, HC071](#)
 - [8.1 HIE Document Sharing & Query](#)
 - [8.2 Public Health Reporting & Immunization](#)

- [Supply Chain & Inventory Management - HC080, HC081](#)
 - [9.1 Medical Supply Procurement](#)
 - [9.2 Inventory & Asset Management](#)
- [Regulatory Compliance & Quality - HC090, HC091, HC092](#)
 - [10.1 Quality Measure Reporting \(HEDIS/MIPS\)](#)
 - [10.2 Regulatory Audit & Compliance](#)
 - [10.3 Incident & Patient Safety Reporting](#)

Test Data

The following test data is used consistently across all sequence diagrams:

Entity	Field	Value
Patient	patient_id	PT-2026-001
	name	John Doe
	dob	1985-04-12
	mrn	MRN-2026-0001
	gender	Male
	phone	555-0100
Provider	provider_id	PROV-101
	name	Dr. Sarah Williams
	specialty	Internal Medicine
	npi	1234567890
	department	Medicine
Encounter	encounter_id	ENC-2026-001
	type	Inpatient
	start_date	2026-06-15
	end_date	2026-06-18
	patient_id	PT-2026-001
	provider_id	PROV-101
Order	order_id	ORD-2026-001
	type	Lab
	status	Active
	patient_id	PT-2026-001
	ordering_provider	PROV-101
Observation	observation_id	OBS-2026-001
	type	Blood Glucose
	value	98
	unit	mg/dL
	status	Final
	patient_id	PT-2026-001
Medication	med_id	MED-2026-001
	name	Metformin 500mg
	dose	500mg
	route	Oral
	frequency	BID
	patient_id	PT-2026-001
Claim	claim_id	CLM-2026-001
	type	Professional
	amount	1250.00
	status	Submitted

Entity	Field	Value
Appointment	patient_id	PT-2026-001
	payer_id	PAYER-001
	appt_id	APT-2026-001
	type	Office Visit
	date	2026-07-10
	status	Scheduled
CarePlan	patient_id	PT-2026-001
	provider_id	PROV-101
	careplan_id	CP-2026-001
	type	Diabetes Management
	status	Active
	patient_id	PT-2026-001
Allergy	start_date	2026-06-01
	allergy_id	ALG-2026-001
	substance	Penicillin
	reaction	Rash
	severity	Moderate
	patient_id	PT-2026-001
Insurance	insurance_id	INS-001
	payer_name	Blue Cross Blue Shield
	plan_type	PPO
	member_id	MBR-2026-001
	group_num	GRP-7890
	patient_id	PT-2026-001
LabResult	lab_result_id	LR-2026-001
	test_name	CBC
	result	Normal
	collected_date	2026-06-15
	resulted_date	2026-06-16
	order_id	ORD-2026-001
ImagingStudy	study_id	IMG-2026-001
	modality	X-Ray
	body_part	Chest
	status	Completed
	patient_id	PT-2026-001
Immunization	imm_id	IMM-2026-001
	vaccine	Influenza
	dose	0.5mL

Entity	Field	Value
Consent	admin_date	2026-09-01
	patient_id	PT-2026-001
	consent_id	CON-2026-001
	type	Treatment Consent
	status	Signed
	signed_date	2026-06-10
Referral	patient_id	PT-2026-001
	referral_id	REF-2026-001
	type	Specialist
	from_provider	PROV-101
	to_specialty	Cardiology
	status	Pending
SupplyItem	patient_id	PT-2026-001
	item_id	SUP-2026-001
	name	Surgical Gloves - Box
	category	PPE
	quantity	500
	reorder_level	100
Incident	incident_id	INC-2026-001
	type	Medication Error
	severity	Moderate
	status	Investigating
	patient_id	PT-2026-001
	measure_id	QM-2026-001
QualityMeasure	name	Diabetes HbA1c Control
	type	HEDIS
	score	85.5
	reporting_period	2026-Q2

Patient Administration - HC001 HC002 HC003 HC004 HC005

End-to-end management of patient identity, registration, consent, and self-service access across the healthcare enterprise.

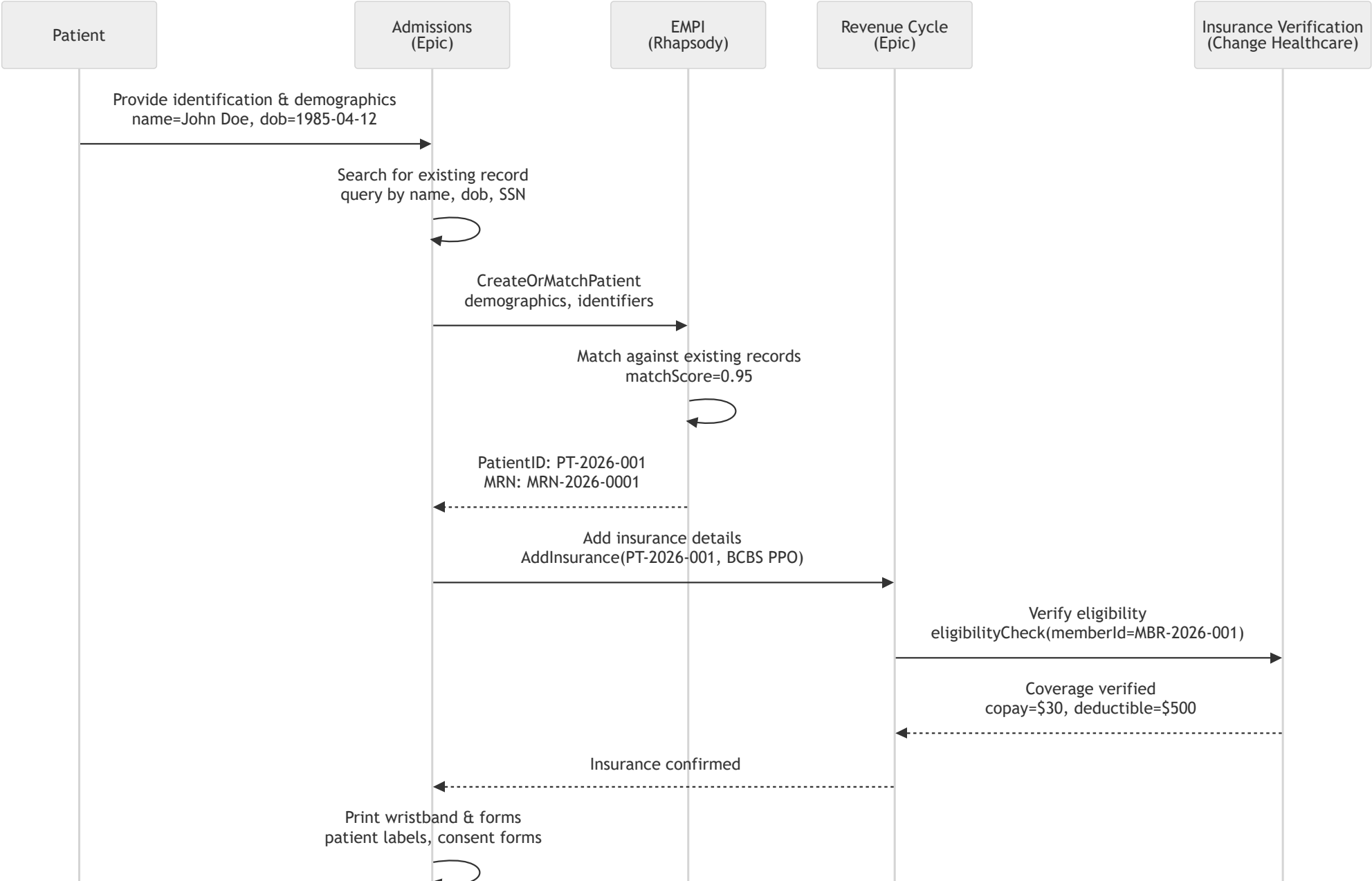
1.1 Patient Registration & Demographics

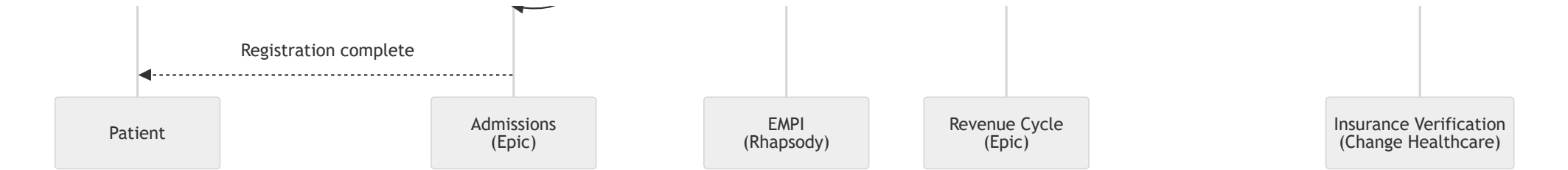
Description: A new patient arrives at the hospital and is registered in the admissions system. Patient demographics are captured, insurance is verified, and a unique medical record number is assigned.

Actors: Patient, Registration Clerk

Systems: Admissions (Epic), Enterprise Master Patient Index (Rhapsody EMPI), Revenue Cycle (Epic), Insurance Verification (Change Healthcare)

Sequence Diagram





Integration Details

Flow	Entity	Info Flow	Source	Target	Source Conn	Target Conn	Pattern	Complexity
Patient Registration	Patient	RegisterPatient	Admissions (Epic)	Enterprise Master Patient Index (Rhapsody EMPI)	Epic REST API (FHIR R5)	EMPI SOAP API (IHE PIX)	API-led (Real-time)	Medium
Patient Registration	Patient	AddInsurance	Admissions (Epic)	Revenue Cycle (Epic)	Epic REST API (FHIR)	Epic REST API (FHIR)	API-led (Real-time)	Simple
Patient Registration	Insurance	VerifyEligibility	Revenue Cycle (Epic)	Insurance Verification (Change Healthcare)	Epic REST API (FHIR)	Change Healthcare REST API (OAuth2)	API-led (Real-time)	Medium
Patient Registration	Patient	PrintWristband	Admissions (Epic)	Label Printing System	Epic API (HL7 v2)	Print Server (RAW TCP)	Event-driven	Simple
Patient Registration	Encounter	CreateEncounter	Admissions (Epic)	EMR (Epic)	Epic REST API (FHIR)	Epic REST API (FHIR)	API-led (Real-time)	Simple
Patient Registration	Guarantor	AddGuarantor	Admissions (Epic)	Revenue Cycle (Epic)	Epic REST API (FHIR)	Epic REST API (FHIR)	API-led (Real-time)	Simple

Acceptance Criteria

AC ID	Description	Related Info Flow	Expected Outcome	Priority
AC-1.1-01	Data integrity: Patient demographic data transferred from Admissions to MPI via RegisterPatient must be complete and consistent	RegisterPatient	Patient records in MPI match source; no data loss; reconciliation report confirms accuracy	Critical
AC-1.1-02	Data integrity: Insurance data transferred from Admissions to Revenue Cycle via AddInsurance must be complete and consistent	AddInsurance	Insurance records in Revenue Cycle match source; no data loss; reconciliation report confirms accuracy	High
AC-1.1-03	Data integrity: Eligibility data transferred from Revenue Cycle to Insurance Verification via VerifyEligibility must be complete and consistent	VerifyEligibility	Eligibility records in Revenue Cycle match source; no data loss; reconciliation report confirms accuracy	High
AC-1.1-04	Error handling: Failed transactions between Admissions and MPI are logged with error context and trigger automatic retry or alert	RegisterPatient	Errors logged with timestamp, HTTP status, request payload excerpt; retry up to 3 times; alert sent after consecutive failures	Critical
AC-1.1-05	Security: All endpoints enforce authentication (Epic REST API (FHIR R5), EMPI SOAP API (IHE PIX), ...) and encrypt data in transit via TLS 1.2+	All flows	Unauthenticated requests rejected with 401; tokens validated per call; invalid/expired tokens logged	Critical

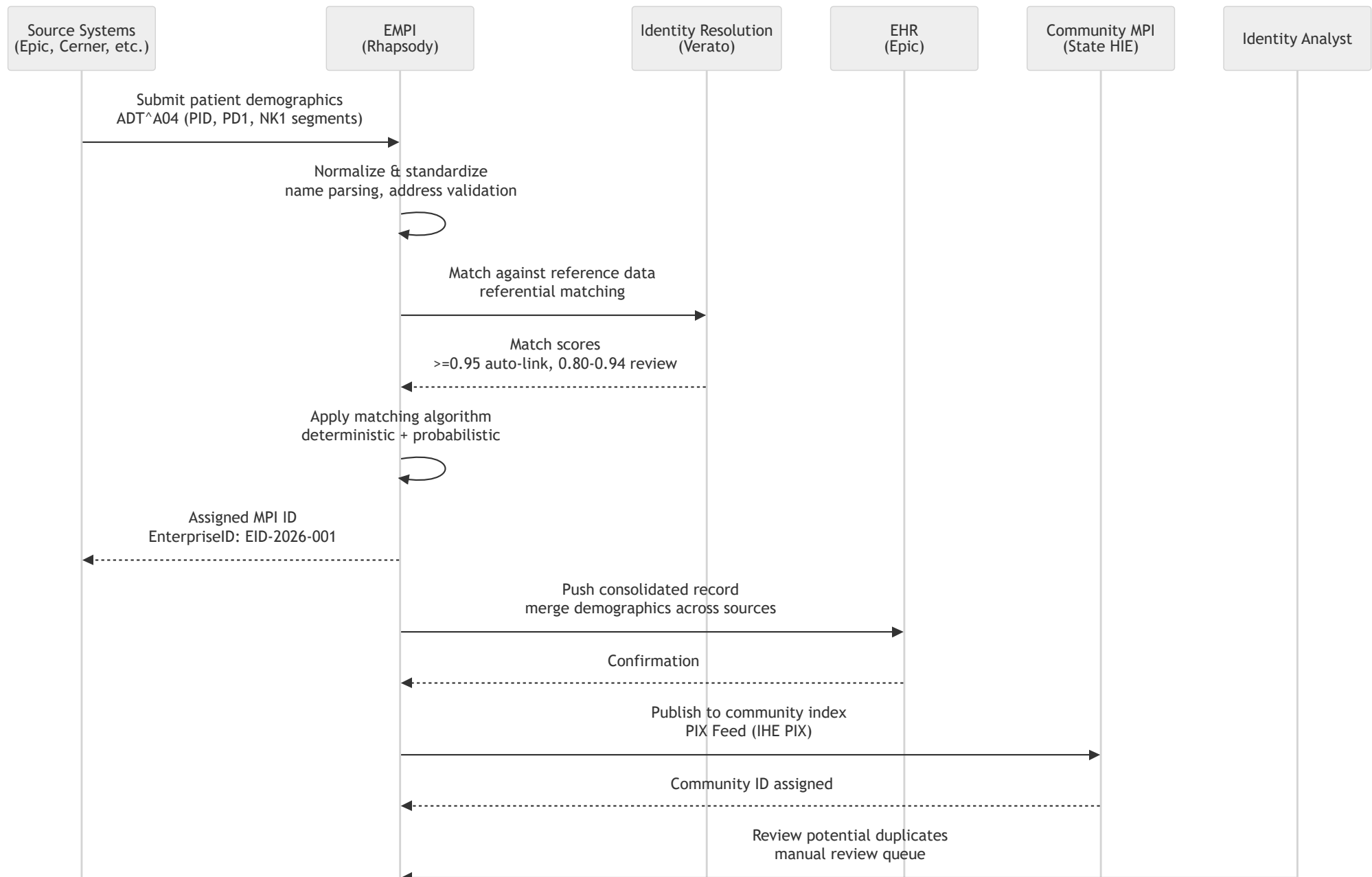
1.2 Patient Identity & MPI Management

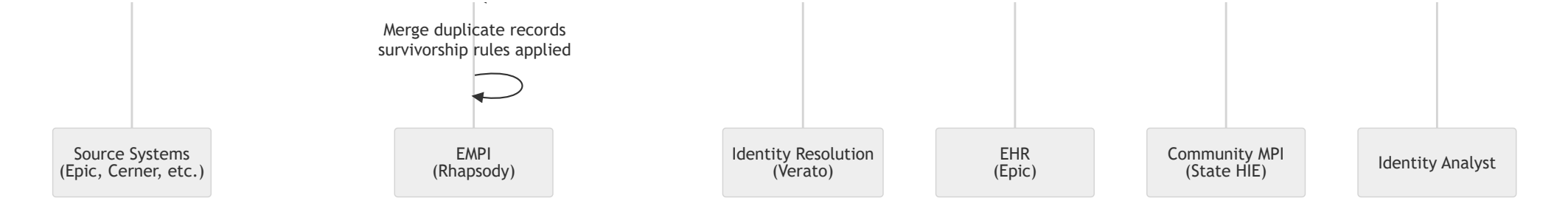
Description: The Enterprise Master Patient Index receives patient data from multiple sources, performs probabilistic matching, and manages duplicate detection and consolidation across the health system.

Actors: Identity Analyst, HIM Specialist

Systems: Enterprise Master Patient Index (Rhapsody EMPI), EHR (Epic), Community MPI (State HIE), Identity Resolution (Verato)

Sequence Diagram





Integration Details

Flow	Entity	Info Flow	Source	Target	Source Conn	Target Conn	Pattern	Complexity
Identity Management	Patient	CreateOrMatchPatient	Source Systems (Epic, Cerner, etc.)	Enterprise Master Patient Index (Rhapsody EMPI)	HL7 v2 ADT (MLLP)	EMPI SOAP API (IHE PIX)	Event-driven	Medium
Identity Management	Patient	ReferentialMatch	Enterprise Master Patient Index (Rhapsody EMPI)	Identity Resolution (Verato)	EMPI REST API	Verato REST API (OAuth2)	API-led (Real-time)	High
Identity Management	Patient	PushConsolidated	Enterprise Master Patient Index (Rhapsody EMPI)	EHR (Epic)	EMPI SOAP API (IHE PDQ)	Epic REST API (FHIR)	Event-driven	Medium
Identity Management	Patient	PublishToCommunity	Enterprise Master Patient Index (Rhapsody EMPI)	Community MPI (State HIE)	EMPI SOAP API (IHE PIX)	State HIE SOAP (IHE PIX)	Batch (Scheduled)	High
Identity Management	Patient	MergeDuplicates	Enterprise Master Patient Index (Rhapsody EMPI)	Enterprise Master Patient Index (Rhapsody EMPI)	EMPI Admin Console	EMPI API	API-led (Real-time)	High
Identity Management	Patient	QueryMatches	Enterprise Master Patient Index (Rhapsody EMPI)	EHR (Epic)	EMPI REST API	Epic REST API (FHIR)	API-led (Real-time)	Medium

Acceptance Criteria

AC ID	Description	Related Info Flow	Expected Outcome	Priority
AC-1.2-01	Data integrity: Patient identity data transferred from Source Systems to EMPI via CreateOrMatchPatient must be complete and consistent	CreateOrMatchPatient	Patient records in EMPI match source; no data loss; reconciliation report confirms accuracy	Critical
AC-1.2-02	Data integrity: Consolidated patient data transferred from EMPI to EHR via PushConsolidated must be complete and consistent	PushConsolidated	Patient records in EHR match EMPI source; no data loss; reconciliation report confirms accuracy	Critical
AC-1.2-03	Error handling: Failed identity matching transactions are logged with error context and trigger manual review or alert	CreateOrMatchPatient	Errors logged with timestamp, match score, request payload; manual review queue populated	Critical

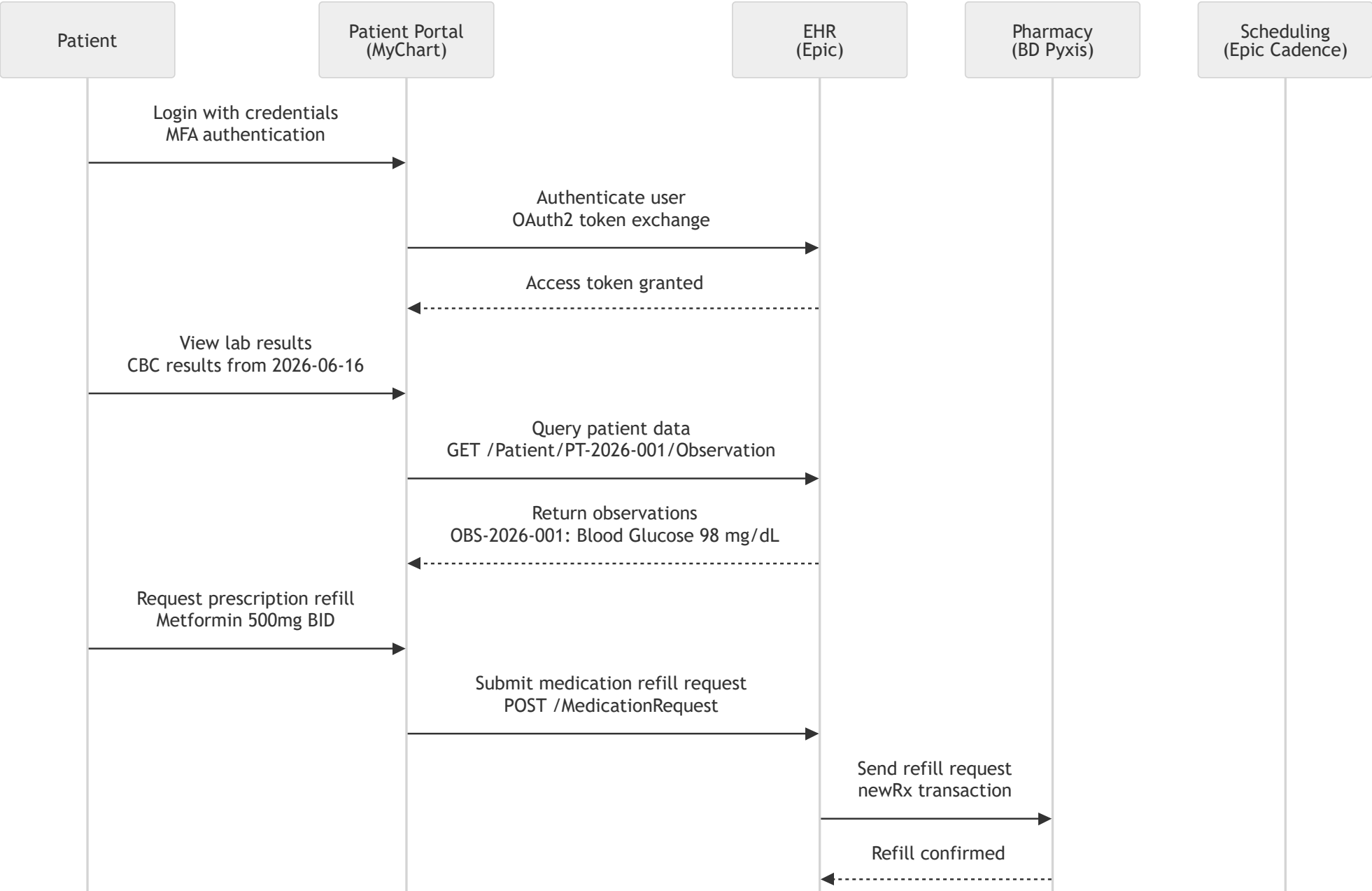
1.3 Patient Portal & Self-Service

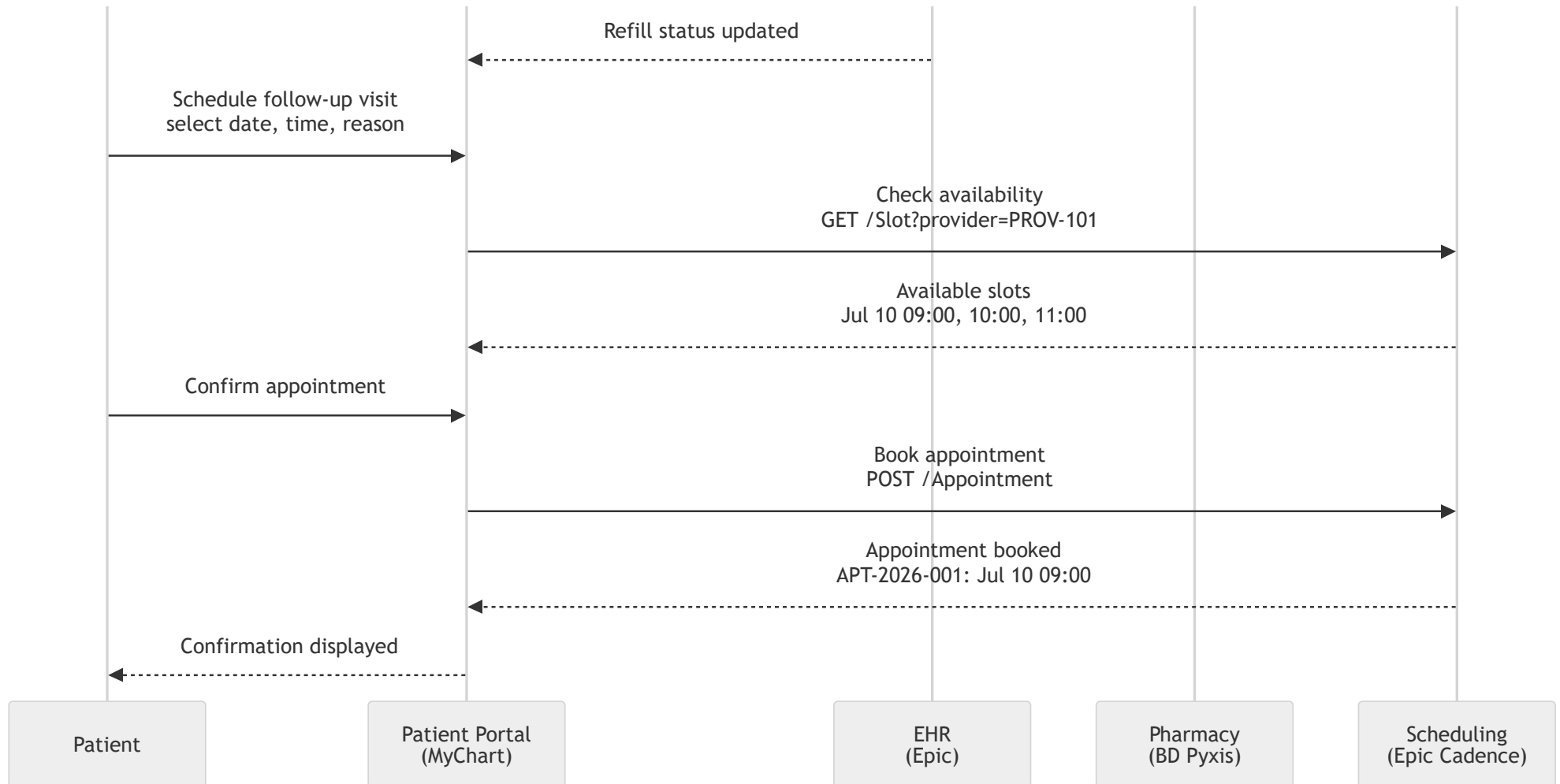
Description: A patient accesses the patient portal to view health records, communicate with providers, request prescription refills, and manage appointments.

Actors: Patient

Systems: Patient Portal (MyChart), EHR (Epic), Pharmacy System (BD Pyxis), Scheduling (Epic Cadence)

Sequence Diagram





Integration Details

Flow	Entity	Info Flow	Source	Target	Source Conn	Target Conn	Pattern	Complexity
Portal Access	Patient	AuthenticateUser	Patient Portal (MyChart)	EHR (Epic)	MyChart REST API	Epic REST API (FHIR)	API-led (Real-time)	Medium
Lab Results	Observation	QueryObservations	Patient Portal (MyChart)	EHR (Epic)	MyChart REST API	Epic REST API (FHIR)	API-led (Real-time)	Simple
Refill Request	Medication	RequestRefill	Patient Portal (MyChart)	EHR (Epic)	MyChart REST API	Epic REST API (FHIR)	API-led (Real-time)	Medium
Refill Notification	Medication	SendRefillRx	EHR (Epic)	Pharmacy System (BD Pyxis)	Epic REST API (FHIR)	Pyxis REST API (NCPDP)	Event-driven	Medium
Appointment Booking	Appointment	BookAppointment	Patient Portal (MyChart)	Scheduling (Epic Cadence)	MyChart REST API	Epic Cadence REST API (FHIR)	API-led (Real-time)	Simple
Slot Availability	Slot	CheckAvailability	Patient Portal (MyChart)	Scheduling (Epic Cadence)	MyChart REST API	Epic Cadence REST API (FHIR)	API-led (Real-time)	Simple

Acceptance Criteria

AC ID	Description	Related Info Flow	Expected Outcome	Priority
AC-1.3-01	Data integrity: Observation data transferred from Patient Portal to EHR via QueryObservations must be complete and consistent	QueryObservations	Observation records returned match source; no data loss	Medium
AC-1.3-02	Data integrity: Refill data transferred from Patient Portal to EHR via RequestRefill must be complete and consistent	RequestRefill	Refill requests in EHR match source; no data loss	High
AC-1.3-03	Data integrity: Appointment data transferred from Patient Portal to Scheduling via BookAppointment must be complete and consistent	BookAppointment	Appointment records in Scheduling match source; no data loss	Medium
AC-1.3-04	Security: Patient Portal enforces MFA and all APIs use OAuth2 with appropriate scopes	All flows	Unauthorized access to patient data rejected with 403; scope violations logged	Critical

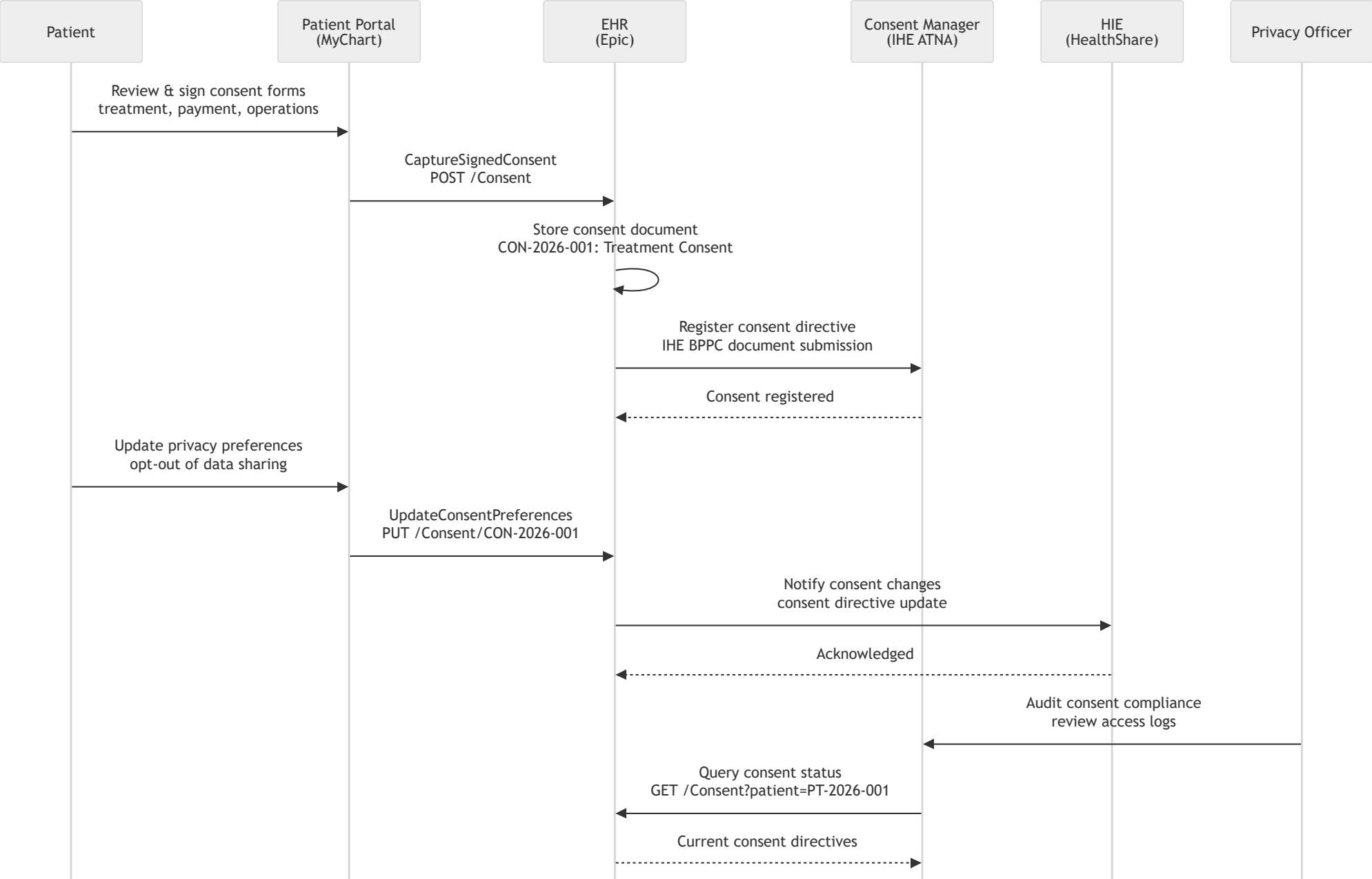
1.4 Consent & Privacy Management

Description: Patient consent directives are captured, stored, and enforced across the health system. HIPAA authorizations are managed and shared with downstream systems.

Actors: Patient, Privacy Officer

Systems: Patient Portal (MyChart), EHR (Epic), Consent Management (IHE ATNA), HIE (HealthShare)

Sequence Diagram





Integration Details

Flow	Entity	Info Flow	Source	Target	Source Conn	Target Conn	Pattern	Complexity
Consent Capture	Consent	CaptureSignedConsent	Patient Portal (MyChart)	EHR (Epic)	MyChart REST API	Epic REST API (FHIR)	API-led (Real-time)	Medium
Consent Registration	Consent	RegisterConsent	EHR (Epic)	Consent Manager (IHE ATNA)	Epic REST API (FHIR)	ATNA SOAP (IHE BPPC)	API-led (Real-time)	High
Privacy Update	Consent	UpdateConsentPreferences	Patient Portal (MyChart)	EHR (Epic)	MyChart REST API	Epic REST API (FHIR)	API-led (Real-time)	Medium
Consent Notification	Consent	NotifyConsentChange	EHR (Epic)	HIE (HealthShare)	Epic REST API (FHIR)	HealthShare REST API (IHE BPPC)	Event-driven	Medium
Consent Audit	Consent	QueryConsentStatus	Consent Manager (IHE ATNA)	EHR (Epic)	ATNA SOAP	Epic REST API (FHIR)	API-led (Real-time)	Medium

Acceptance Criteria

AC ID	Description	Related Info Flow	Expected Outcome	Priority
AC-1.4-01	Data integrity: Consent data transferred from Patient Portal to EHR via CaptureSignedConsent must be complete and consistent	CaptureSignedConsent	Consent records in EHR match source; no data loss; reconciliation report confirms accuracy	Critical
AC-1.4-02	Data integrity: Consent directive data transferred from EHR to Consent Manager via RegisterConsent must be complete and consistent	RegisterConsent	Consent directives in Consent Manager match source; no data loss; reconciliation report confirms accuracy	High
AC-1.4-03	Error handling: Failed consent registration transactions are logged with error context and trigger manual review	RegisterConsent	Errors logged with timestamp, HTTP status, request payload excerpt; manual review queue populated	Critical

AC ID	Description	Related Info Flow	Expected Outcome	Priority
-------	-------------	-------------------	------------------	----------

Clinical Documentation & EHR - HC010 HC011 HC012

Comprehensive clinical documentation management including structured notes, computerized provider order entry, and patient problem lists.

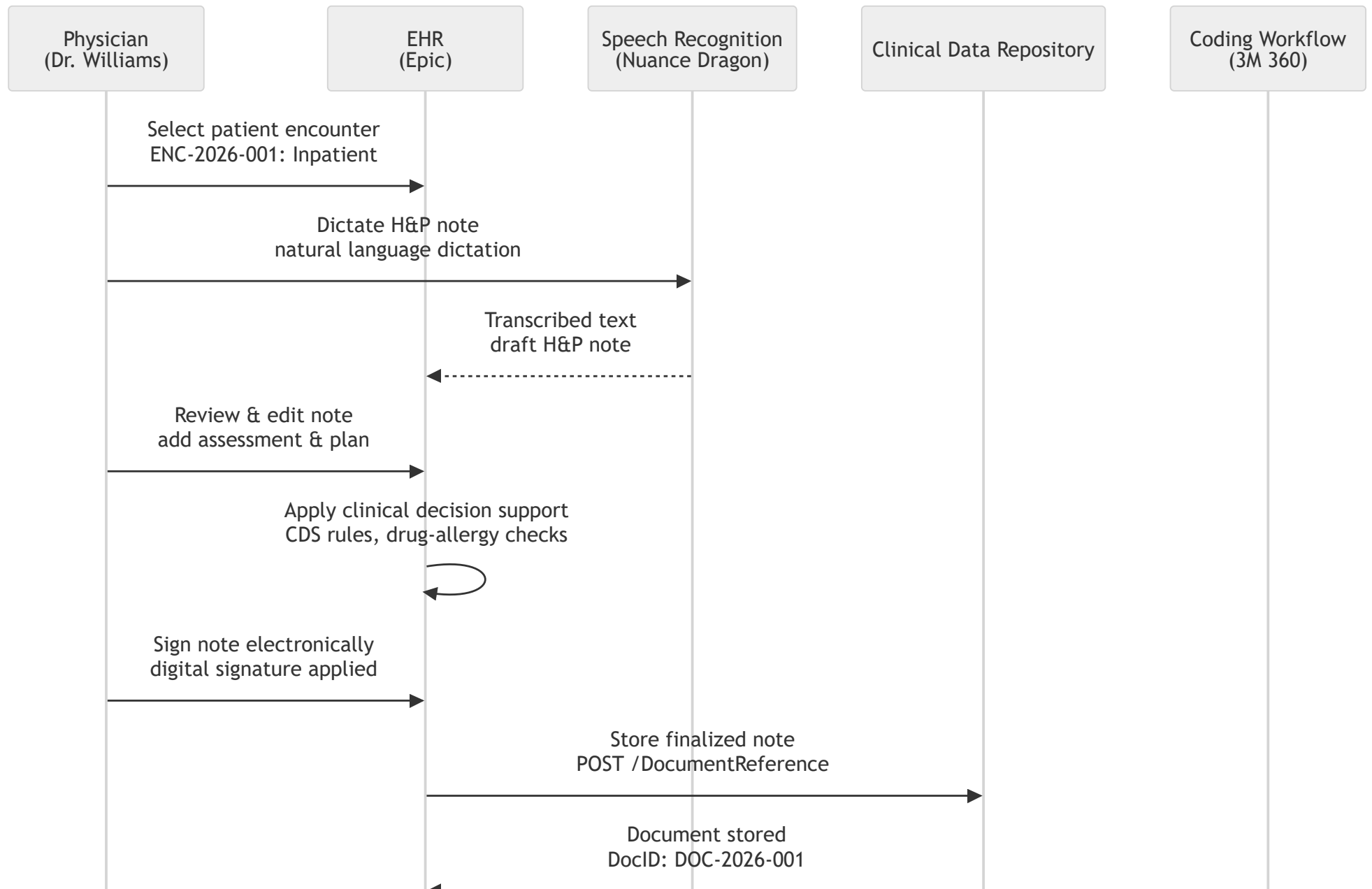
2.1 Clinical Notes & H&P Documentation

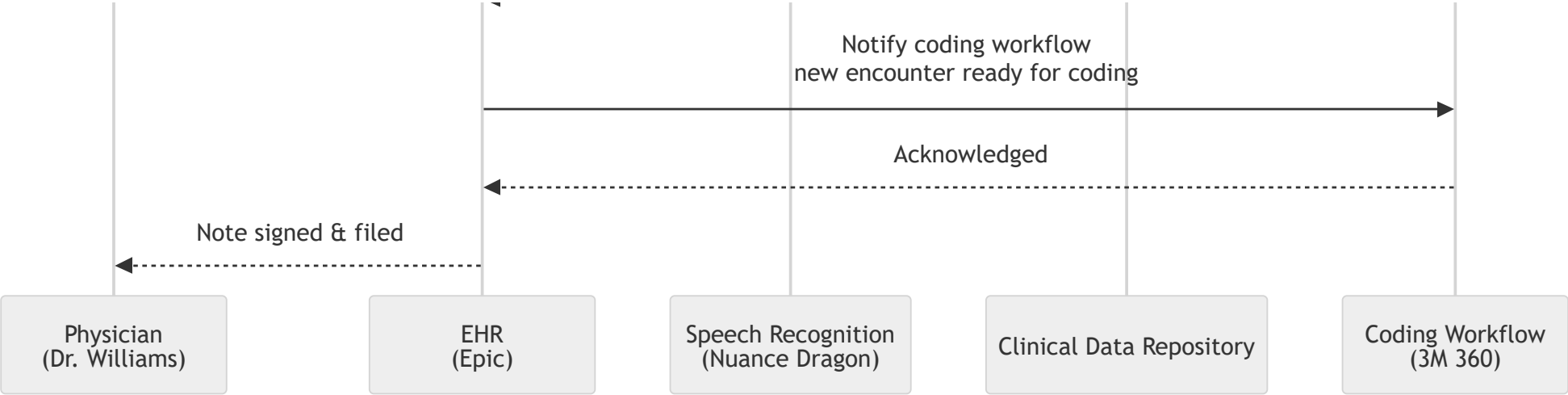
Description: A physician documents a patient encounter including history and physical examination findings, assessment, and plan. The note is signed and stored in the clinical data repository.

Actors: Physician (Dr. Sarah Williams), Medical Coder

Systems: EHR (Epic), Clinical Data Repository (CDR), Speech Recognition (Nuance Dragon), Coding Workflow (3M 360)

Sequence Diagram





Integration Details

Flow	Entity	Info Flow	Source	Target	Source Conn	Target Conn	Pattern	Complexity
Clinical Documentation	Encounter	DictateNote	Speech Recognition (Nuance Dragon)	EHR (Epic)	Dragon REST API	Epic REST API (FHIR)	API-led (Real-time)	Medium
Clinical Documentation	Document	StoreNote	EHR (Epic)	Clinical Data Repository (CDR)	Epic REST API (FHIR)	CDR REST API (IHE XDS)	API-led (Real-time)	Medium
Clinical Documentation	Encounter	SignNote	EHR (Epic)	EHR (Epic)	Epic REST API (FHIR)	Epic REST API (FHIR)	API-led (Real-time)	Simple
Clinical Documentation	Encounter	NotifyCoding	EHR (Epic)	Coding Workflow (3M 360)	Epic REST API (FHIR)	3M REST API (HL7 FHIR)	Event-driven	Medium
Clinical Documentation	Encounter	ApplyCDS	EHR (Epic)	EHR (Epic)	Epic Clinical Rules Engine	Epic REST API (FHIR)	API-led (Real-time)	Medium

Acceptance Criteria

AC ID	Description	Related Info Flow	Expected Outcome	Priority
AC-2.1-01	Data integrity: Clinical note data transferred from EHR to CDR via StoreNote must be complete and consistent	StoreNote	Clinical notes in CDR match source; no data loss; reconciliation report confirms accuracy	Critical
AC-2.1-02	Data integrity: Dictated note data transferred from Dragon to EHR via DictateNote must be complete and consistent	DictateNote	Dictated notes in EHR match source; no data loss; reconciliation report confirms accuracy	High
AC-2.1-03	Error handling: Failed clinical note storage transactions are logged with error context and trigger automatic retry or alert	StoreNote	Errors logged with timestamp, HTTP status, request payload excerpt; retry up to 3 times; alert sent after consecutive failures	Critical

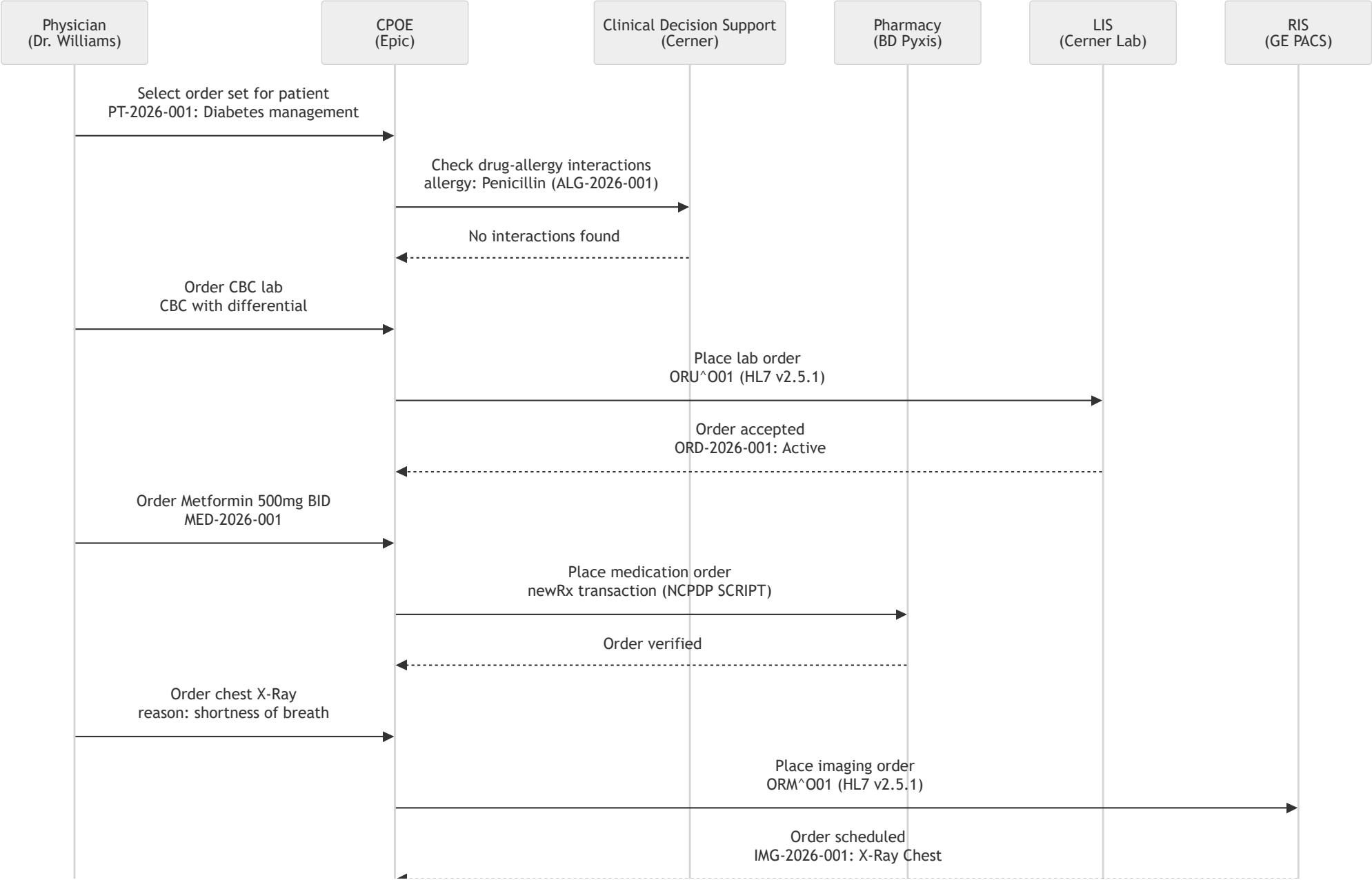
2.2 Order Entry & Management (CPOE)

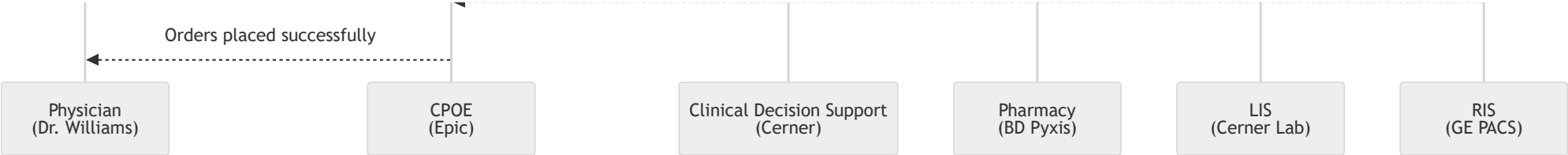
Description: A physician enters a set of orders for an inpatient including lab tests, medications, and imaging studies through the computerized provider order entry system.

Actors: Physician (Dr. Sarah Williams), Nurse

Systems: CPOE (Epic), Pharmacy System (BD Pyxis), LIS (Cerner Lab), RIS (GE PACS), Clinical Decision Support (Cerner CDS)

Sequence Diagram





Integration Details

Flow	Entity	Info Flow	Source	Target	Source Conn	Target Conn	Pattern	Complexity
CPOE	Order	CheckDrugAllergy	CPOE (Epic)	Clinical Decision Support (Cerner CDS)	Epic REST API (FHIR)	Cerner CDS REST API (CDS Hooks)	API-led (Real-time)	Medium
CPOE	Order	PlaceLabOrder	CPOE (Epic)	LIS (Cerner Lab)	Epic REST API (FHIR)	Cerner Lab HL7 v2.5.1 (MLLP)	API-led (Real-time)	Medium
CPOE	Medication	PlaceMedicationOrder	CPOE (Epic)	Pharmacy System (BD Pyxis)	Epic REST API (FHIR)	Pyxis NCPDP SCRIPT (SOAP)	API-led (Real-time)	Medium
CPOE	Order	PlaceImagingOrder	CPOE (Epic)	RIS (GE PACS)	Epic REST API (FHIR)	GE RIS HL7 v2.5.1 (MLLP)	API-led (Real-time)	Medium
CPOE	Order	OrderSetSelection	CPOE (Epic)	CPOE (Epic)	Epic REST API (FHIR)	Epic REST API (FHIR)	API-led (Real-time)	Simple

Acceptance Criteria

AC ID	Description	Related Info Flow	Expected Outcome	Priority
AC-2.2-01	Data integrity: Lab order data transferred from CPOE to LIS via PlaceLabOrder must be complete and consistent	PlaceLabOrder	Lab orders in LIS match source; no data loss; reconciliation report confirms accuracy	Critical
AC-2.2-02	Data integrity: Medication order data transferred from CPOE to Pharmacy via PlaceMedicationOrder must be complete and consistent	PlaceMedicationOrder	Medication orders in Pharmacy match source; no data loss; reconciliation report confirms accuracy	Critical
AC-2.2-03	Data integrity: Imaging order data transferred from CPOE to RIS via PlaceImagingOrder must be complete and consistent	PlaceImagingOrder	Imaging orders in RIS match source; no data loss; reconciliation report confirms accuracy	Critical
AC-2.2-04	Security: All CPOE transactions enforce provider authentication and electronic signature	All flows	Unauthenticated orders rejected; orders require valid provider digital signature	Critical

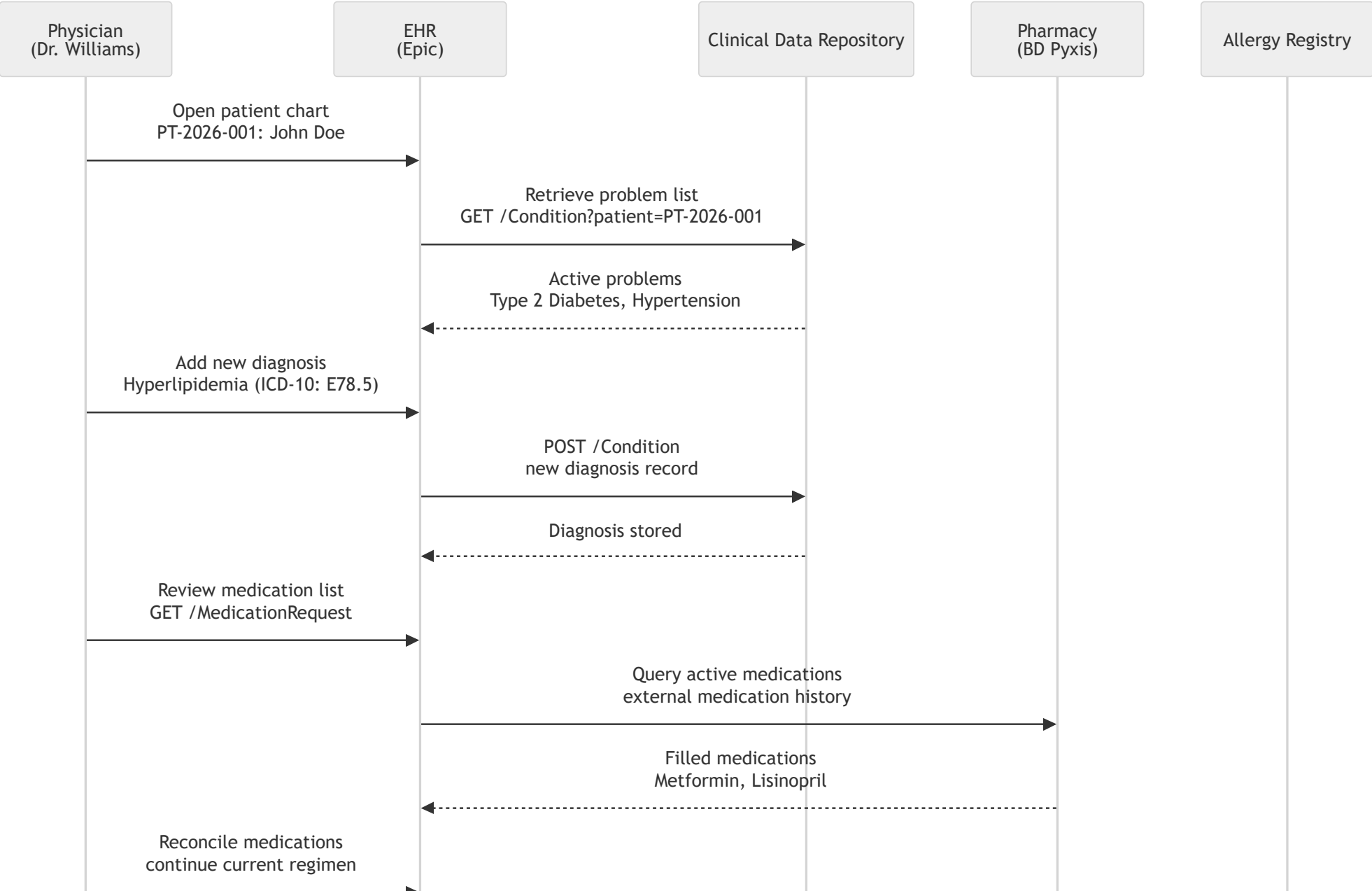
2.3 Problem List & Medical History

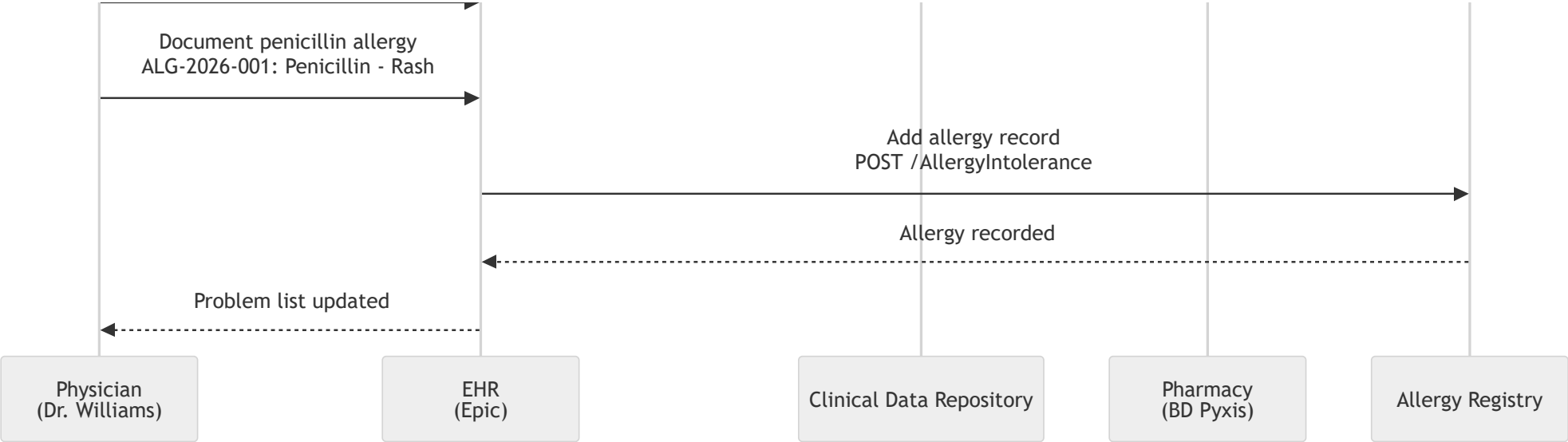
Description: The physician updates the patient's problem list, adds new diagnoses, reconciles medications, and documents allergies during an encounter.

Actors: Physician (Dr. Sarah Williams), Clinical Informaticist

Systems: EHR (Epic), Clinical Data Repository (CDR), Pharmacy System (BD Pyxis), Allergy Registry

Sequence Diagram





Integration Details

Flow	Entity	Info Flow	Source	Target	Source Conn	Target Conn	Pattern	Complexity
Problem List	Condition	RetrieveProblemList	EHR (Epic)	Clinical Data Repository (CDR)	Epic REST API (FHIR)	CDR REST API (FHIR)	API-led (Real-time)	Simple
Problem List	Condition	AddDiagnosis	EHR (Epic)	Clinical Data Repository (CDR)	Epic REST API (FHIR)	CDR REST API (FHIR)	API-led (Real-time)	Simple
Medication Reconcile	Medication	QueryActiveMeds	EHR (Epic)	Pharmacy System (BD Pyxis)	Epic REST API (FHIR)	Pyxis REST API (NCPDP)	API-led (Real-time)	Medium
Allergy	AllergyIntolerance	DocumentAllergy	EHR (Epic)	Allergy Registry	Epic REST API (FHIR)	Allergy Registry REST API	API-led (Real-time)	Medium
Medication Reconcile	Medication	ReconcileMeds	EHR (Epic)	EHR (Epic)	Epic REST API (FHIR)	Epic REST API (FHIR)	API-led (Real-time)	Medium

Acceptance Criteria

AC ID	Description	Related Info Flow	Expected Outcome	Priority
AC-2.3-01	Data integrity: Diagnosis data transferred from EHR to CDR via AddDiagnosis must be complete and consistent	AddDiagnosis	Diagnosis records in CDR match source; no data loss; reconciliation report confirms accuracy	High
AC-2.3-02	Data integrity: Medication data transferred from EHR to Pharmacy via QueryActiveMeds must be complete and consistent	QueryActiveMeds	Medication records in EHR match Pharmacy source; no data loss; reconciliation report confirms accuracy	High
AC-2.3-03	Data integrity: Allergy data transferred from EHR to Allergy Registry via DocumentAllergy must be complete and consistent	DocumentAllergy	Allergy records in Registry match source; no data loss; reconciliation r	High

AC ID	Description	Related Info Flow	Expected Outcome	Priority
			eport confirms accuracy	

Laboratory & Diagnostics - HC020 HC021 HC022

End-to-end laboratory and diagnostic imaging management from order entry through results reporting and clinical interpretation.

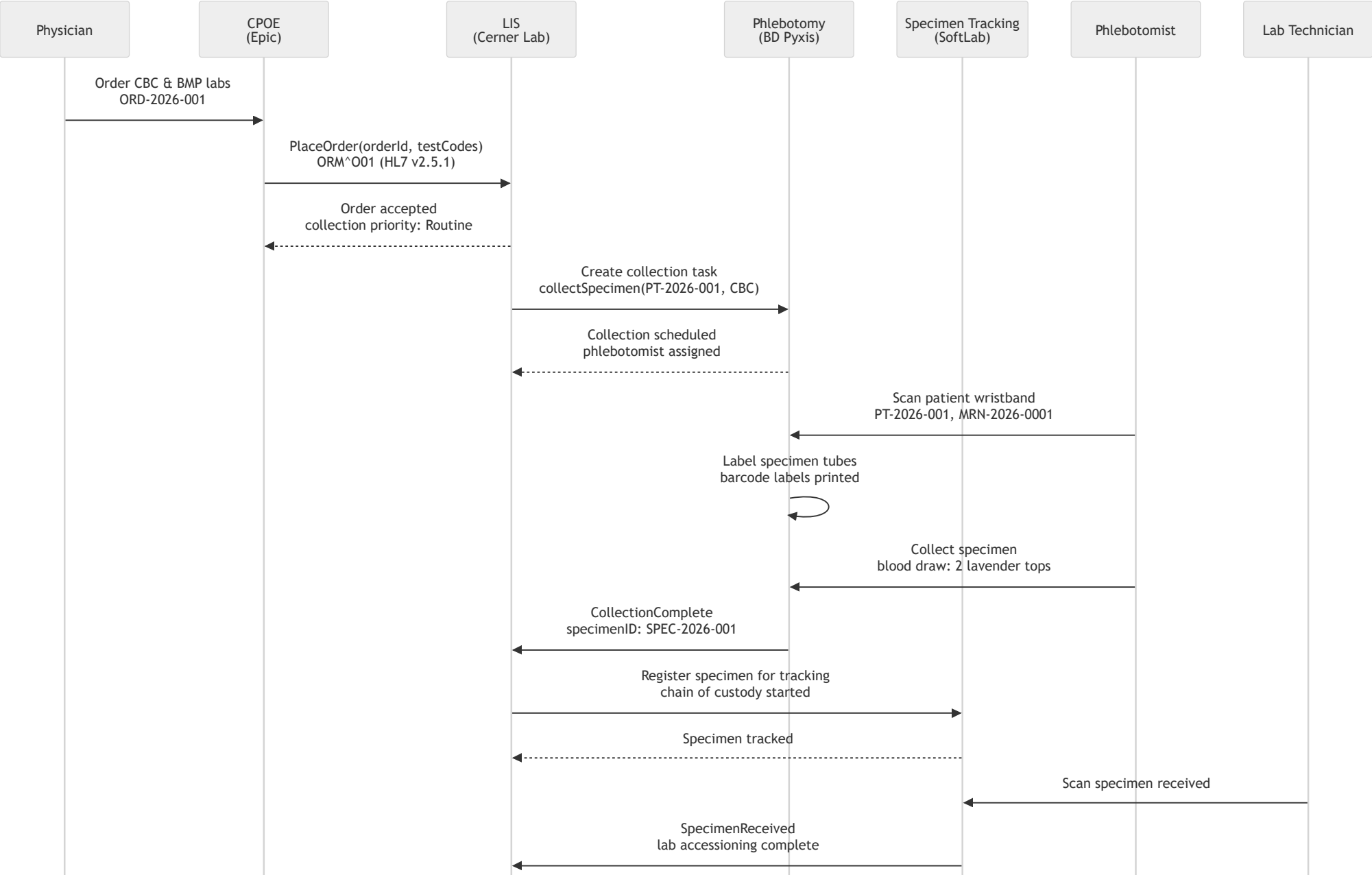
3.1 Lab Order & Specimen Collection

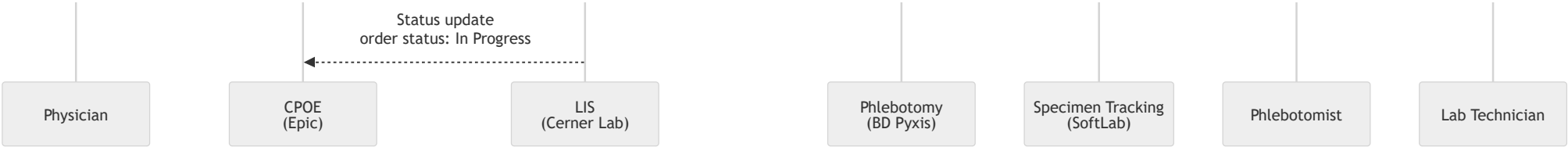
Description: A lab order is placed, specimen is collected from the patient, labeled, and transported to the lab for processing.

Actors: Physician, Phlebotomist, Lab Technician

Systems: CPOE (Epic), LIS (Cerner Lab), Phlebotomy (BD Pyxis), Specimen Tracking (SCC SoftLab)

Sequence Diagram





Integration Details

Flow	Entity	Info Flow	Source	Target	Source Conn	Target Conn	Pattern	Complexity
Lab Order	Order	PlaceOrder	CPOE (Epic)	LIS (Cerner Lab)	Epic REST API (FHIR)	Cerner Lab HL7 v2.5.1 (MLLP)	API-led (Real-time)	Medium
Lab Order	Order	CreateCollectionTask	LIS (Cerner Lab)	Phlebotomy (BD Pyxis)	Cerner Lab HL7 v2.5.1 (MLLP)	Pyxis REST API	Event-driven	Medium
Lab Order	Specimen	CollectionComplete	Phlebotomy (BD Pyxis)	LIS (Cerner Lab)	Pyxis REST API	Cerner Lab HL7 v2.5.1 (MLLP)	API-led (Real-time)	Medium
Lab Order	Specimen	RegisterSpecimen	LIS (Cerner Lab)	Specimen Tracking (SCC SoftLab)	Cerner Lab HL7 v2.5.1 (MLLP)	SoftLab REST API (IHE LSWF)	Event-driven	Medium
Lab Order	Order	UpdateStatus	LIS (Cerner Lab)	CPOE (Epic)	Cerner Lab HL7 v2.5.1 (MLLP)	Epic REST API (FHIR)	Event-driven	Medium

Acceptance Criteria

AC ID	Description	Related Info Flow	Expected Outcome	Priority
AC-3.1-01	Data integrity: Lab order data transferred from CPOE to LIS via PlaceOrder must be complete and consistent	PlaceOrder	Lab orders in LIS match source; no data loss; reconciliation report confirms accuracy	Critical
AC-3.1-02	Data integrity: Collection task data transferred from LIS to Phlebotomy via CreateCollectionTask must be complete and consistent	CreateCollectionTask	Collection tasks in Phlebotomy match source; no data loss	High
AC-3.1-03	Error handling: Failed lab order transactions are logged with error context and trigger automatic retry or alert	PlaceOrder	Errors logged with timestamp, HTTP status, request payload excerpt; retry up to 3 times; alert sent after consecutive failures	Critical

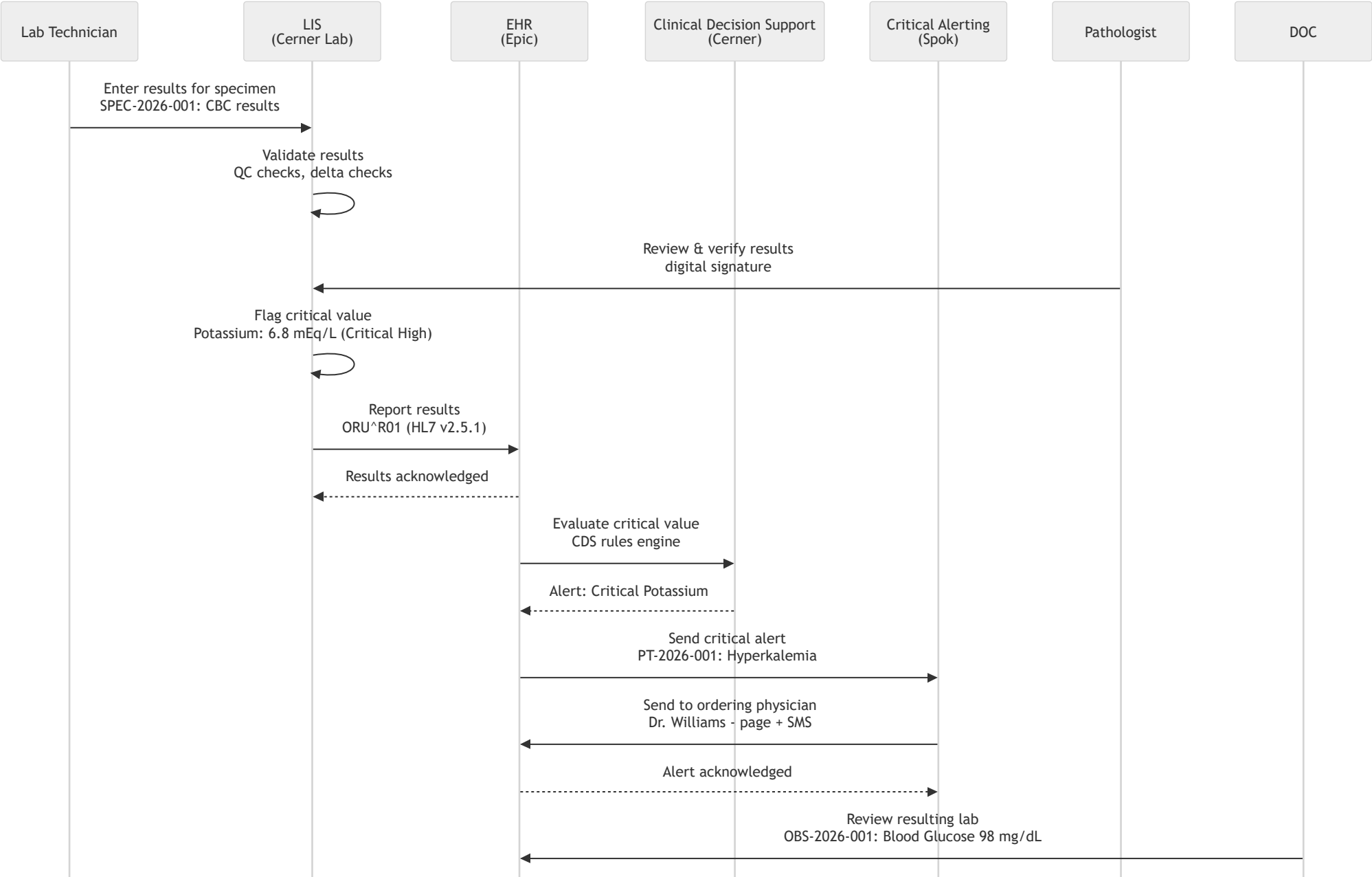
3.2 Lab Results Reporting

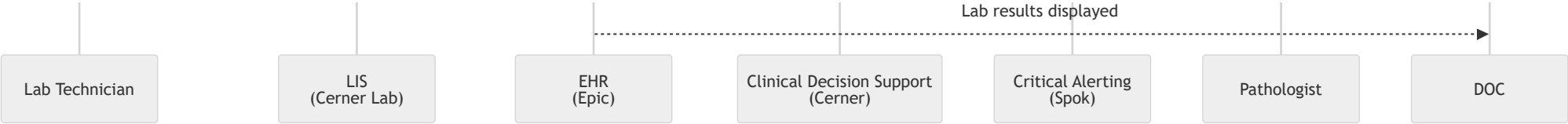
Description: Lab results are resulted, validated, and reported back to the ordering provider. Critical values trigger immediate alerts.

Actors: Lab Technician, Pathologist, Physician

Systems: LIS (Cerner Lab), EHR (Epic), Clinical Decision Support (Cerner CDS), Critical Alerting (Spok)

Sequence Diagram





Integration Details

Flow	Entity	Info Flow	Source	Target	Source Conn	Target Conn	Pattern	Complexity
Lab Results	Observation	ReportResults	LIS (Cerner Lab)	EHR (Epic)	Cerner Lab HL7 v2.5.1 (MLLP)	Epic REST API (FHIR)	Event-driven	Medium
Lab Results	Observation	EvaluateCriticalValue	EHR (Epic)	Clinical Decision Support (Cerner CDS)	Epic REST API (FHIR)	Cerner CDS REST API (CDS Hooks)	API-led (Real-time)	Medium
Lab Results	Observation	SendCriticalAlert	EHR (Epic)	Critical Alerting (Spok)	Epic REST API (FHIR)	Spok REST API (HL7 FHIR)	Event-driven	High
Lab Results	Observation	AcknowledgeAlert	Critical Alerting (Spok)	EHR (Epic)	Spok REST API (HL7 FHIR)	Epic REST API (FHIR)	API-led (Real-time)	Medium
Lab Results	Observation	ReviewResults	Physician	EHR (Epic)	Epic UI (Web)	Epic REST API (FHIR)	API-led (Real-time)	Simple

Acceptance Criteria

AC ID	Description	Related Info Flow	Expected Outcome	Priority
AC-3.2-01	Data integrity: Lab result data transferred from LIS to EHR via ReportResults must be complete and consistent	ReportResults	Lab results in EHR match source; no data loss; reconciliation report confirms accuracy	Critical
AC-3.2-02	Data integrity: Critical alert data transferred from EHR to Alerting via SendCriticalAlert must be complete and consistent	SendCriticalAlert	Critical alerts delivered to correct provider within 5 minutes	Critical
AC-3.2-03	Security: Lab results are only accessible to authorized providers; patient consent verified before disclosure	All flows	Unauthorized access to lab results rejected with 403; audit trail logged	Critical

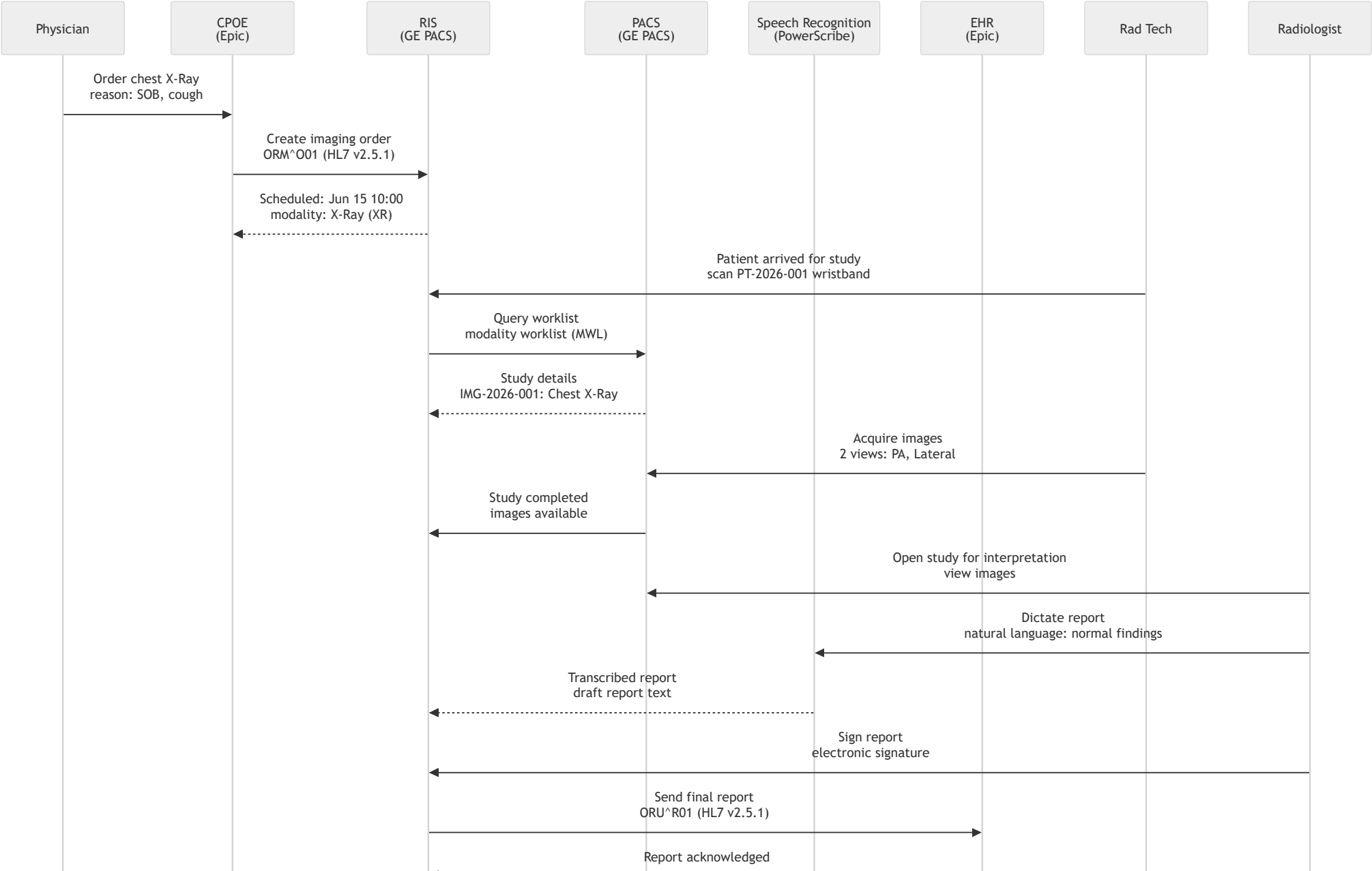
3.3 Radiology & Imaging Management

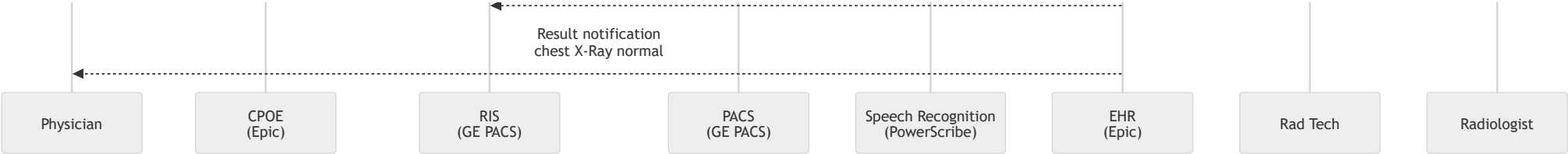
Description: An imaging order is scheduled, the study is performed, images are interpreted by a radiologist, and the report is distributed to the ordering provider.

Actors: Physician, Radiologist, Radiology Technician

Systems: CPOE (Epic), RIS (GE PACS), PACS (GE PACS), Speech Recognition (Nuance PowerScribe), EHR (Epic)

Sequence Diagram





Integration Details

Flow	Entity	Info Flow	Source	Target	Source Conn	Target Conn	Pattern	Complexity
Imaging	Order	CreateImagingOrder	CPOE (Epic)	RIS (GE PACS)	Epic REST API (FHIR)	GE RIS HL7 v2.5.1 (MLLP)	API-led (Real-time)	Medium
Imaging	Study	QueryWorklist	RIS (GE PACS)	PACS (GE PACS)	GE RIS HL7 v2.5.1 (MLLP)	DICOM MWL (TCP)	API-led (Real-time)	Medium
Imaging	Study	AcquireImages	Radiology Technician	PACS (GE PACS)	DICOM Modality (TCP)	DICOM Storage (TCP)	Event-driven	High
Imaging	Report	DictateReport	Speech Recognition (Nuance PowerScribe)	RIS (GE PACS)	PowerScribe REST API	GE RIS HL7 v2.5.1 (MLLP)	API-led (Real-time)	Medium
Imaging	Report	SendFinalReport	RIS (GE PACS)	EHR (Epic)	GE RIS HL7 v2.5.1 (MLLP)	Epic REST API (FHIR)	Event-driven	Medium

Acceptance Criteria

AC ID	Description	Related Info Flow	Expected Outcome	Priority
AC-3.3-01	Data integrity: Imaging order data transferred from CPOE to RIS via CreateImagingOrder must be complete and consistent	CreateImagingOrder	Imaging orders in RIS match source; no data loss; reconciliation report confirms accuracy	Critical
AC-3.3-02	Data integrity: Report data transferred from RIS to EHR via SendFinalReport must be complete and consistent	SendFinalReport	Report records in EHR match source; no data loss; reconciliation report confirms accuracy	High
AC-3.3-03	Error handling: Failed imaging report transactions are logged with error context and trigger automatic retry or alert	SendFinalReport	Errors logged with timestamp, HTTP status, request payload excerpt	High

AC ID	Description	Related Info Flow	Expected Outcome	Priority
			; retry up to 3 times; alert sent after consecutive failures	

Pharmacy & Medication Management - HC030 HC031 HC032

Comprehensive medication management including ordering, dispensing, verification, and administration with barcode scanning and safety checks.

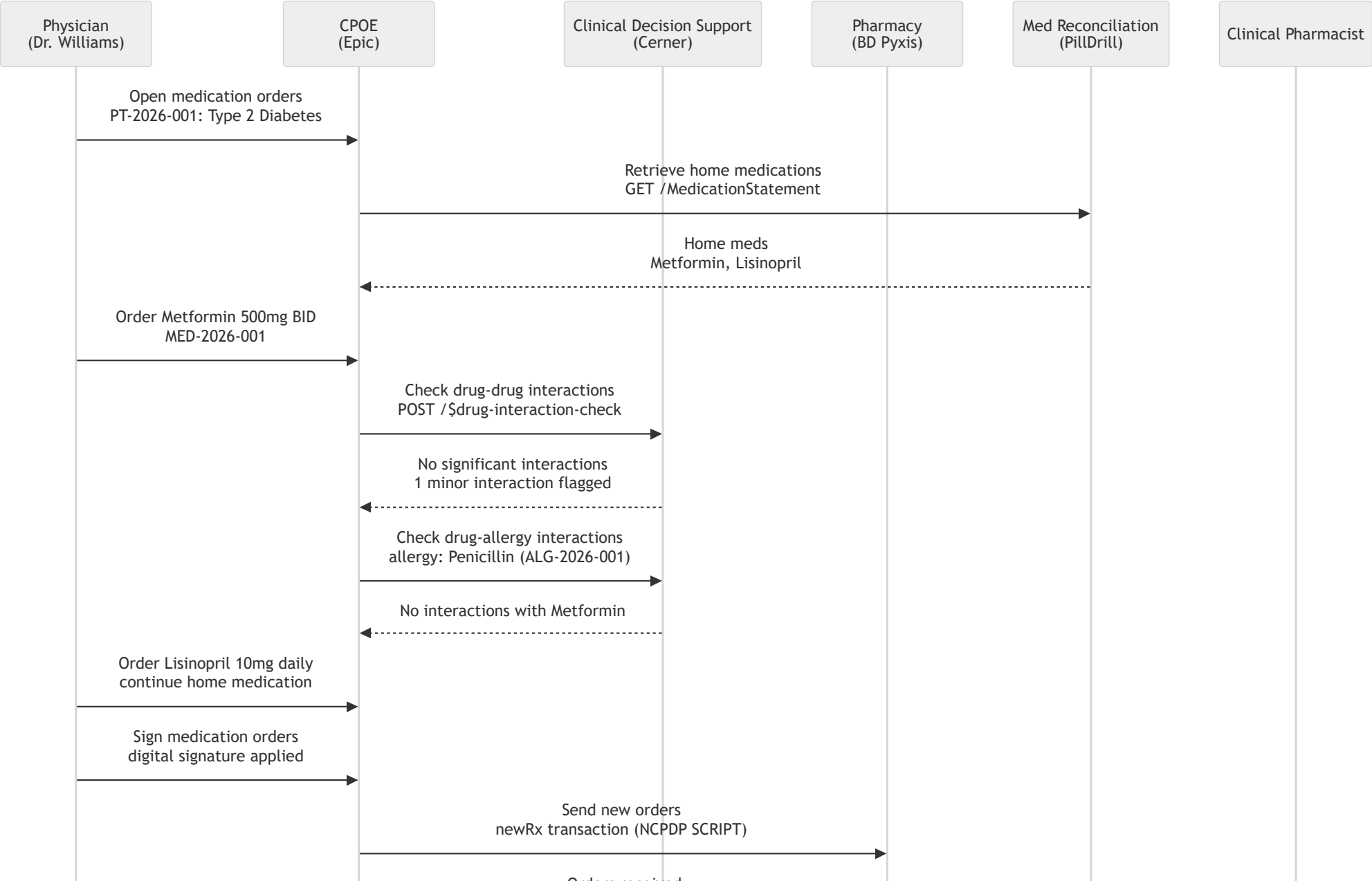
4.1 Medication Ordering & Reconciliation

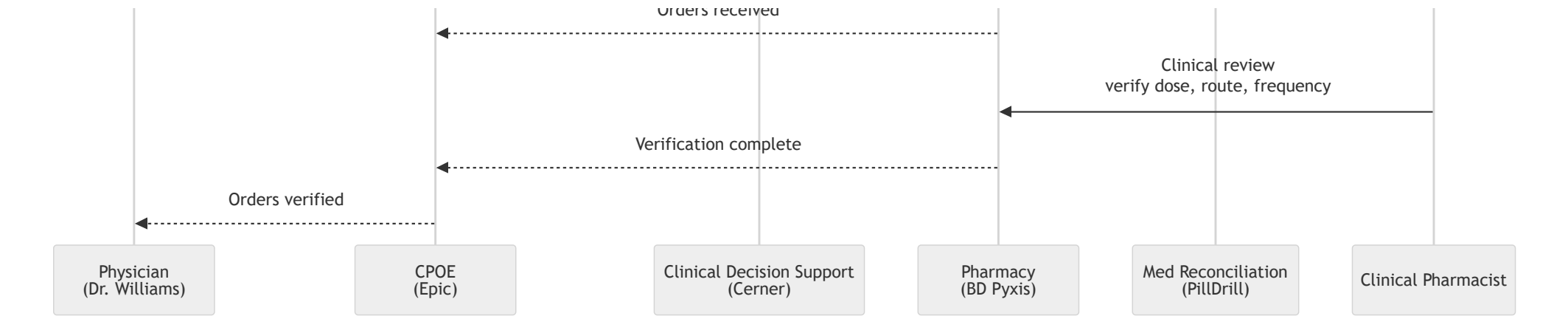
Description: A physician orders medications, performs drug interaction checks, reconciles home medications, and reviews allergy information before signing.

Actors: Physician (Dr. Sarah Williams), Clinical Pharmacist

Systems: CPOE (Epic), Pharmacy System (BD Pyxis), Clinical Decision Support (Cerner CDS), Medication Reconciliation (PillDrill)

Sequence Diagram





Integration Details

Flow	Entity	Info Flow	Source	Target	Source Conn	Target Conn	Pattern	Complexity
Med Order	Medication	RetrieveHomeMeds	CPOE (Epic)	Medication Reconciliation (PillDrill)	Epic REST API (FHIR)	PillDrill REST API (FHIR)	API-led (Real-time)	Medium
Med Order	Medication	CheckDrugInteractions	CPOE (Epic)	Clinical Decision Support (Cerner CDS)	Epic REST API (FHIR)	Cerner CDS REST API (CDS Hooks)	API-led (Real-time)	Medium
Med Order	Medication	CheckDrugAllergy	CPOE (Epic)	Clinical Decision Support (Cerner CDS)	Epic REST API (FHIR)	Cerner CDS REST API (CDS Hooks)	API-led (Real-time)	Medium
Med Order	Medication	SendNewOrders	CPOE (Epic)	Pharmacy System (BD Pyxis)	Epic REST API (FHIR)	Pyxis NCPDP SCRIPT (SOAP)	API-led (Real-time)	Medium
Med Order	Medication	ClinicalReview	Clinical Pharmacist	Pharmacy System (BD Pyxis)	Pyxis Admin UI	Pyxis REST API	API-led (Real-time)	Medium

Acceptance Criteria

AC ID	Description	Related Info Flow	Expected Outcome	Priority
AC-4.1-01	Data integrity: Medication order data transferred from CPOE to Pharmacy via SendNewOrders must be complete and consistent	SendNewOrders	Medication orders in Pharmacy match source; no data loss; reconciliation report confirms accuracy	Critical
AC-4.1-02	Data integrity: Interaction check results transferred from CDS to CPOE via CheckDrugInteractions must be complete and consistent	CheckDrugInteractions	Interaction check results in CPOE match source; no data loss; reconciliation report confirms accuracy	Critical
AC-4.1-03	Error handling: Failed medication order transactions are logged with error context and trigger automatic retry or alert	SendNewOrders	Errors logged with timestamp, HTTP status, request payload excerpt; retry up to 3 times; alert sent after consecutive failures	Critical

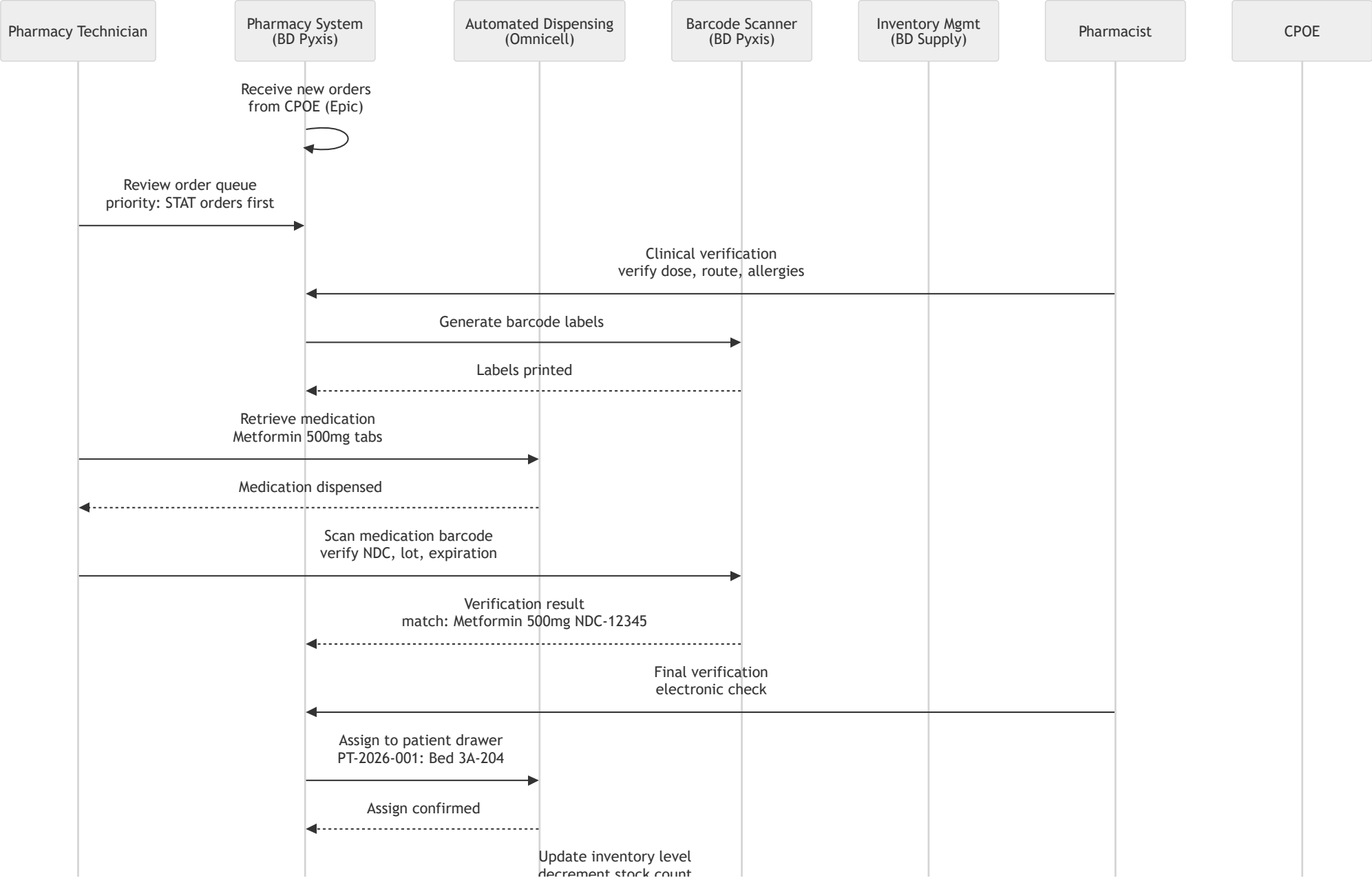
4.2 Pharmacy Dispensing & Verification

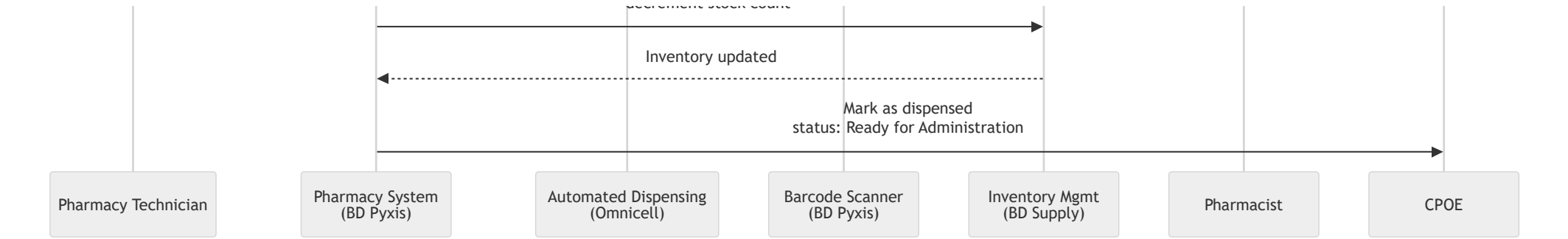
Description: The pharmacy receives medication orders, performs verification, prepares doses, and manages the dispensing process with barcode verification.

Actors: Pharmacist, Pharmacy Technician

Systems: Pharmacy System (BD Pyxis), ADM (Omnicell), Barcode Scanning (BD Pyxis), Inventory Management (BD Supply)

Sequence Diagram





Integration Details

Flow	Entity	Info Flow	Source	Target	Source Conn	Target Conn	Pattern	Complexity
Dispensing	Medication	GenerateLabel	Pharmacy System (BD Pyxis)	Barcode Scanner (BD Pyxis)	Pyxis REST API	Pyxis Printing Service (RAW)	Event-driven	Simple
Dispensing	Medication	DispenseFromADC	Pharmacy Technician	Automated Dispensing (OmniceII)	OmniceII UI	OmniceII REST API	API-led (Real-time)	Medium
Dispensing	Medication	ScanAndVerify	Barcode Scanner (BD Pyxis)	Pharmacy System (BD Pyxis)	Pyxis Barcode API	Pyxis REST API	API-led (Real-time)	Medium
Dispensing	Medication	AssignToPatient	Pharmacy System (BD Pyxis)	Automated Dispensing (OmniceII)	Pyxis REST API	OmniceII REST API	Event-driven	Medium
Dispensing	Medication	UpdateInventory	Pharmacy System (BD Pyxis)	Inventory Mgmt (BD Supply)	Pyxis REST API	BD Supply REST API (EDI)	Event-driven	Medium

Acceptance Criteria

AC ID	Description	Related Info Flow	Expected Outcome	Priority
AC-4.2-01	Data integrity: Barcode verification data transferred from Scanner to Pharmacy via ScanAndVerify must be complete and consistent	ScanAndVerify	Barcode verification results match; incorrect medication rejected with alert	Critical
AC-4.2-02	Data integrity: Inventory data transferred from Pharmacy to Inventory via UpdateInventory must be complete and consistent	UpdateInventory	Inventory levels accurately reflect dispensed quantities; no stock discrepancies	High
AC-4.2-03	Error handling: Failed barcode verification triggers immediate alert and prevents dispensing	ScanAndVerify	Mismatched barcode triggers visual/audio alert; dispensing blocked until resolved	Critical

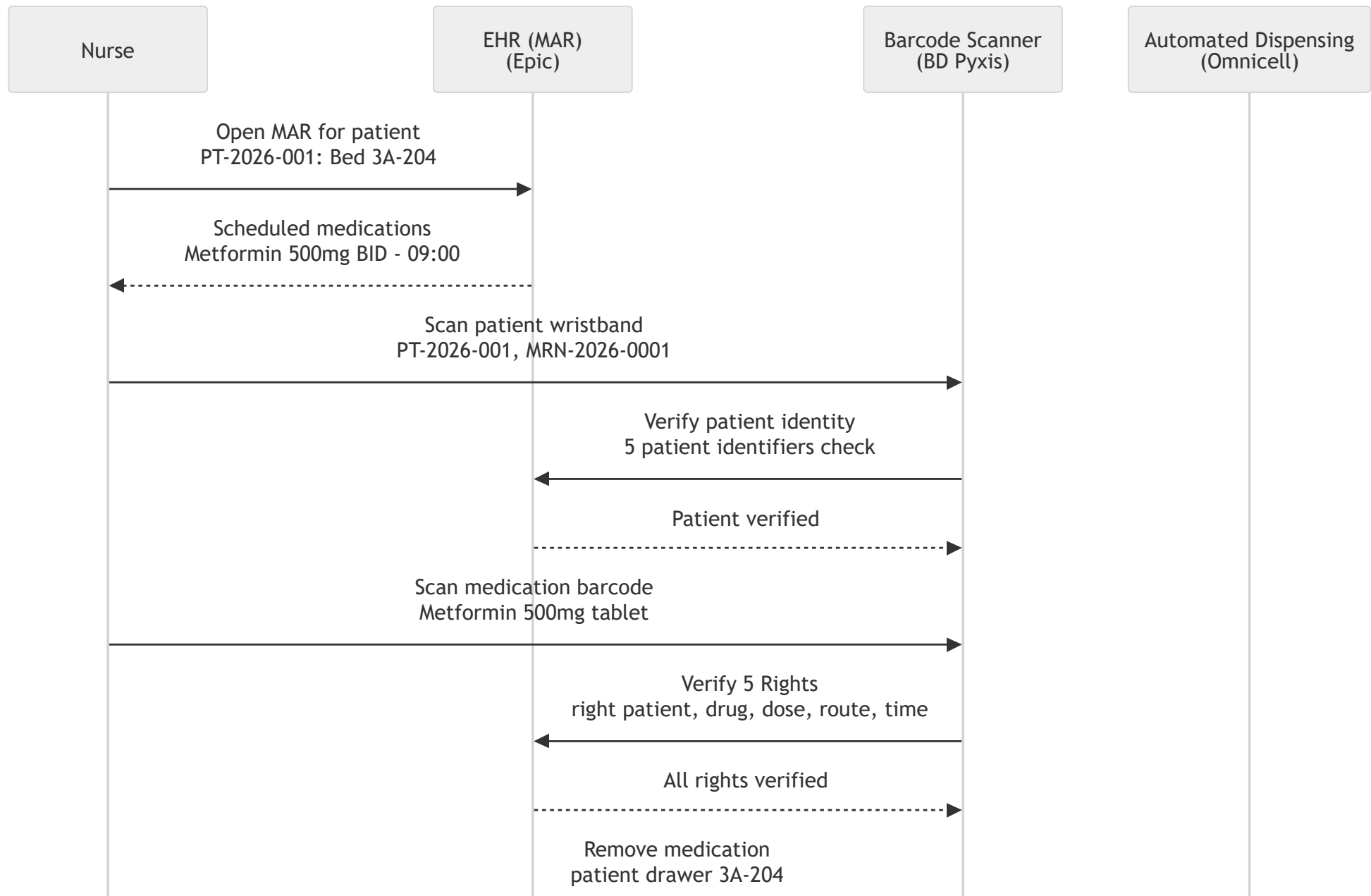
4.3 Medication Administration (MAR)

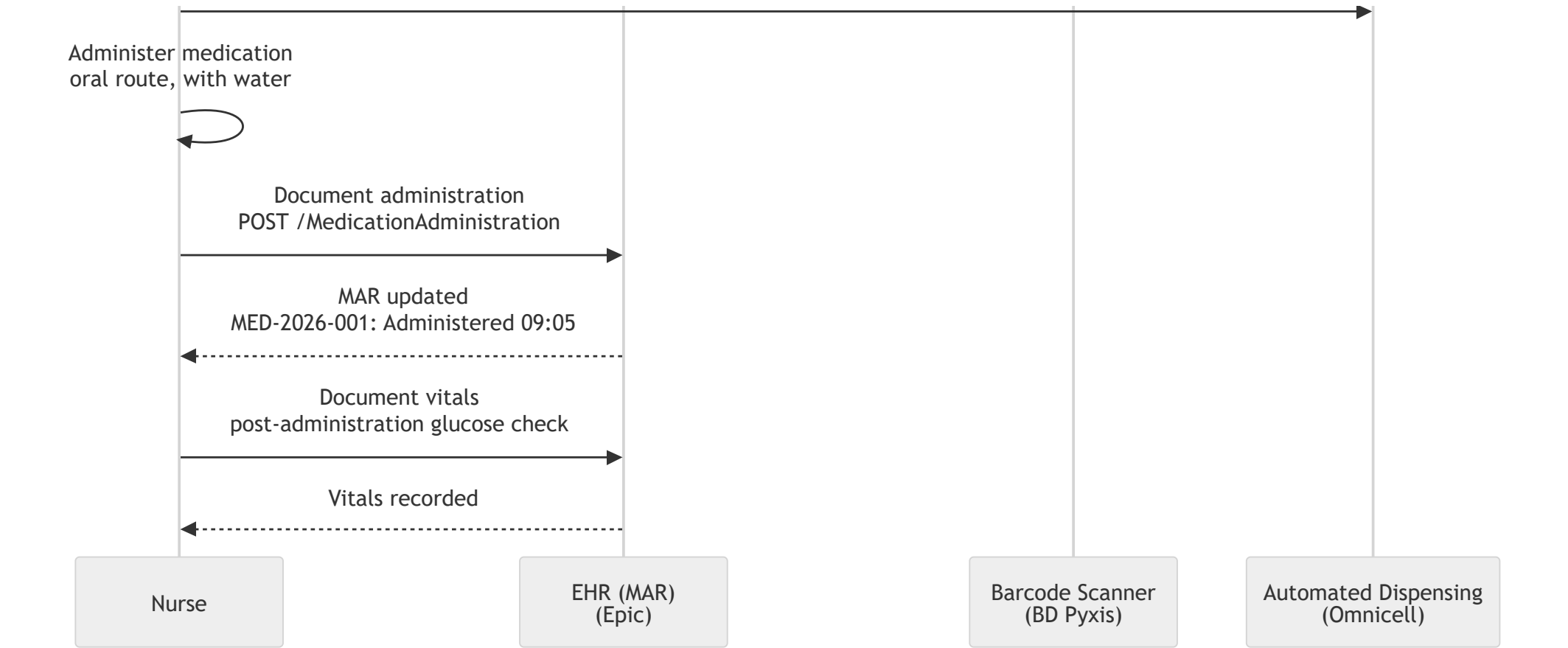
Description: A nurse administers medications to the patient using barcode-assisted medication administration (BCMA) and documents the MAR.

Actors: Nurse

Systems: EHR (Epic), Barcode Scanner (BD Pyxis), ADM (OmniceII), MAR (Epic)

Sequence Diagram





Integration Details

Flow	Entity	Info Flow	Source	Target	Source Conn	Target Conn	Pattern	Complexity
BCMA	Patient	ScanPatientWristband	Barcode Scanner (BD Pyxis)	EHR (Epic)	Pyxis Barcode API	Epic REST API (FHIR)	API-led (Real-time)	Medium
BCMA	Medication	ScanMedicationBarcode	Barcode Scanner (BD Pyxis)	EHR (Epic)	Pyxis Barcode API	Epic REST API (FHIR)	API-led (Real-time)	Medium
BCMA	Medication	Verify5Rights	EHR (Epic)	EHR (Epic)	Epic REST API (FHIR)	Epic REST API (FHIR)	API-led (Real-time)	Medium
BCMA	Medication	DocumentAdministration	EHR (Epic)	EHR (Epic)	Epic REST API (FHIR)	Epic REST API (FHIR)	API-led (Real-time)	Simple
BCMA	Observation	DocumentVitals	EHR (Epic)	EHR (Epic)	Epic REST API (FHIR)	Epic REST API (FHIR)	API-led (Real-time)	Simple
BCMA	Medication	RetrieveFromADC	Nurse	Automated Dispensing (Omniceil)	Omniceil UI	Omniceil REST API	API-led (Real-time)	Simple

Acceptance Criteria

AC ID	Description	Related Info Flow	Expected Outcome	Priority
AC-4.3-01	Data integrity: Barcode scan data transferred from Scanner to EHR via ScanPatientWristband must be complete and consistent	ScanPatientWristband	Patient identity verified against 5 identifiers; mismatch blocks administration	Critical
AC-4.3-02	Data integrity: Medication administration data transferred via DocumentAdministration must be complete and consistent	DocumentAdministration	MAR records match administered dose; no data loss; reconciliation report confirms accuracy	Critical
AC-4.3-03	Error handling: Failed barcode verification during BCMA triggers immediate alert and prevents administration	ScanMedicationBarcode	Mismatched barcode triggers visual/audio alert; administration blocked	Critical
AC-4.3-04	Security: BCMA ensures 5 Rights verification before administration is a	All flows	Administration blocked if any of 5 Rights fails validation	Critical

AC ID	Description	Related Info Flow	Expected Outcome	Priority
	llowed			

Billing & Revenue Cycle Management - HC040 HC041 HC042

End-to-end revenue cycle management from charge capture through claim submission, payment posting, and denial management.

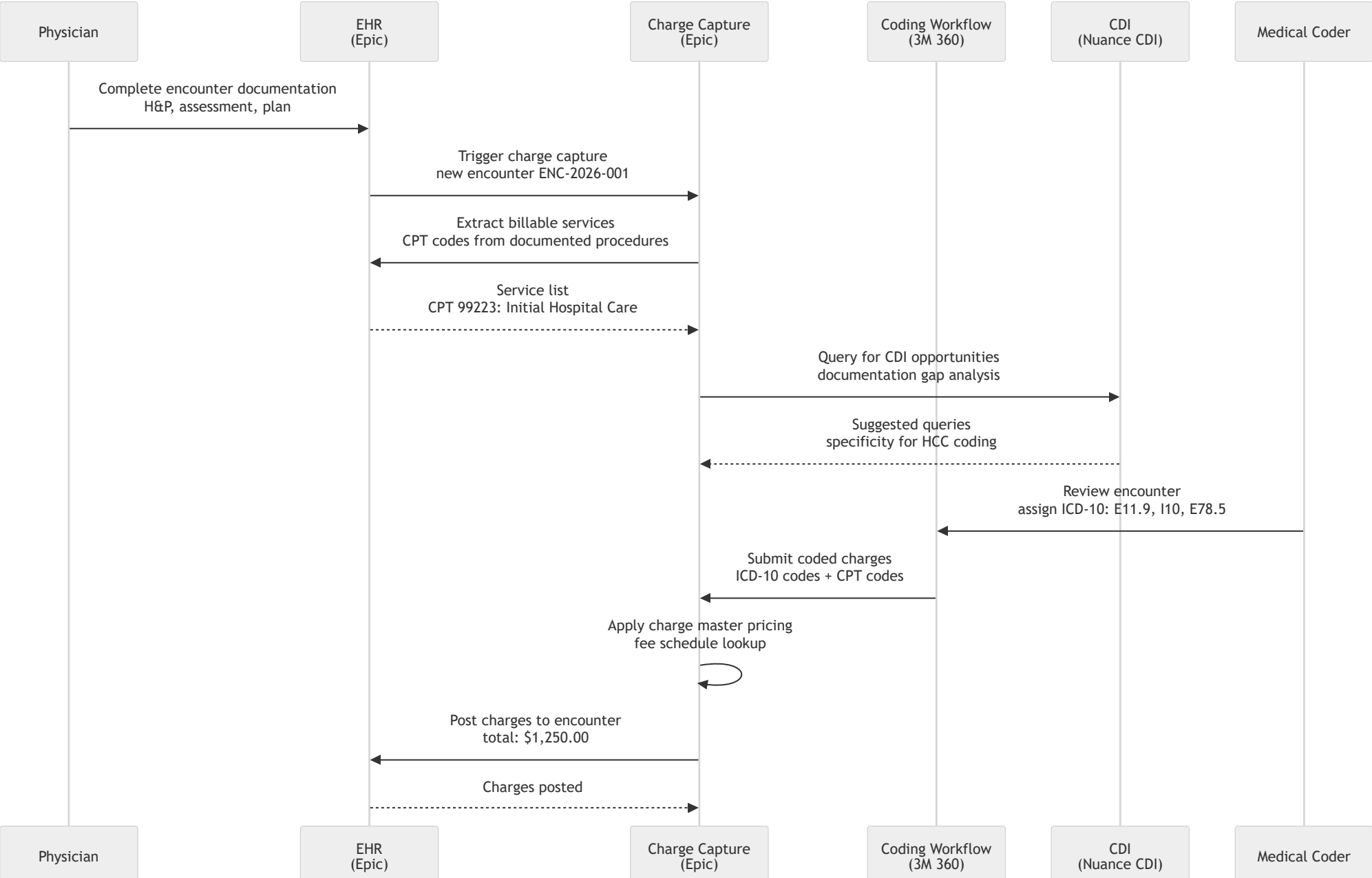
5.1 Charge Capture & Coding

Description: Charges are captured from clinical documentation, coded with appropriate ICD-10/CPT codes, and validated before claim submission.

Actors: Physician (Dr. Sarah Williams), Medical Coder, Charge Capture Specialist

Systems: EHR (Epic), Coding Workflow (3M 360), Charge Capture (Epic), CDI (Nuance CDI)

Sequence Diagram



Integration Details

Flow	Entity	Info Flow	Source	Target	Source Conn	Target Conn	Pattern	Complexity
Charge Capture	Encounter	ExtractBillableServices	Charge Capture (Epic)	EHR (Epic)	Epic Charge Services API	Epic REST API (FHIR)	API-led (Real-time)	Medium
Charge Capture	Encounter	QueryCDIOpportunities	Charge Capture (Epic)	CDI (Nuance CDI)	Epic REST API	Nuance CDI REST API (FHIR)	API-led (Real-time)	Medium
Charge Capture	Encounter	SubmitCodedCharges	Coding Workflow (3M 360)	Charge Capture (Epic)	3M REST API	Epic REST API	API-led (Real-time)	Medium
Charge Capture	Encounter	PostCharges	Charge Capture (Epic)	EHR (Epic)	Epic Charge Services API	Epic REST API (FHIR)	API-led (Real-time)	Simple
Charge Capture	Encounter	ApplyChargeMaster	Charge Capture (Epic)	Charge Capture (Epic)	Epic Fee Schedule DB	Epic Charge Services API	API-led (Real-time)	Medium

Acceptance Criteria

AC ID	Description	Related Info Flow	Expected Outcome	Priority
AC-5.1-01	Data integrity: Charge data transferred from EHR to Charge Capture via ExtractBillableServices must be complete and consistent	ExtractBillableServices	Charge records match documented services; no unbilled services identified	Critical
AC-5.1-02	Data integrity: Coding data transferred from Coding Workflow to Charge Capture via SubmitCodedCharges must be complete and consistent	SubmitCodedCharges	Coded charges match clinical documentation; audit trail confirms accuracy	High
AC-5.1-03	Error handling: Failed charge capture transactions are logged with error context and trigger automatic retry or alert	ExtractBillableServices	Errors logged with timestamp, HTTP status, request payload excerpt; retry up to 3 times; alert sent after consecutive failures	Critical

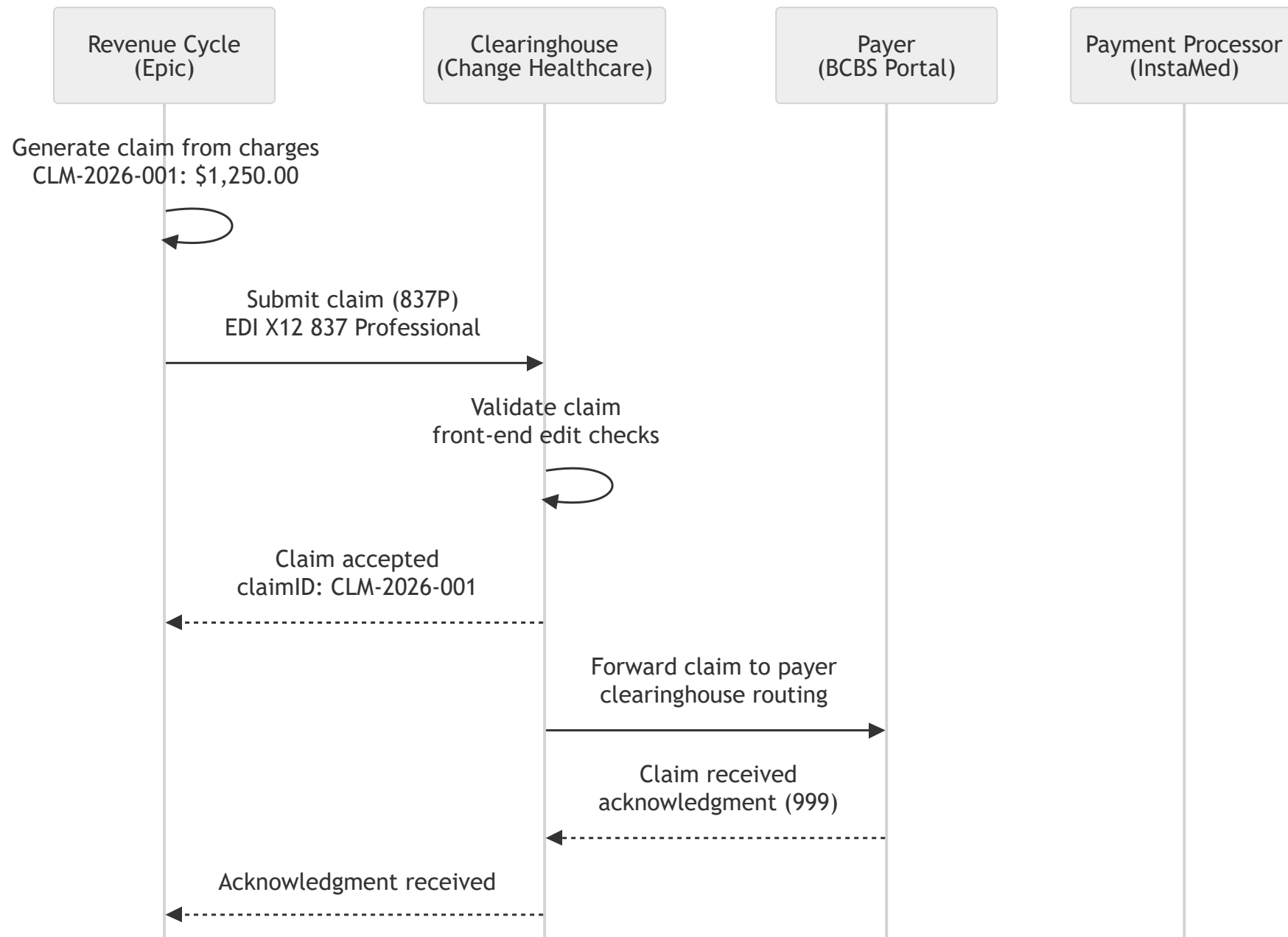
5.2 Claims Submission & Adjudication

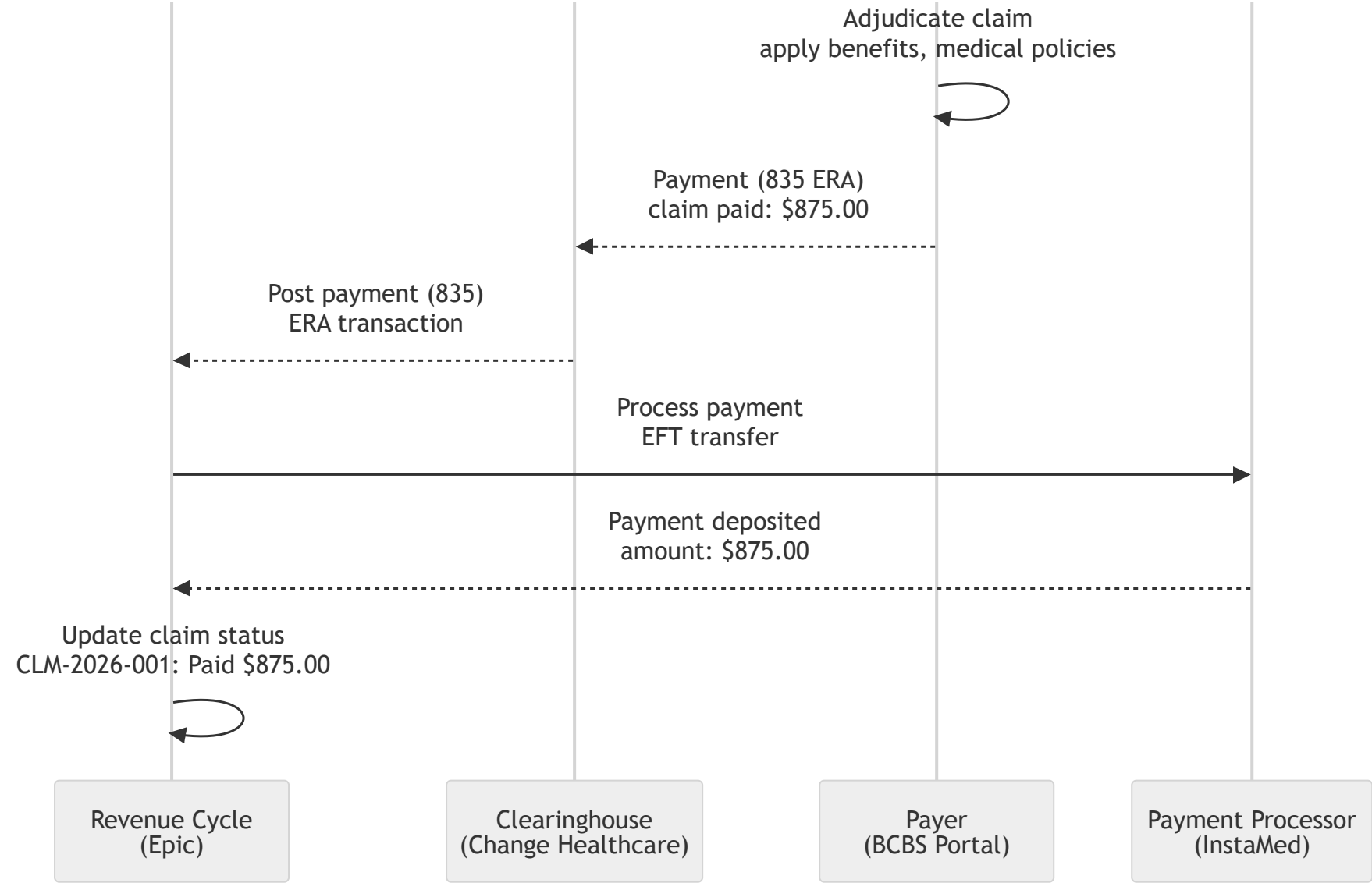
Description: Claims are generated from posted charges, validated through clearinghouse edits, submitted to payers, and adjudication results are received.

Actors: Billing Specialist, Payer

Systems: Revenue Cycle (Epic), Clearinghouse (Change Healthcare), Payer Portal (BCBS), Payment Processor (InstaMed)

Sequence Diagram





Integration Details

Flow	Entity	Info Flow	Source	Target	Source Conn	Target Conn	Pattern	Complexity
Claims	Claim	SubmitClaim (837P)	Revenue Cycle (Epic)	Clearinghouse (Change Healthcare)	Epic EDI Gateway (SFTP)	Change Healthcare EDI (SFTP/AS2)	Batch (Scheduled)	High
Claims	Claim	ValidateClaim	Clearinghouse (Change Healthcare)	Clearinghouse (Change Healthcare)	Change Healthcare EDI Engine	Change Healthcare EDI Engine	API-led (Real-time)	Medium
Claims	Claim	RouteToPayer	Clearinghouse (Change Healthcare)	Payer (BCBS Portal)	Change Healthcare EDI (AS2)	BCBS EDI (AS2)	Batch (Scheduled)	High
Claims	Claim	AdjudicateClaim	Payer (BCBS Portal)	Payer (BCBS Portal)	BCBS Adjudication Engine	BCBS Adjudication Engine	Batch (Scheduled)	High
Claims	Claim	PostPayment (835)	Clearinghouse (Change Healthcare)	Revenue Cycle (Epic)	Change Healthcare EDI (SFTP)	Epic EDI Gateway (SFTP)	Batch (Scheduled)	High
Claims	Payment	ProcessEFT	Revenue Cycle (Epic)	Payment Processor (InstaMed)	Epic REST API	InstaMed REST API (ACH)	API-led (Real-time)	Medium

Acceptance Criteria

AC ID	Description	Related Info Flow	Expected Outcome	Priority
AC-5.2-01	Data integrity: Claim data transferred from Revenue Cycle to Clearinghouse via SubmitClaim must be complete and consistent	SubmitClaim	Claims in Clearinghouse match source; no data loss; reconciliation report confirms accuracy	Critical
AC-5.2-02	Data integrity: Payment data transferred from Clearinghouse to Revenue Cycle via PostPayment must be complete and consistent	PostPayment	Payment records in Revenue Cycle match payer source; no data loss; reconciliation report confirms accuracy	Critical
AC-5.2-03	Error handling: Claim submission failures are logged with error context and trigger automatic retry or alert	SubmitClaim	Errors logged with timestamp, EDI status, claim payload excerpt; retry up to 3 times; alert sent after consecutive failures	Critical

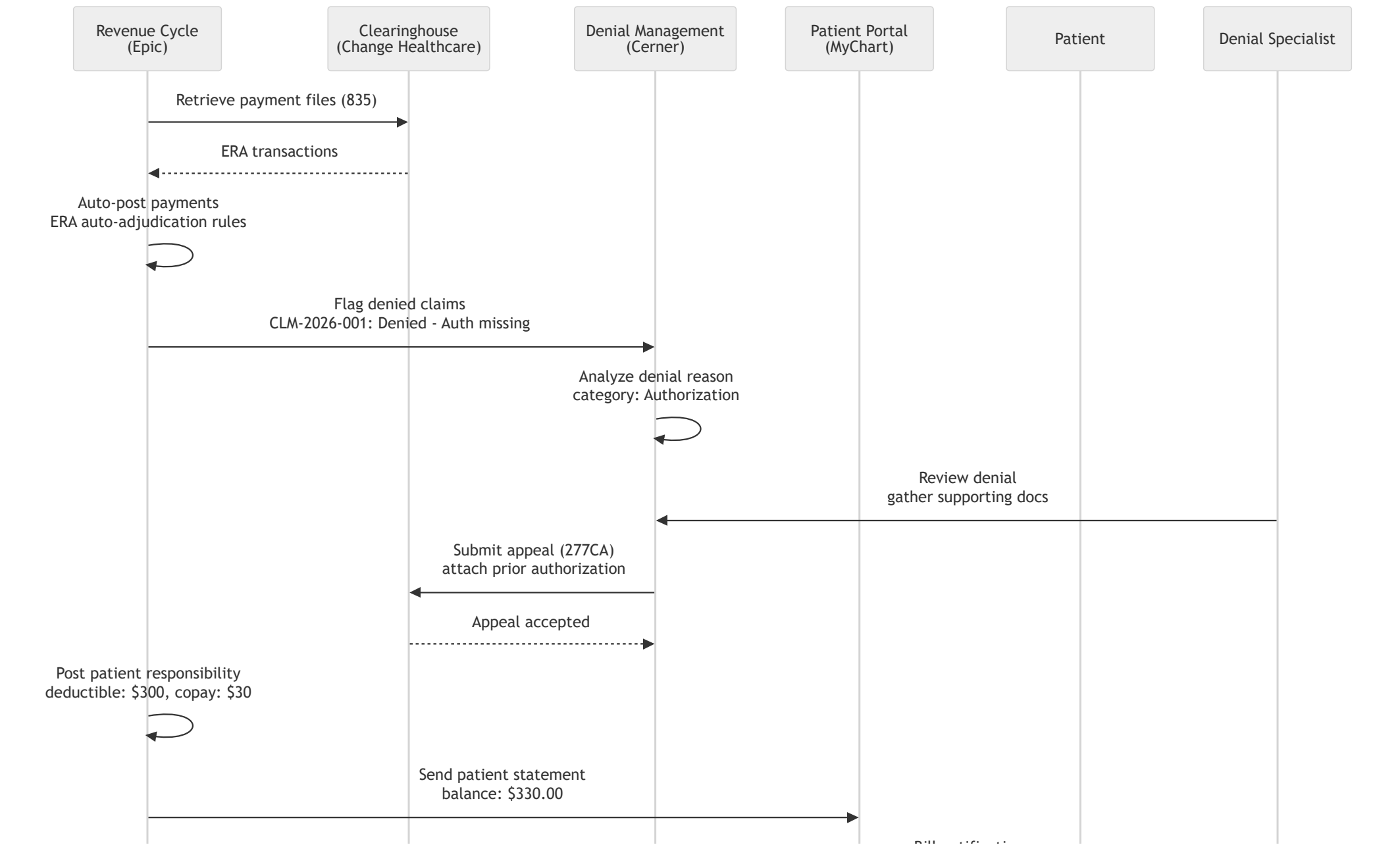
5.3 Payment Posting & Denial Management

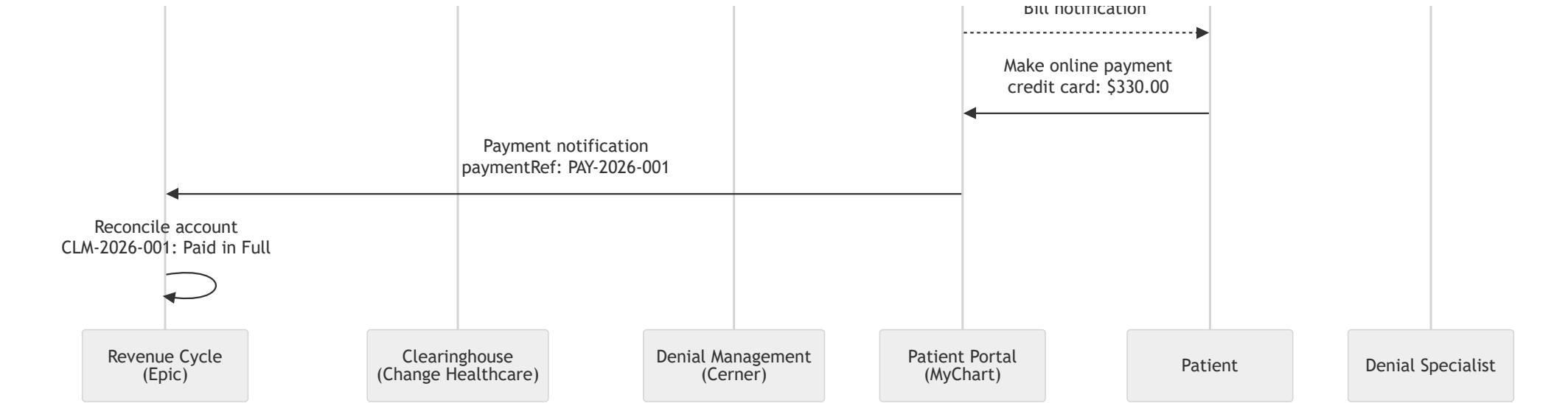
Description: Payments are posted to patient accounts, denials are analyzed and appealed, and accounts are reconciled.

Actors: Payment Poster, Denial Specialist, Patient

Systems: Revenue Cycle (Epic), Clearinghouse (Change Healthcare), Denial Management (Cerner), Patient Portal (MyChart)

Sequence Diagram





Integration Details

Flow	Entity	Info Flow	Source	Target	Source Conn	Target Conn	Pattern	Complexity
Payment Posting	Claim	RetrievePaymentFiles (835)	Revenue Cycle (Epic)	Clearinghouse (Change Healthcare)	Epic EDI Gateway (SFTP)	Change Healthcare EDI (SFTP/AS2)	Batch (Scheduled)	High
Denial Mgmt	Claim	FlagDeniedClaim	Revenue Cycle (Epic)	Denial Management (Cerner)	Epic REST API (FHIR)	Cerner REST API	Event-driven	Medium
Denial Mgmt	Claim	SubmitAppeal (277CA)	Denial Management (Cerner)	Clearinghouse (Change Healthcare)	Cerner REST API	Change Healthcare EDI (SFTP)	API-led (Real-time)	High
Payment Posting	Claim	PostPatientResponsibility	Revenue Cycle (Epic)	Revenue Cycle (Epic)	Epic ERA Auto-post	Epic REST API	Batch (Scheduled)	Medium
Patient Billing	Claim	SendPatientStatement	Revenue Cycle (Epic)	Patient Portal (MyChart)	Epic REST API (FHIR)	MyChart REST API	Batch (Scheduled)	Medium
Patient Billing	Payment	ProcessPatientPayment	Patient Portal (MyChart)	Revenue Cycle (Epic)	MyChart REST API	Epic Payment API	API-led (Real-time)	Medium

Acceptance Criteria

AC ID	Description	Related Info Flow	Expected Outcome	Priority
AC-5.3-01	Data integrity: Payment data transferred from Clearinghouse to Revenue Cycle via RetrievePaymentFiles must be complete and consistent	RetrievePaymentFiles	Payment records match ERA source; no data loss; reconciliation report confirms accuracy	Critical
AC-5.3-02	Data integrity: Denial data transferred from Revenue Cycle to Denial Management via FlagDeniedClaim must be complete and consistent	FlagDeniedClaim	Denial records in Denial Management match source; no data loss; reconciliation report confirms accuracy	High
AC-5.3-03	Error handling: Failed appeal submissions are logged with error context and trigger automatic retry or alert	SubmitAppealtd>	Errors logged with timestamp, HTTP status, request payload excerpt; retry up to 3 times; alert sent after consecutive failures	High

Appointment Scheduling & Access - HC050 HC051

Patient access management including appointment scheduling, check-in/check-out, referral management, and prior authorization.

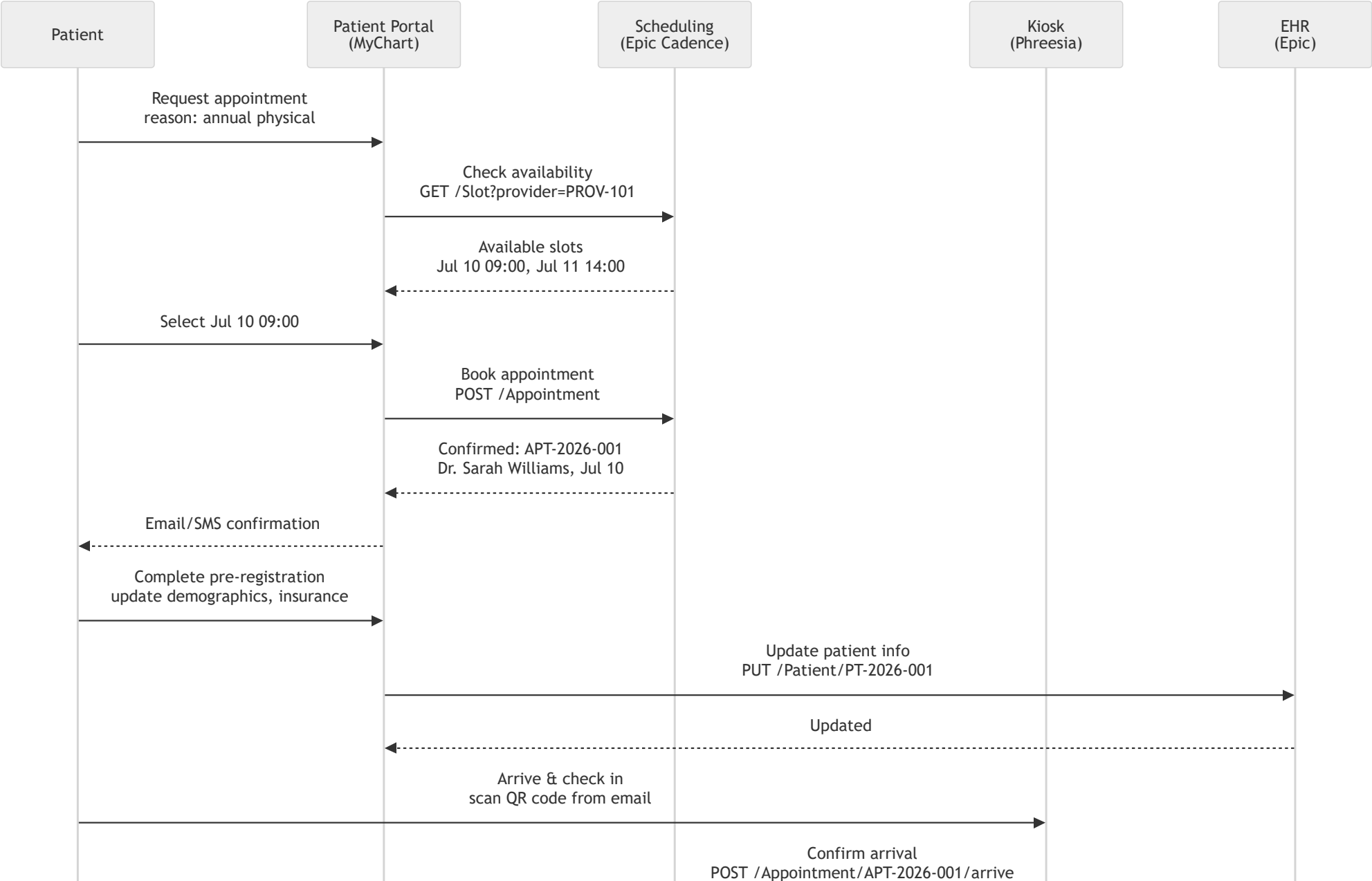
6.1 Patient Scheduling & Check-in

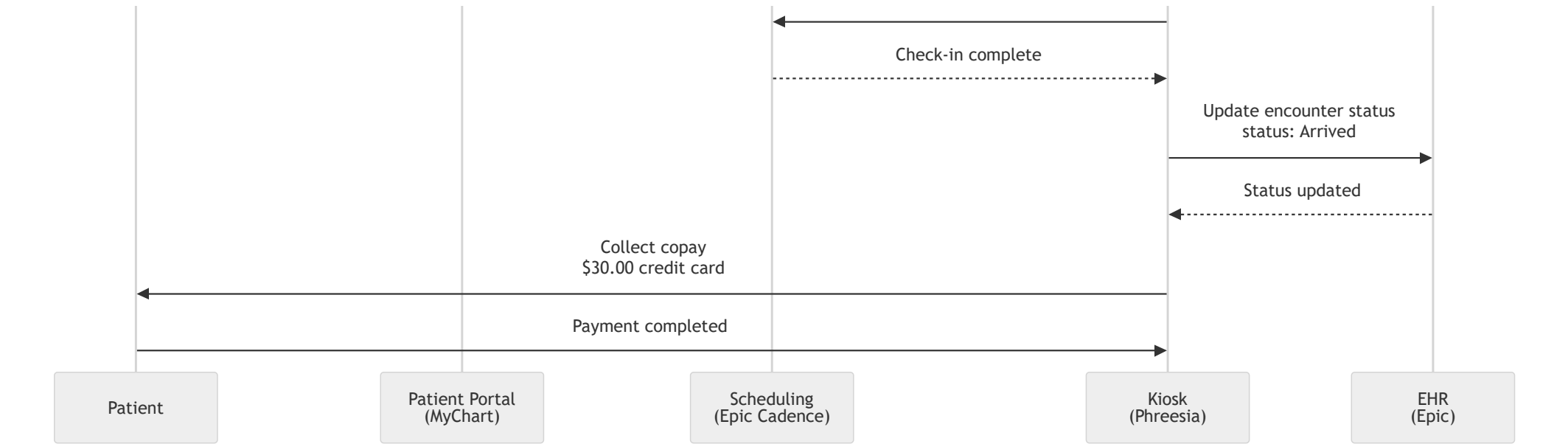
Description: A patient schedules an appointment online, receives confirmation, checks in at the facility, and completes pre-visit registration.

Actors: Patient, Front Desk Staff

Systems: Patient Portal (MyChart), Scheduling (Epic Cadence), EHR (Epic), Kiosk (Phreesia)

Sequence Diagram





Integration Details

Flow	Entity	Info Flow	Source	Target	Source Conn	Target Conn	Pattern	Complexity
Scheduling	Appointment	CheckAvailability	Patient Portal (MyChart)	Scheduling (Epic Cadence)	MyChart REST API	Epic Cadence REST API (FHIR)	API-led (Real-time)	Simple
Scheduling	Appointment	BookAppointment	Patient Portal (MyChart)	Scheduling (Epic Cadence)	MyChart REST API	Epic Cadence REST API (FHIR)	API-led (Real-time)	Simple
Pre-registration	Patient	UpdateDemographics	Patient Portal (MyChart)	EHR (Epic)	MyChart REST API	Epic REST API (FHIR)	API-led (Real-time)	Simple
Check-in	Appointment	ConfirmArrival	Kiosk (Phreesia)	Scheduling (Epic Cadence)	Phreesia REST API	Epic Cadence REST API (FHIR)	API-led (Real-time)	Medium
Check-in	Encounter	UpdateStatus	Kiosk (Phreesia)	EHR (Epic)	Phreesia REST API	Epic REST API (FHIR)	API-led (Real-time)	Medium
Check-in	Payment	CollectCopay	Kiosk (Phreesia)	Payment Processor	Phreesia Payment API	Payment Gateway REST API	API-led (Real-time)	Medium

Acceptance Criteria

AC ID	Description	Related Info Flow	Expected Outcome	Priority
AC-6.1-01	Data integrity: Appointment data transferred from Patient Portal to Scheduling via BookAppointment must be complete and consistent	BookAppointment	Appointment records in Scheduling match source; no data loss; reconciliation report confirms accuracy	Medium
AC-6.1-02	Data integrity: Check-in data transferred from Kiosk to EHR via ConfirmArrival must be complete and consistent	ConfirmArrival	Check-in records in EHR match source; no data loss; reconciliation report confirms accuracy	High
AC-6.1-03	Error handling: Failed scheduling transactions are logged with error context and trigger automatic retry or alert	BookAppointment	Errors logged with timestamp, HTTP status, request payload excerpt; retry up to 3 times; alert sent after consecutive failures	High

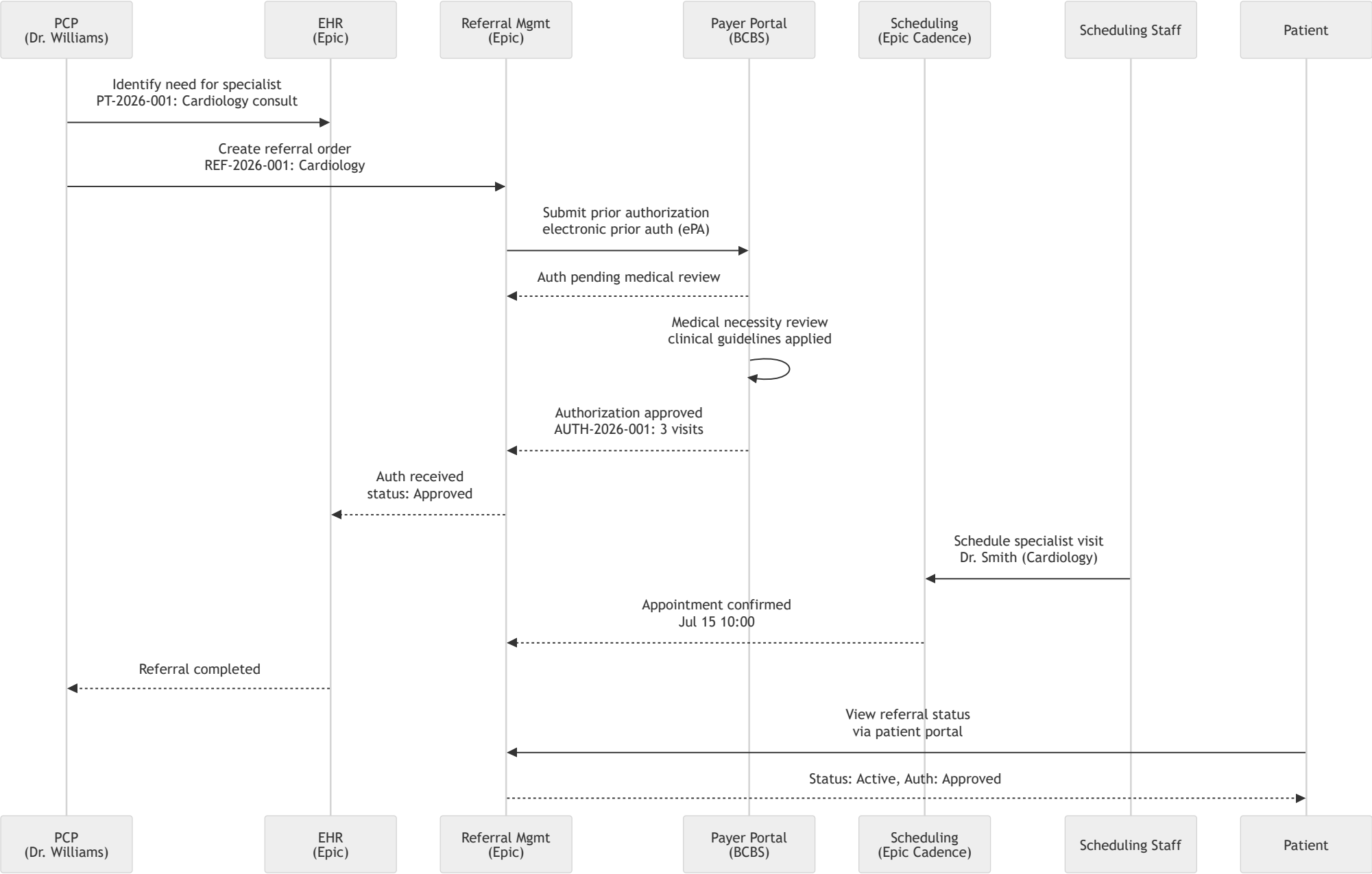
6.2 Referral Management & Authorizations

Description: A primary care physician refers a patient to a specialist. The referral is submitted, prior authorization is obtained, and the specialist appointment is scheduled.

Actors: PCP (Dr. Sarah Williams), Patient, Specialist Office, Payer

Systems: EHR (Epic), Referral Management (Epic), Payer Portal (BCBS), Scheduling (Epic Cadence)

Sequence Diagram



Integration Details

Flow	Entity	Info Flow	Source	Target	Source Conn	Target Conn	Pattern	Complexity
Referral	Referral	CreateReferral	EHR (Epic)	Referral Mgmt (Epic)	Epic REST API (FHIR)	Epic REST API (FHIR)	API-led (Real-time)	Simple
Referral	Referral	SubmitPriorAuth	Referral Mgmt (Epic)	Payer Portal (BCBS)	Epic REST API (FHIR)	BCBS ePA REST API (X12 278)	API-led (Real-time)	High
Referral	Referral	ReceiveAuthResponse	Payer Portal (BCBS)	Referral Mgmt (Epic)	BCBS ePA REST API (X12 278)	Epic REST API (FHIR)	API-led (Real-time)	High
Referral	Appointment	ScheduleSpecialist	Scheduling (Epic Cadence)	Referral Mgmt (Epic)	Epic Cadence REST API (FHIR)	Epic REST API (FHIR)	API-led (Real-time)	Medium
Referral	Referral	ViewReferralStatus	Patient Portal (MyChart)	Referral Mgmt (Epic)	MyChart REST API	Epic REST API (FHIR)	API-led (Real-time)	Simple

Acceptance Criteria

AC ID	Description	Related Info Flow	Expected Outcome	Priority
AC-6.2-01	Data integrity: Referral data transferred from EHR to Referral Management via CreateReferral must be complete and consistent	CreateReferral	Referral records in Referral Management match source; no data loss; reconciliation report confirms accuracy	High
AC-6.2-02	Data integrity: Authorization data transferred from Payer Portal to Referral Management via ReceiveAuthResponse must be complete and consistent	ReceiveAuthResponse	Authorization records in Referral Management match payer source; no data loss; reconciliation report confirms accuracy	Critical
AC-6.2-03	Error handling: Failed prior authorization submissions are logged with error context and trigger automatic retry or	SubmitPriorAuth	Errors logged with timestamp, HTTP status, request payload excerpt; retry up to 3 times; alert sent after consecutive failures	Critical

AC ID	Description	Related Info Flow	Expected Outcome	Priority
	alert			

Care Coordination & Population Health - HC060 HC061 HC062

Integrated care coordination including care plan management, transition of care, discharge planning, and population health risk stratification.

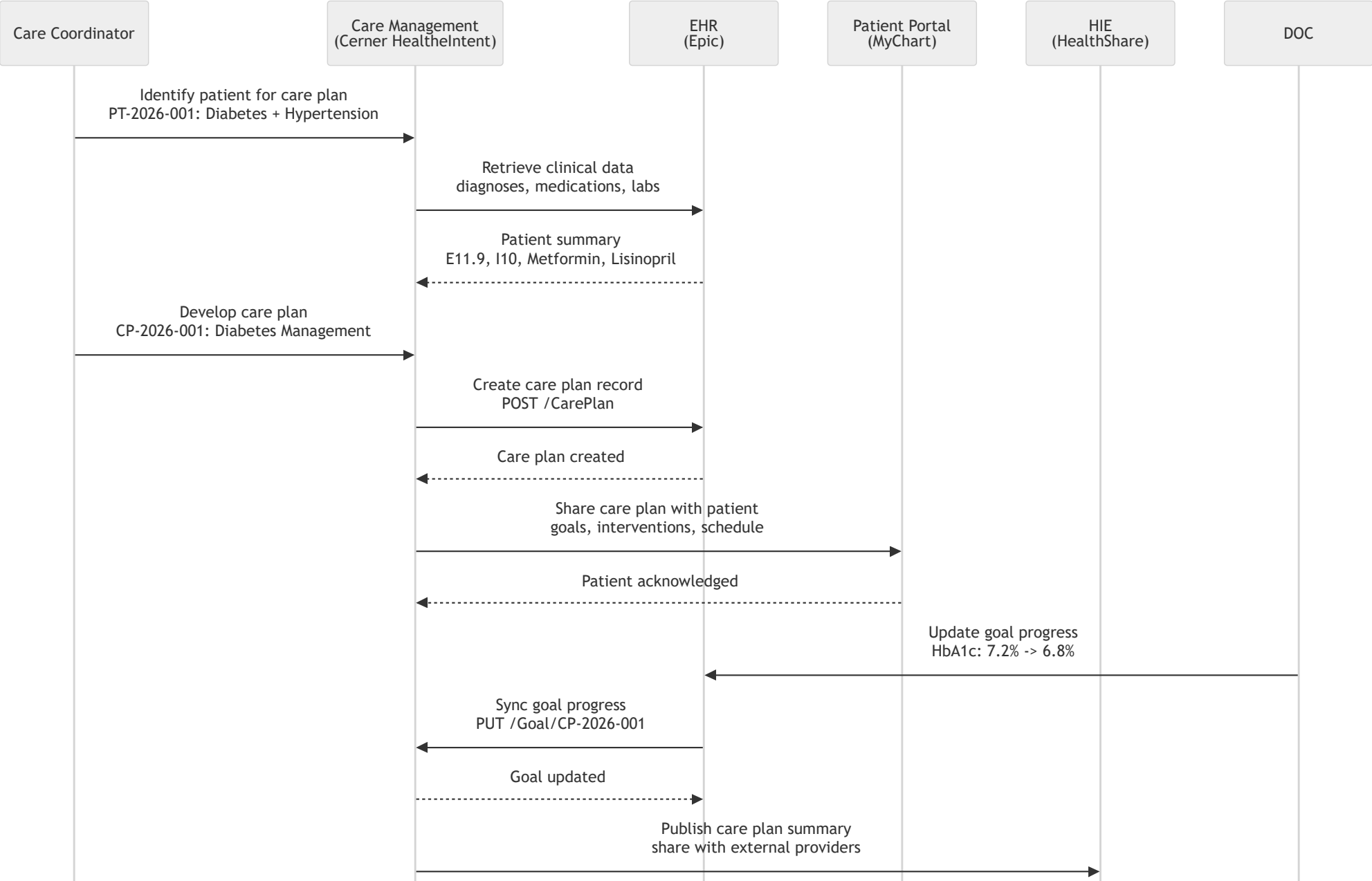
7.1 Care Plan Development & Management

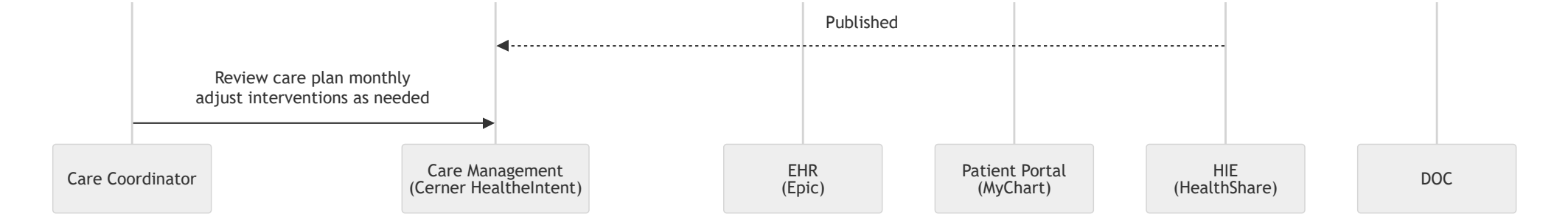
Description: An interdisciplinary team develops a comprehensive care plan for a patient with complex chronic conditions, sets goals, and tracks progress.

Actors: Care Coordinator, Physician (Dr. Sarah Williams), Patient

Systems: Care Management (Cerner HealtheIntent), EHR (Epic), Patient Portal (MyChart), HIE (HealthShare)

Sequence Diagram





Integration Details

Flow	Entity	Info Flow	Source	Target	Source Conn	Target Conn	Pattern	Complexity
Care Plan	Patient	RetrieveClinicalData	Care Management (Cerner HealthIntent)	EHR (Epic)	Cerner REST API	Epic REST API (FHIR)	API-led (Real-time)	Medium
Care Plan	CarePlan	CreateCarePlan	Care Management (Cerner HealthIntent)	EHR (Epic)	Cerner REST API	Epic REST API (FHIR)	API-led (Real-time)	Medium
Care Plan	CarePlan	ShareWithPatient	Care Management (Cerner HealthIntent)	Patient Portal (MyChart)	Cerner REST API	MyChart REST API	API-led (Real-time)	Medium
Care Plan	Goal	SyncGoalProgress	EHR (Epic)	Care Management (Cerner HealthIntent)	Epic REST API (FHIR)	Cerner REST API	Event-driven	Medium
Care Plan	CarePlan	PublishToHIE	Care Management (Cerner HealthIntent)	HIE (HealthShare)	Cerner REST API	HealthShare REST API (IHE XDS)	Batch (Scheduled)	High

Acceptance Criteria

AC ID	Description	Related Info Flow	Expected Outcome	Priority
AC-7.1-01	Data integrity: Care Plan data transferred from Care Management to EHR via CreateCarePlan must be complete and consistent	CreateCarePlan	Care Plan records in EHR match source; no data loss; reconciliation report confirms accuracy	High
AC-7.1-02	Data integrity: Goal progress data transferred from EHR to Care Management via SyncGoalProgress must be complete and consistent	SyncGoalProgress	Goal progress records in Care Management match source; no data loss; reconciliation report confirms accuracy	Medium
AC-7.1-03	Error handling: Failed care plan transactions are logged with error context and trigger automatic retry or alert	CreateCarePlan	Errors logged with timestamp, HTTP status, request payload excerpt; retry up to 3 times; alert sent after consecutive failures	High

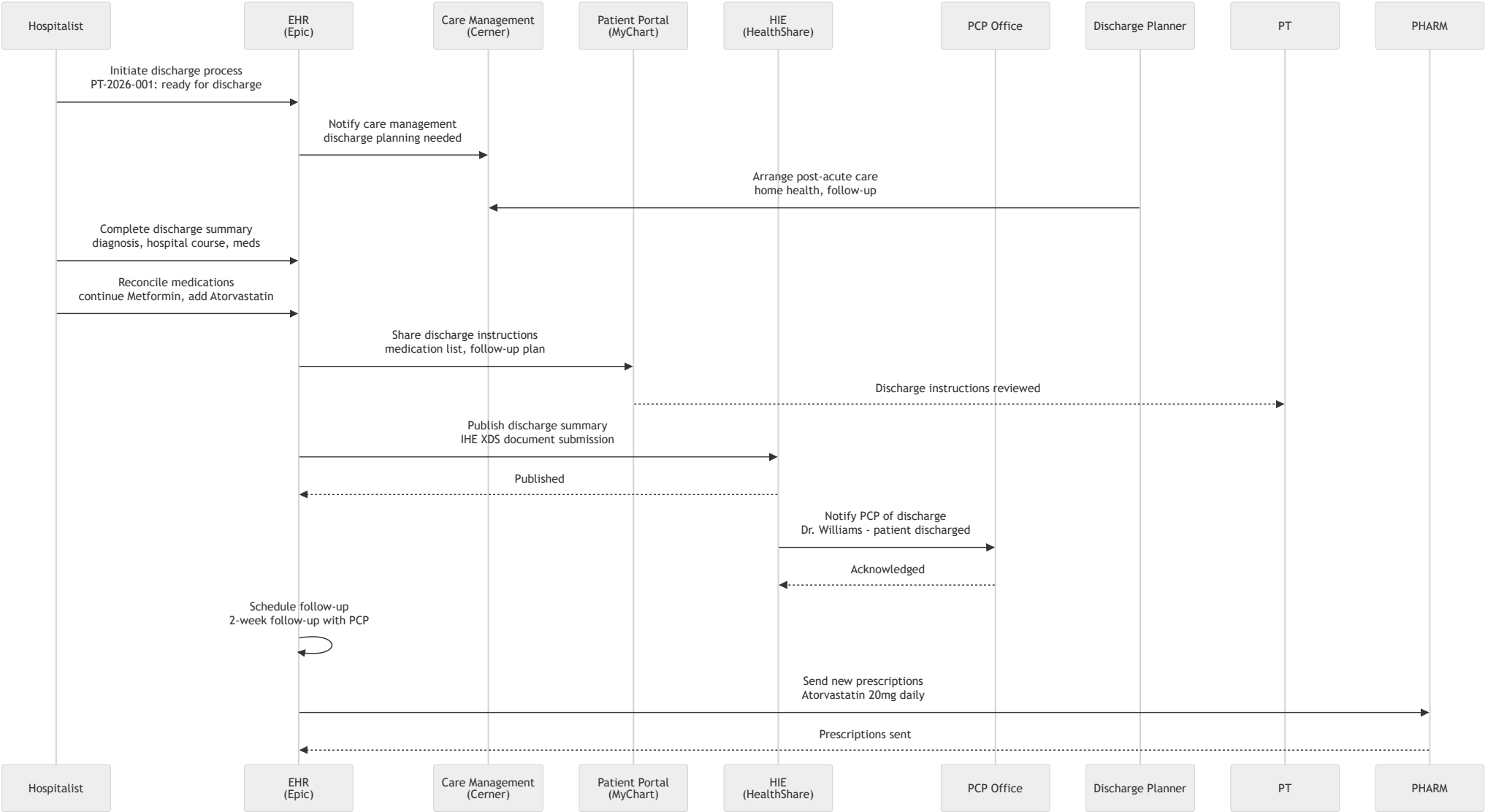
7.2 Transition of Care & Discharge

Description: A patient is discharged from inpatient care. Discharge summary is created, medications are reconciled, follow-up is scheduled, and the summary is shared with the PCP.

Actors: Hospitalist, Patient, PCP, Discharge Planner

Systems: EHR (Epic), Care Management (Cerner HealthIntent), Patient Portal (MyChart), HIE (HealthShare), Pharmacy (BD Pyxis)

Sequence Diagram



Integration Details

Flow	Entity	Info Flow	Source	Target	Source Conn	Target Conn	Pattern	Complexity
Discharge	Encounter	NotifyDischargePlanning	EHR (Epic)	Care Management (Cerner HealthIntent)	Epic REST API (FHIR)	Cerner REST API	Event-driven	Medium
Discharge	Document	CompleteDischargeSummary	EHR (Epic)	EHR (Epic)	Epic REST API (FHIR)	Epic REST API (FHIR)	API-led (Real-time)	Simple
Discharge	Medication	ReconcileMedications	EHR (Epic)	EHR (Epic)	Epic REST API (FHIR)	Epic REST API (FHIR)	API-led (Real-time)	Medium
Discharge	Document	PublishDischargeSummary	EHR (Epic)	HIE (HealthShare)	Epic REST API (FHIR)	HealthShare REST API (IHE XDS)	API-led (Real-time)	High
Discharge	Document	NotifyPCP	HIE (HealthShare)	PCP Office	HealthShare REST API (IHE XDS)	PCP EHR REST API (FHIR)	Event-driven	High
Discharge	Medication	SendNewPrescriptions	EHR (Epic)	Pharmacy System (BD Pyxis)	Epic REST API (FHIR)	Pyxis NCPDP SCRIPT (SOAP)	API-led (Real-time)	Medium

Acceptance Criteria

AC ID	Description	Related Info Flow	Expected Outcome	Priority
AC-7.2-01	Data integrity: Discharge summary data transferred from EHR to HIE via PublishDischargeSummary must be complete and consistent	PublishDischargeSummary	Discharge summaries in HIE match source; no data loss; reconciliation report confirms accuracy	Critical
AC-7.2-02	Data integrity: Medication reconciliation data must be complete and consistent across all systems	ReconcileMedications	Medication list matches at discharge; discrepancies flagged for review	Critical
AC-7.2-03	Error handling: Failed discharge summary publication transactions are logged with error context and trigger automatic retry or alert	PublishDischargeSummary	Errors logged with timestamp, HTTP status, request payload excerpt; retry up to 3 times; alert sent after consecutive failures	High

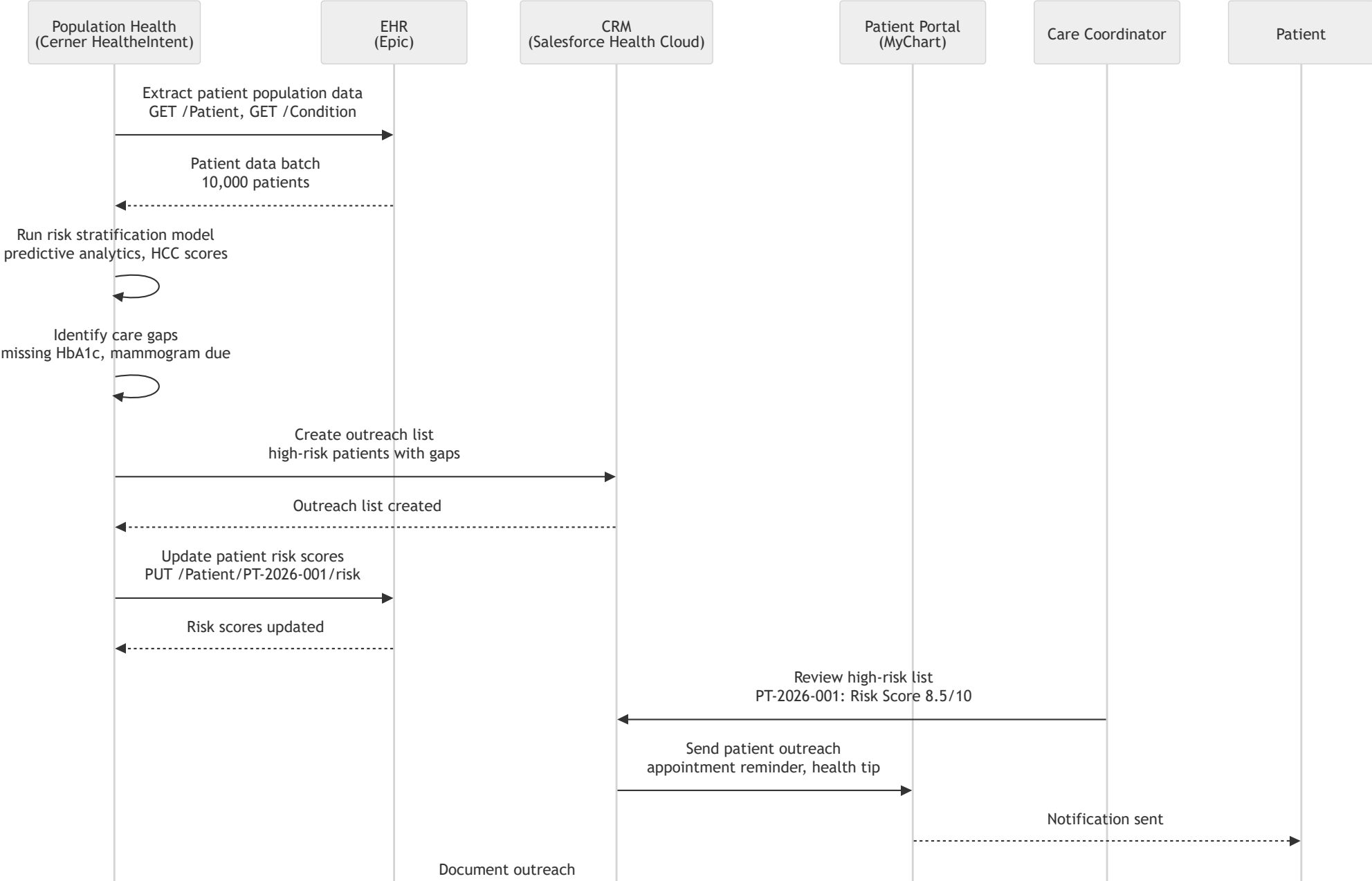
7.3 Population Health Risk Stratification

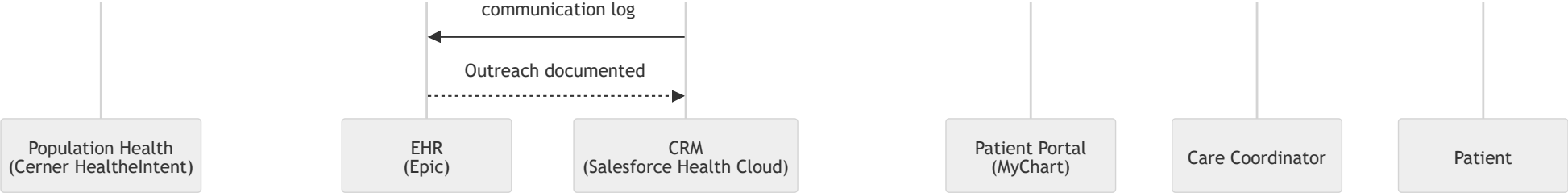
Description: The population health platform analyzes patient data to stratify risk, identify gaps in care, and trigger outreach for high-risk patients.

Actors: Population Health Analyst, Care Coordinator

Systems: Population Health (Cerner HealthIntent), EHR (Epic), CRM (Salesforce Health Cloud), Patient Portal (MyChart)

Sequence Diagram





Integration Details

Flow	Entity	Info Flow	Source	Target	Source Conn	Target Conn	Pattern	Complexity
Population Health	Patient	ExtractPatientData	Population Health (Cerner HealtheIntent)	EHR (Epic)	Cerner REST API	Epic REST API (FHIR)	Batch (Scheduled)	High
Population Health	Patient	RunRiskStratification	Population Health (Cerner HealtheIntent)	Population Health (Cerner HealtheIntent)	Cerner Analytics Engine	Cerner Analytics Engine	Batch (Scheduled)	High
Population Health	Patient	CreateOutreachList	Population Health (Cerner HealtheIntent)	CRM (Salesforce Health Cloud)	Cerner REST API	Salesforce REST API (OAuth2)	Event-driven	Medium
Population Health	Patient	UpdateRiskScores	Population Health (Cerner HealtheIntent)	EHR (Epic)	Cerner REST API	Epic REST API (FHIR)	Batch (Scheduled)	Medium
Population Health	Patient	SendPatientOutreach	CRM (Salesforce Health Cloud)	Patient Portal (MyChart)	Salesforce REST API	MyChart REST API	Event-driven	Medium
Population Health	Patient	DocumentOutreach	CRM (Salesforce Health Cloud)	EHR (Epic)	Salesforce REST API	Epic REST API (FHIR)	API-led (Real-time)	Medium

Acceptance Criteria

AC ID	Description	Related Info Flow	Expected Outcome	Priority
AC-7.3-01	Data integrity: Patient population data transferred from EHR to Population Health via ExtractPatientData must be complete and consistent	ExtractPatientData	Patient data in Population Health matches source; no data loss; reconciliation report confirms accuracy	Critical
AC-7.3-02	Data integrity: Risk score data transferred from Population Health to EHR via UpdateRiskScores must be complete and consistent	UpdateRiskScores	Risk scores in EHR match source; no data loss; reconciliation report confirms accuracy	High
AC-7.3-03	Error handling: Failed population health data extraction transactions are logged with error	ExtractPatientData	Errors logged with timestamp, HTTP status, request payload excerpt; retry up to 3 times; alert sent after consecutive failures	High

AC ID	Description	Related Info Flow	Expected Outcome	Priority
	context and trigger automatic retry or alert			

Health Information Exchange - HC070 HC071

Interoperability services including clinical document sharing, query across organizational boundaries, and public health reporting.

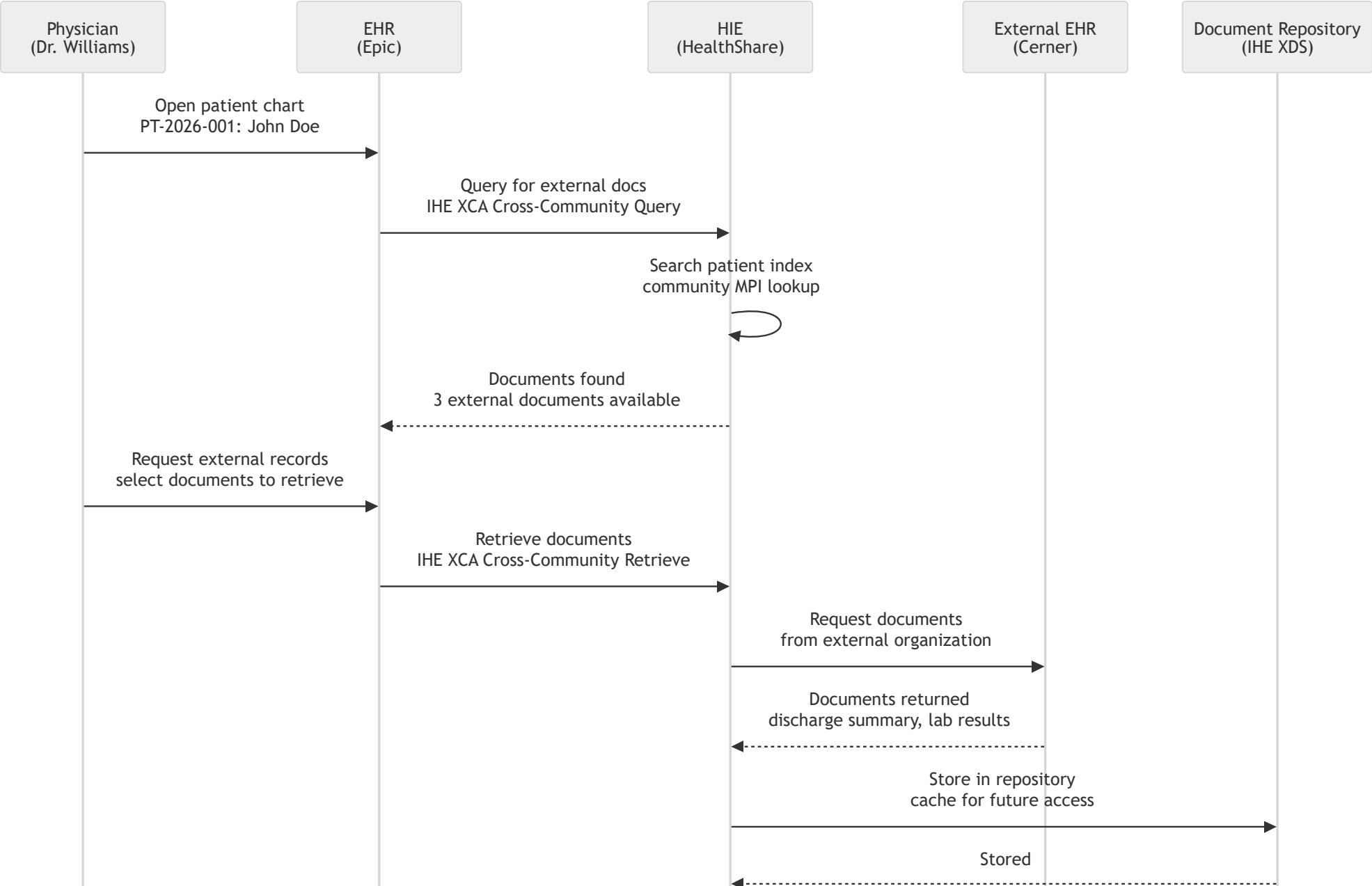
8.1 HIE Document Sharing & Query

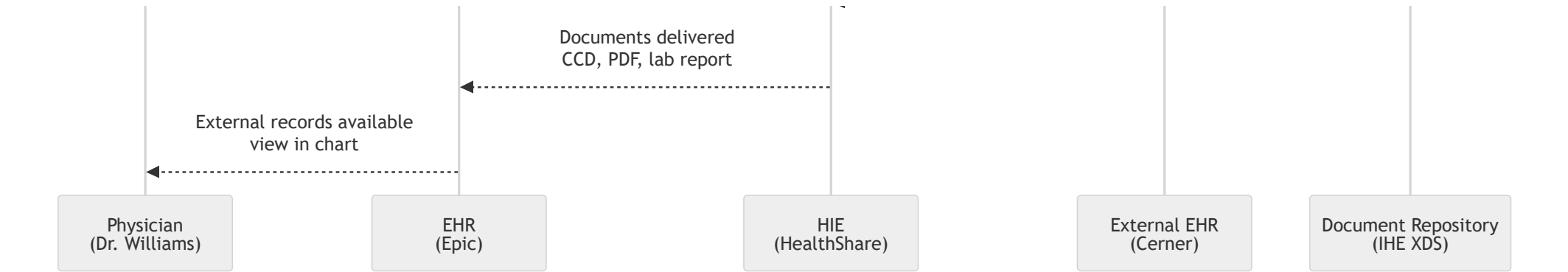
Description: A provider queries the health information exchange for a patient's external records. Documents are retrieved from other organizations and incorporated into the local EHR.

Actors: Physician (Dr. Sarah Williams), HIE Operator

Systems: EHR (Epic), HIE (HealthShare), External EHR (Cerner), Document Repository (IHE XDS Registry)

Sequence Diagram





Integration Details

Flow	Entity	Info Flow	Source	Target	Source Conn	Target Conn	Pattern	Complexity
HIE	Document	QueryExternalDocuments	EHR (Epic)	HIE (HealthShare)	Epic REST API (FHIR)	HealthShare REST API (IHE XCA)	API-led (Real-time)	High
HIE	Document	RetrieveDocuments	EHR (Epic)	HIE (HealthShare)	Epic REST API (FHIR)	HealthShare REST API (IHE XCA)	API-led (Real-time)	High
HIE	Document	RequestFromExternal	HIE (HealthShare)	External EHR (Cerner)	HealthShare REST API (IHE XCA)	Cerner REST API (FHIR)	API-led (Real-time)	High
HIE	Document	StoreInRepository	HIE (HealthShare)	Document Repository (IHE XDS)	HealthShare REST API	XDS SOAP (IHE XDS)	API-led (Real-time)	Medium
HIE	Document	DeliverToEHR	HIE (HealthShare)	EHR (Epic)	HealthShare REST API (IHE XCA)	Epic REST API (FHIR)	API-led (Real-time)	Medium

Acceptance Criteria

AC ID	Description	Related Info Flow	Expected Outcome	Priority
AC-8.1-01	Data integrity: Document data transferred from HIE to EHR via DeliverToEHR must be complete and consistent	DeliverToEHR	Document records in EHR match source; no data loss; reconciliation report confirms accuracy	Critical
AC-8.1-02	Data integrity: Query results transferred from HIE to EHR via QueryExternalDocuments must be complete and consistent	QueryExternalDocuments	Query results in EHR match source; no data loss; reconciliation report confirms accuracy	High
AC-8.1-03	Security: All endpoints enforce authentication (HealthShare REST API (IHE XCA), Epic REST API (FHIR), ...) and encrypt data in transit via TLS 1.2+	All flows	Unauthenticated requests rejected with 401; tokens validated per call; invalid/expired tokens logged	Critical
AC-8.1-04	Privacy: Patient consent verified before disclosure of external records	All flows	Consent directive checked; disclosure blocked if opt-out status detected	Critical

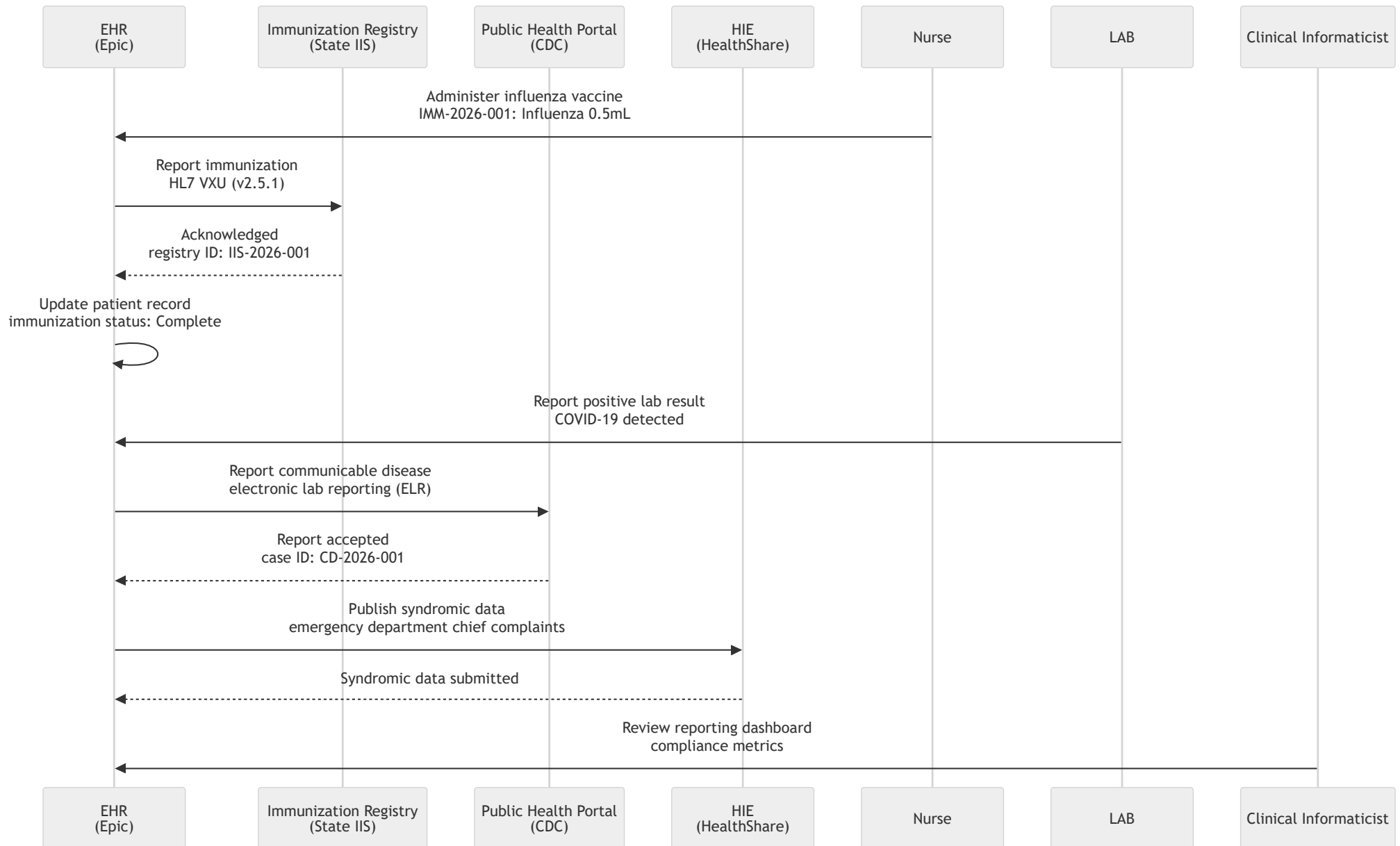
8.2 Public Health Reporting & Immunization

Description: Immunization records are reported to the state immunization registry and communicable disease reports are submitted to public health authorities.

Actors: Clinical Informaticist, Public Health Agency

Systems: EHR (Epic), Immunization Registry (State IIS), Public Health Portal (CDC), HIE (HealthShare)

Sequence Diagram



Integration Details

Flow	Entity	Info Flow	Source	Target	Source Conn	Target Conn	Pattern	Complexity
Immunization	Immunization	ReportImmunization	EHR (Epic)	Immunization Registry (State IIS)	Epic REST API (FHIR)	State IIS HL7 VXU (SOAP)	API-led (Real-time)	Medium
Public Health	Observation	ReportCommunicableDisease	EHR (Epic)	Public Health Portal (CDC)	Epic REST API (FHIR)	CDC ELR REST API (HL7 FHIR)	Event-driven	High
Public Health	Encounter	SubmitSyndromicData	EHR (Epic)	HIE (HealthShare)	Epic REST API (FHIR)	HealthShare REST API (IHE SD)	Batch (Scheduled)	Medium
Public Health	Immunization	ReviewCompliance	Clinical Informaticist	EHR (Epic)	Epic UI (Web)	Epic REST API (FHIR)	API-led (Real-time)	Simple
Immunization	Immunization	QueryRegistry	EHR (Epic)	Immunization Registry (State IIS)	Epic REST API (FHIR)	State IIS HL7 VXQ (SOAP)	API-led (Real-time)	Medium

Acceptance Criteria

AC ID	Description	Related Info Flow	Expected Outcome	Priority
AC-8.2-01	Data integrity: Immunization data transferred from EHR to Immunization Registry via ReportImmunization must be complete and consistent	ReportImmunization	Immunization records in Registry match source; no data loss; reconciliation report confirms accuracy	High
AC-8.2-02	Data integrity: Communicable disease data transferred from EHR to CDC via ReportCommunicableDisease must be complete and consistent	ReportCommunicableDisease	Disease reports in CDC match source; no data loss; reconciliation report confirms accuracy	Critical
AC-8.2-03	Error handling: Failed public health reporting transacti	ReportCommunicableDisease	Errors logged with timestamp, HTTP status, request payload excerpt; retry up to 3 times; alert sent after consecutive failures	Critical

AC ID	Description	Related Info Flow	Expected Outcome	Priority
	ons are logged with error context and trigger automatic retry or alert			

Supply Chain & Inventory Management - HC080 HC081

Medical supply chain management including procurement, inventory tracking, asset management, and vendor coordination.

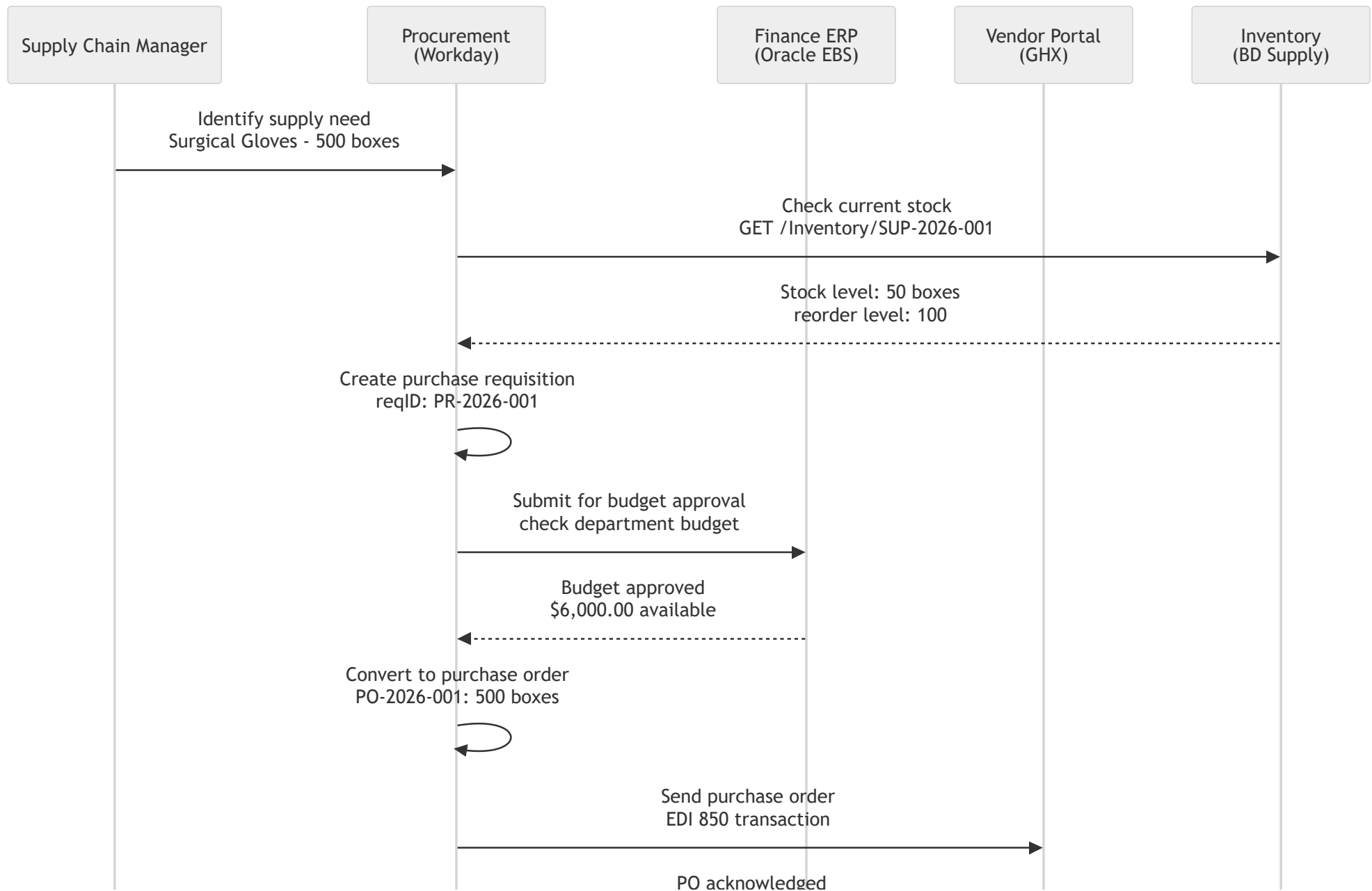
9.1 Medical Supply Procurement

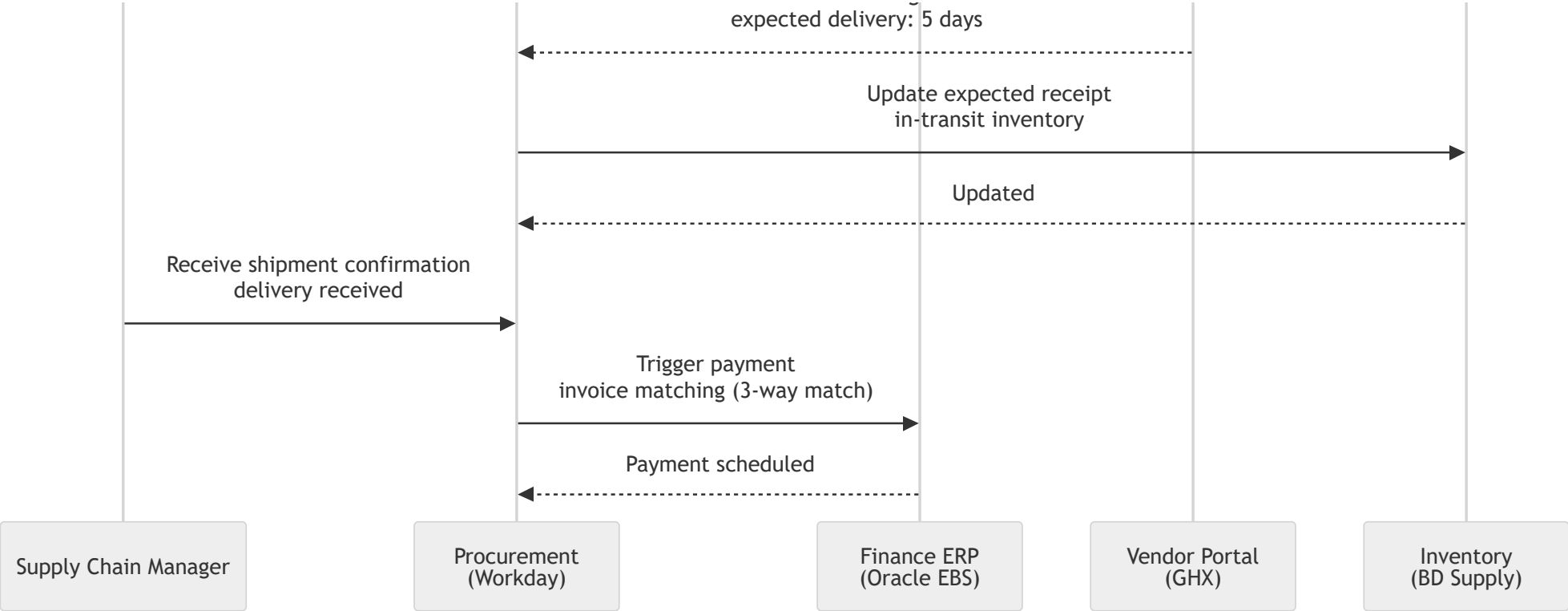
Description: The supply chain department identifies a need for medical supplies, creates a purchase order, receives vendor approval, and manages the procurement lifecycle.

Actors: Supply Chain Manager, Vendor

Systems: Procurement (Workday), Finance ERP (Oracle EBS), Vendor Portal (Global Healthcare Exchange), Inventory (BD Supply)

Sequence Diagram





Integration Details

Flow	Entity	Info Flow	Source	Target	Source Conn	Target Conn	Pattern	Complexity
Procurement	SupplyItem	CheckStockLevel	Procurement (Workday)	Inventory (BD Supply)	Workday REST API	BD Supply REST API (GS1)	API-led (Real-time)	Simple
Procurement	PurchaseOrder	SubmitBudgetApproval	Procurement (Workday)	Finance ERP (Oracle EBS)	Workday REST API	Oracle EBS SOAP (EDI)	API-led (Real-time)	Medium
Procurement	PurchaseOrder	SendPO (EDI 850)	Procurement (Workday)	Vendor Portal (GHX)	Workday EDI Gateway	GHX EDI (AS2)	Batch (Scheduled)	High
Procurement	PurchaseOrder	UpdateInventoryExpected	Procurement (Workday)	Inventory (BD Supply)	Workday REST API	BD Supply REST API (GS1)	Event-driven	Medium
Procurement	PurchaseOrder	TriggerPayment	Procurement (Workday)	Finance ERP (Oracle EBS)	Workday REST API	Oracle EBS SOAP (EDI)	Event-driven	Medium

Acceptance Criteria

AC ID	Description	Related Info Flow	Expected Outcome	Priority
AC-9.1-01	Data integrity: Purchase Order data transferred from Procurement to Vendor via SendPO must be complete and consistent	SendPO	Purchase Orders in Vendor Portal match source; no data loss; reconciliation report confirms accuracy	High
AC-9.1-02	Data integrity: Budget approval data transferred from Procurement to Finance via SubmitBudgetApproval must be complete and consistent	SubmitBudgetApproval	Budget records in Finance match source; no data loss; reconciliation report confirms accuracy	Critical
AC-9.1-03	Error handling: Failed procurement transactions are logged with error context and trigger automatic retry or alert	SendPO	Errors logged with timestamp, HTTP status, EDI status; retry up to 3 times; alert sent after consecutive failures	High

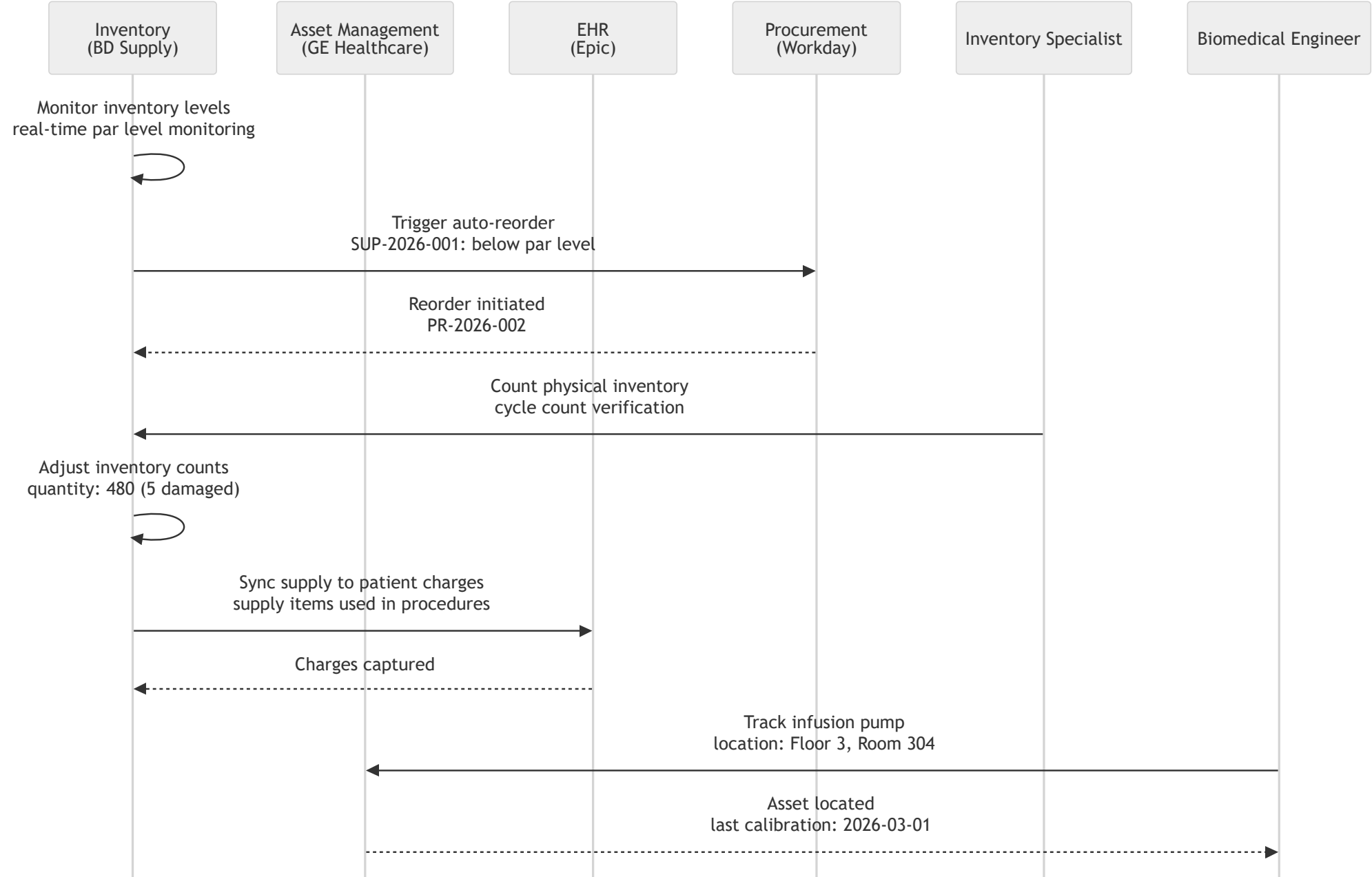
9.2 Inventory & Asset Management

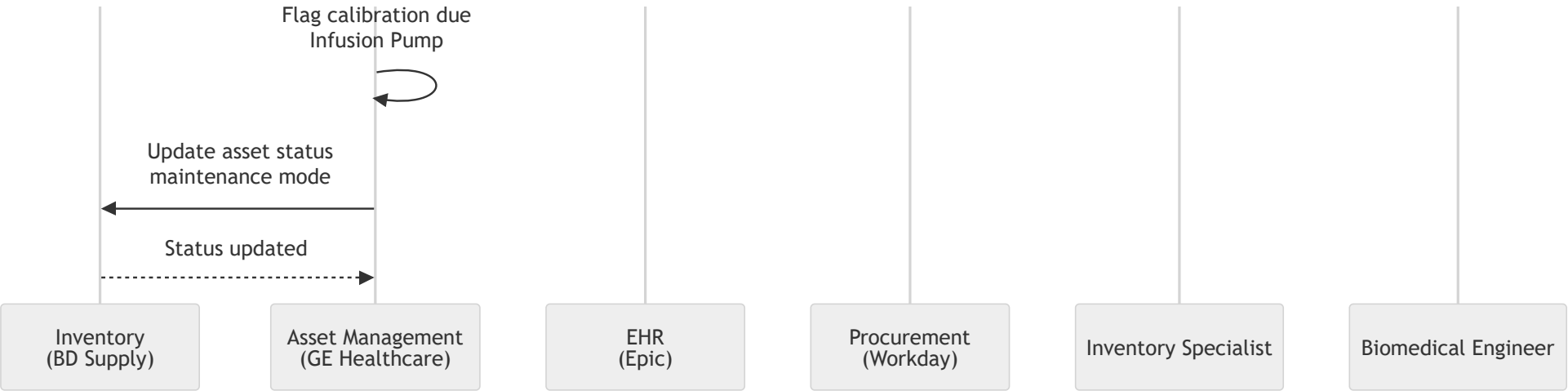
Description: Medical inventory levels are monitored, par levels are maintained, expiring items are flagged, and medical equipment assets are tracked across the facility.

Actors: Inventory Specialist, Biomedical Engineer

Systems: Inventory (BD Supply), Asset Management (GE Healthcare), EHR (Epic), Procurement (Workday)

Sequence Diagram





Integration Details

Flow	Entity	Info Flow	Source	Target	Source Conn	Target Conn	Pattern	Complexity
Inventory	SupplyItem	MonitorParLevels	Inventory (BD Supply)	Inventory (BD Supply)	BD Supply REST API	BD Supply REST API	API-led (Real-time)	Medium
Inventory	SupplyItem	TriggerAutoReorder	Inventory (BD Supply)	Procurement (Workday)	BD Supply REST API (GS1)	Workday REST API	Event-driven	Medium
Inventory	SupplyItem	SyncSupplyToCharges	Inventory (BD Supply)	EHR (Epic)	BD Supply REST API	Epic REST API (FHIR)	Batch (Scheduled)	Medium
Asset Mgmt	Asset	TrackAssetLocation	Biomedical Engineer	Asset Management (GE Healthcare)	GE Asset UI	GE Healthcare REST API	API-led (Real-time)	Simple
Asset Mgmt	Asset	FlagCalibrationDue	Asset Management (GE Healthcare)	Asset Management (GE Healthcare)	GE Healthcare REST API	GE Healthcare REST API	Event-driven	Medium
Asset Mgmt	Asset	UpdateAssetStatus	Asset Management (GE Healthcare)	Inventory (BD Supply)	GE Healthcare REST API	BD Supply REST API (GS1)	Event-driven	Medium

Acceptance Criteria

AC ID	Description	Related Info Flow	Expected Outcome	Priority
AC-9.2-01	Data integrity: Inventory level data monitored via MonitorParLevels must be accurate and up-to-date	MonitorParLevels	Inventory levels match physical counts within tolerance; discrepancies flagged for review	High
AC-9.2-02	Data integrity: Supply charge data transferred from Inventory to EHR via SyncSupplyToCharges must be complete and consistent	SyncSupplyToCharges	Supply charges in EHR match inventory usage; no data loss; reconciliation report confirms accur	High

AC ID	Description	Related Info Flow	Expected Outcome	Priority
			acy	
AC-9.2-03	Error handling: Failed auto-reorder transactions are logged with error context and trigger manual review	TriggerAutoReorder	Errors logged with timestamp, stock level, item ID; manual review queue populated	Medium

Regulatory Compliance & Quality - HC090 HC091 HC092

Comprehensive regulatory compliance, quality measurement, patient safety incident reporting, and accreditation management.

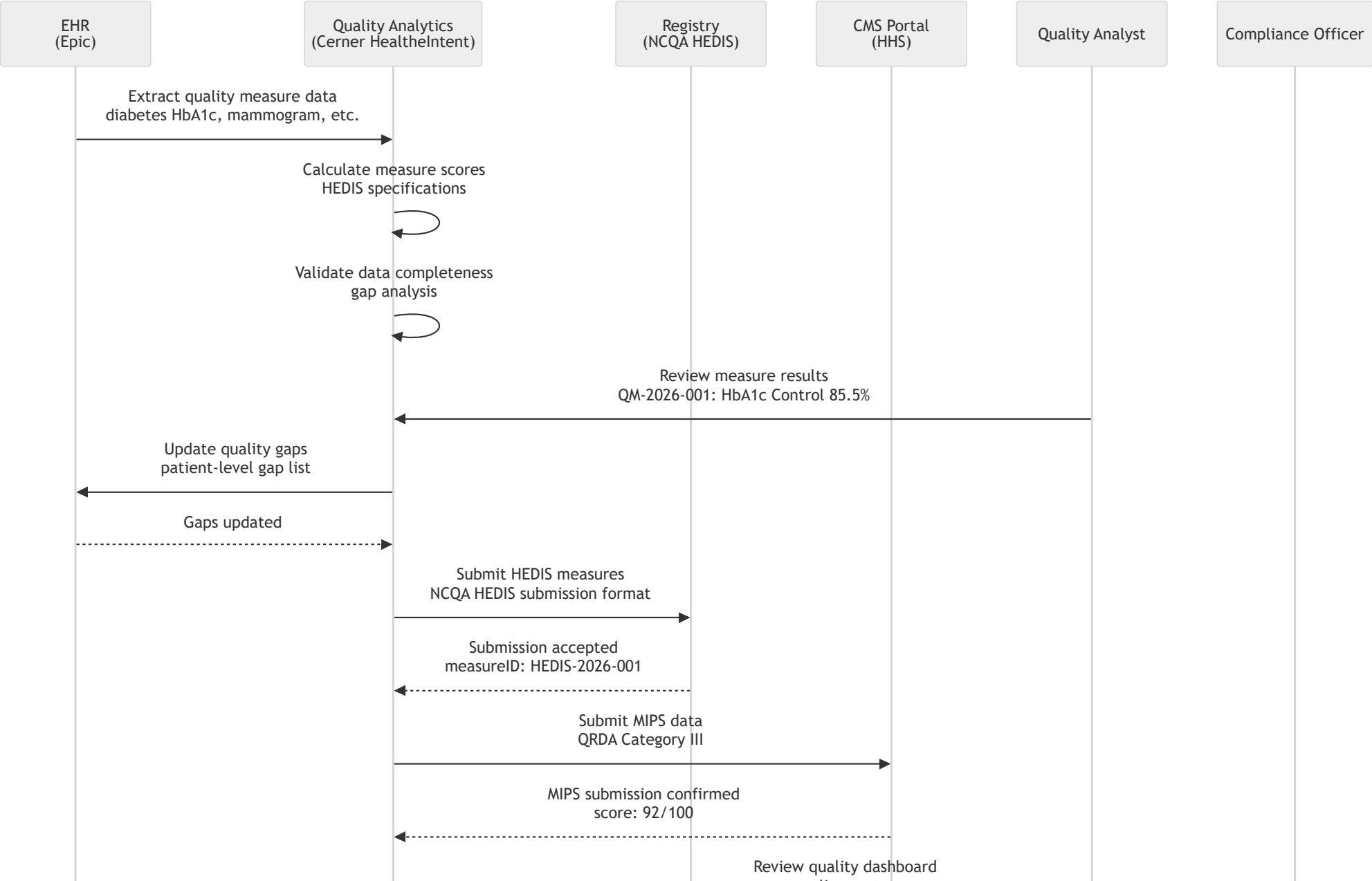
10.1 Quality Measure Reporting (HEDIS/MIPS)

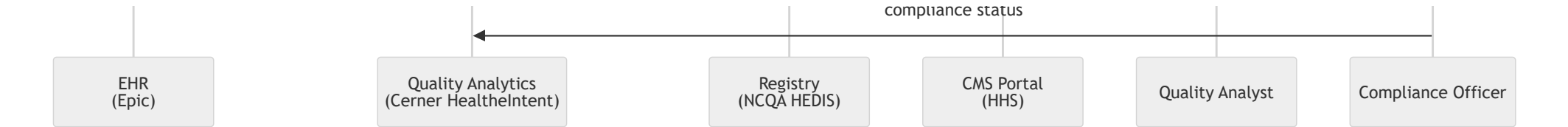
Description: Quality measure data is extracted from the EHR, calculated against HEDIS and MIPS specifications, validated, and submitted to regulatory bodies.

Actors: Quality Analyst, Compliance Officer

Systems: EHR (Epic), Quality Analytics (Cerner HealtheIntent), Registry (NCQA), CMS Portal (HHS)

Sequence Diagram





Integration Details

Flow	Entity	Info Flow	Source	Target	Source Conn	Target Conn	Pattern	Complexity
Quality	Measure	ExtractMeasureData	EHR (Epic)	Quality Analytics (Cerner HealthIntent)	Epic REST API (FHIR)	Cerner REST API	Batch (ETL)	High
Quality	Measure	CalculateMeasureScores	Quality Analytics (Cerner HealthIntent)	Quality Analytics (Cerner HealthIntent)	Cerner Analytics Engine	Cerner Analytics Engine	Batch (Scheduled)	High
Quality	Measure	UpdateQualityGaps	Quality Analytics (Cerner HealthIntent)	EHR (Epic)	Cerner REST API	Epic REST API (FHIR)	Batch (Scheduled)	Medium
Quality	Measure	SubmitHEDIS	Quality Analytics (Cerner HealthIntent)	Registry (NCQA HEDIS)	Cerner REST API	NCQA SFTP (EDI)	Batch (Scheduled)	High
Quality	Measure	SubmitMIPS	Quality Analytics (Cerner HealthIntent)	CMS Portal (HHS)	Cerner REST API	CMS REST API (QRDA)	Batch (Scheduled)	High

Acceptance Criteria

AC ID	Description	Related Info Flow	Expected Outcome	Priority
AC-10.1-01	Data integrity: Quality measure data transferred from EHR to Quality Analytics via ExtractMeasureData must be complete and consistent	ExtractMeasureData	Quality measure records in Quality Analytics match source; no data loss; reconciliation report confirms accuracy	Critical
AC-10.1-02	Data integrity: HEDIS submission data transferred from Quality Analytics to NCQA via SubmitHEDIS must be complete and consistent	SubmitHEDIS	HEDIS submission accepted by NCQA; no data loss; submission report confirms accuracy	Critical
AC-10.1-03	Error handling: Failed quality measure reporting transactions are logged with error context and trigger automatic retry or alert	SubmitHEDIS	Errors logged with timestamp, HTTP status, measure payload excerpt; retry up to 3 times; alert sent after consecutive failures	Critical

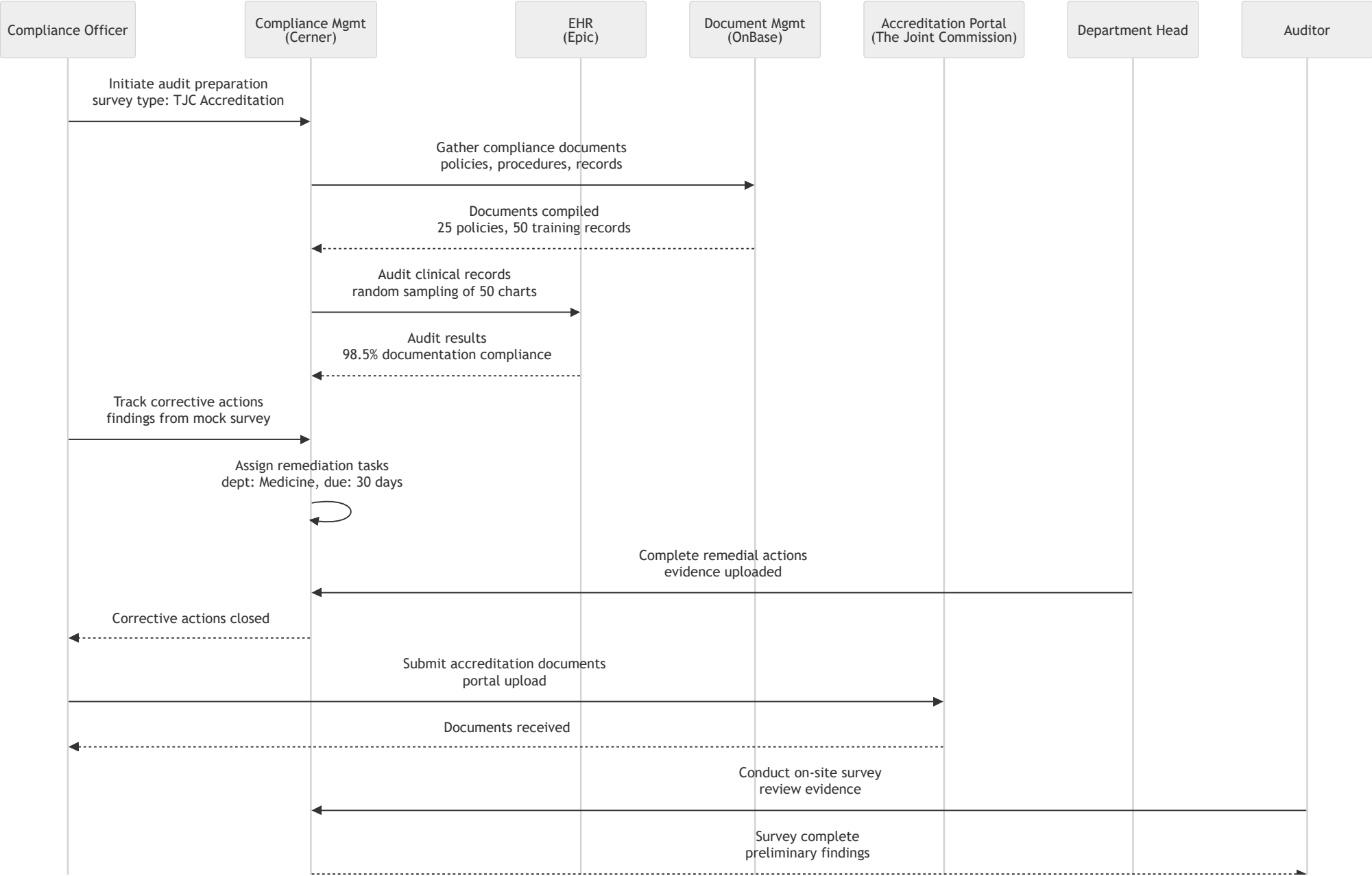
10.2 Regulatory Audit & Compliance

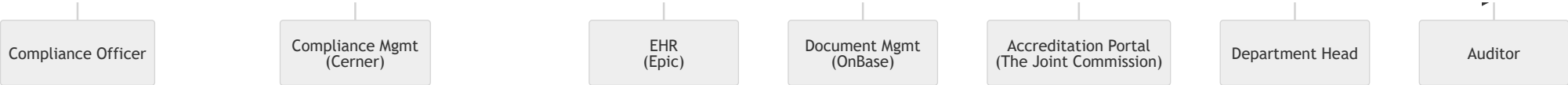
Description: The organization prepares for a regulatory audit by gathering compliance documentation, tracking corrective actions, and managing accreditation surveys.

Actors: Compliance Officer, Auditor, Department Head

Systems: Compliance Management (Cerner), EHR (Epic), Document Management (OnBase), Accreditation Portal (The Joint Commission)

Sequence Diagram





Integration Details

Flow	Entity	Info Flow	Source	Target	Source Conn	Target Conn	Pattern	Complexity
Audit	Document	GatherComplianceDocs	Compliance Mgmt (Cerner)	Document Mgmt (OnBase)	Cerner REST API	OnBase REST API (CMIS)	API-led (Real-time)	Medium
Audit	Encounter	AuditClinicalRecords	Compliance Mgmt (Cerner)	EHR (Epic)	Cerner REST API	Epic REST API (FHIR)	API-led (Real-time)	Medium
Audit	Incident	TrackCorrectiveActions	Compliance Mgmt (Cerner)	Compliance Mgmt (Cerner)	Cerner REST API	Cerner REST API	API-led (Real-time)	Medium
Audit	Document	SubmitAccreditationDocs	Compliance Officer	Accreditation Portal (The Joint Commission)	TJC Portal Web UI	TJC REST API	Batch (Scheduled)	Medium
Audit	Incident	ConductOnSiteSurvey	Auditor	Compliance Mgmt (Cerner)	Auditor Mobile App	Cerner REST API	API-led (Real-time)	Medium

Acceptance Criteria

AC ID	Description	Related Info Flow	Expected Outcome	Priority
AC-10.2-01	Data integrity: Compliance document data transferred from Document Mgmt to Compliance Mgmt via GatherComplianceDocs must be complete and consistent	GatherComplianceDocs	Compliance documents indexed correctly; no data loss; audit trail confirms accuracy	High
AC-10.2-02	Data integrity: Clinical audit data transferred from EHR to Compliance Mgmt via AuditClinicalRecords must be complete and consistent	AuditClinicalRecords	Audit results accurately reflect chart data; no data loss; reconciliation report confirms accuracy	Critical
AC-10.2-03	Error handling: Failed accreditation document submission transactions are logged with error context and trigger manual intervention	SubmitAccreditationDocs	Errors logged with timestamp, document ID, portal status; manual review triggered	High

10.3 Incident & Patient Safety Reporting

Description: A patient safety incident is reported, investigated, and tracked through resolution. Root cause analysis is performed and preventive actions are implemented.

Actors: Nurse, Risk Manager, Patient Safety Officer

Systems: Incident Management (RL Solutions), EHR (Epic), Risk Management (Origami Risk), Quality Analytics (Cerner HealtheIntent)

Sequence Diagram





Integration Details

Flow	Entity	Info Flow	Source	Target	Source Conn	Target Conn	Pattern	Complexity
Incident	Incident	ReportIncident	Nurse	Incident Mgmt (RL Solutions)	RL Solutions Web UI	RL Solutions REST API	API-led (Real-time)	Simple
Incident	Incident	FlagPatientRecord	Incident Mgmt (RL Solutions)	EHR (Epic)	RL Solutions REST API	Epic REST API (FHIR)	Event-driven	Medium
Incident	Incident	PerformRootCauseAnalysis	Incident Mgmt (RL Solutions)	Risk Management (Origami Risk)	RL Solutions REST API	Origami Risk REST API	API-led (Real-time)	High
Incident	Incident	UpdatePatientSafetyPlan	Incident Mgmt (RL Solutions)	EHR (Epic)	RL Solutions REST API	Epic REST API (FHIR)	API-led (Real-time)	Medium
Incident	Incident	SubmitForTrending	Incident Mgmt (RL Solutions)	Quality Analytics (Cerner HealthIntent)	RL Solutions REST API	Cerner REST API	Batch (Scheduled)	Medium

Acceptance Criteria

AC ID	Description	Related Info Flow	Expected Outcome	Priority
AC-10.3-01	Data integrity: Incident data transferred from Nurse to Incident Mgmt via ReportIncident must be complete and consistent	ReportIncident	Incident records in RL Solutions match source; no data loss; reconciliation report confirms accuracy	High
AC-10.3-02	Data integrity: RCA data transferred from Incident Mgmt to Risk Management via PerformRootCauseAnalysis must be complete and consistent	PerformRootCauseAnalysis	RCA records in Risk Management match source; no data loss; reconciliation report confirms accuracy	High
AC-10.3-03	Error handling: Failed incident reporting transactions are logged with error context and trigger manual review	ReportIncident	Errors logged with timestamp, HTTP status, incident details; manual review queue populated	Critical